SweeperCAD

User manual

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Welcome

Everything here is a **sweep**: a cross-section pushed along a path. You draw the parts in 2D graphs and the 3D shape follows.

New here? **Help ▸ First steps tour** walks you through your first object in about two minutes - one spot at a time, each with one thing to try. It also opens at startup until you tick *don't show this tour at startup again*.

The floating panels - placement settings, section, measure - can be **dragged by their title**, and **double-clicking the title folds** them to a single line. They remember where you put them.

An imported reference **reloads when its file changes on disk**. Export over the same file from Blender, ZBrush, a scanner, anything - and it updates here within a couple of seconds, with your section planes and fits still where you left them.

When a scene goes over its budget a small **detail reduced** note appears in the 3D view; click it to change the budget. It is not there at full quality.

The **rebuild delay** in Settings defaults to 250 ms and goes as low as 20 - lower means the model follows your edits more closely, at the cost of rebuilding more often.

Display detail in Settings sets how many triangles the VIEW may use for the whole scene. As a scene fills up everything is drawn a little coarser to stay near that number, and gets its detail back when the scene empties. Raise it on a fast machine.

It is the view only: **exported files are always built at full quality**, whatever this is set to.

The **rebuild delay** (250 ms by default) - how long the app waits after an edit before rebuilding - lives in Settings. **Pause** stays in the 3D view, where you can reach it while a rebuild is running.

Settings > Geometry kernel holds the kernel's switches, all on by default; turning one off or on rebuilds everything:

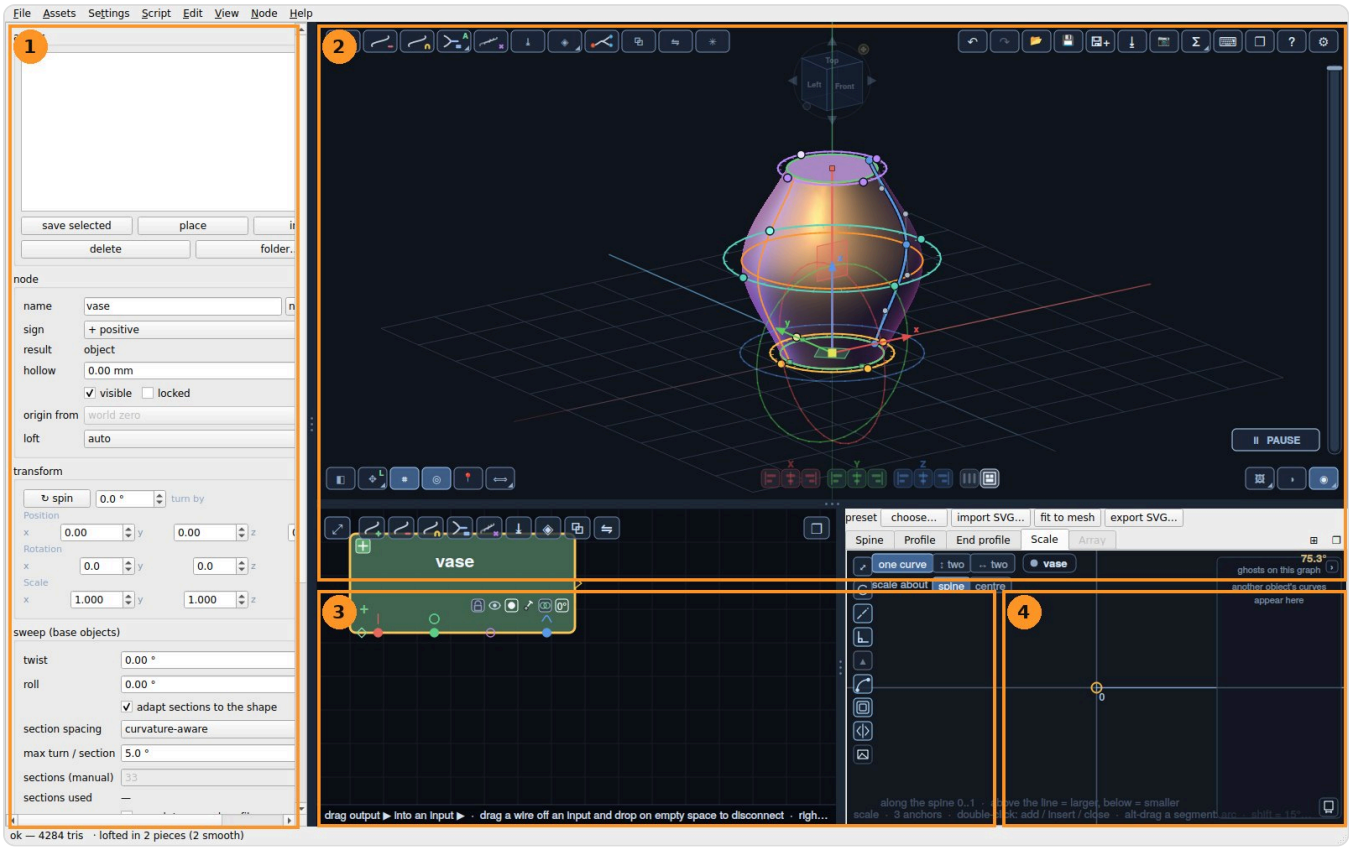
- **use all cores** - the scene's objects are built side by side, each by one of a few helper processes; every boolean is spread over the cores; and a fillet that has to look for a radius that fits tries several at once. The result is the same as on one core.
- **retry failed booleans with a tiny tolerance** - a join or cut that fails, or whose answer is impossible, is tried again treating parts a hair apart or a hair inside each other as touching. Said in the status line, and the seams can still be rounded.

- **make straight surfaces exact planes and cylinders** - an imported STEP or IGES whose faces are all splines gets the ones that are really planes, cylinders, cones or spheres made so (said on import); joins and fillets on them are quicker and fail less.
- **say why a boolean failed** - which part has the problem, what it is (it crosses itself, an edge too small, a face that cannot be rebuilt) and roughly where.
- **mesh export: join on the mesh if the exact join fails** - an STL of parts that overlap without being joined (a combine that could not join them) comes out as one watertight piece; STEP stays exact.
- **quick look: heavy cuts done on the mesh** - while you edit, the quick look leaves added parts unjoined; a cut through such a pile of pieces is the exact kernel's slowest case, so the quick look does it on the mesh instead (the teapot's cavity 12.5 → 1.5 s, the whole quick look 24 → 10 s). The full build that follows cuts exactly, as always.
- **glue parts that only touch** (off by default) - a part set on another's face is joined without intersecting every pair of faces. On a few parts it gains nothing measurable; it is there for assemblies of many touching parts. Used only when the two can only touch.
- **merge faces a combine split** (off by default) - parts set flush leave the shared side split along the seam; this merges it back into one face. A merged face takes one part's name and colour over both.

Off by default are the two that change how a result looks or only pay off in special cases; the rest are on because they make builds faster or harder to break.

Exported STL and STEP files carry a line in their header saying they were made with SweeperCAD, and where to find it. Header only - nothing reads it as data, slicers ignore it, the geometry is untouched.

The window



1 the panel: the selected object's settings · **2** the 3D view · **3** the node editor: how objects combine · **4** the graphs: spine, profile, end profile, scale

1. Getting started

The first-run screen

The first time the app opens it shows its terms — what you may do, that it is early software, no warranty, no liability, and that nothing is sent anywhere. The full text is in the app; there is nothing to read elsewhere. Continuing means you accept it. If a problem does occur, it is written to `~/.sweepercad/crash.log`.

The splash screen

Point at it to keep it on screen — move the pointer away and it carries on. The support and sponsorship links on it are clickable.

The terms, updates, and what this is for

The first time it opens, SweeperCAD shows its terms; **Help ▸ Terms** brings them back any time. They are in the app rather than on a website on purpose — you can read them on a machine with no network.

Three things in them worth knowing without reading the whole page:

- **Not for medical use.** Nothing designed here should become an implant, surgical guide, prosthesis, orthosis or instrument. Medical devices need validated software and documented processes; this has neither.
- **Please don't pass the app on** — send people to sweepercad.com. A copy handed over never learns there is a newer one, and this is young software that is still being fixed.
- **Nothing is sent anywhere.** There is no analytics, no licence check and no update check, which is why **Help ▸ Check for a newer version…** simply opens the website: the visit is yours to make.

It is also not trying to replace the CAD you already use. It does one thing — sweeping a profile along a spine — and is meant to sit alongside the rest of your tools.

Two more things the terms say, because they matter to anyone deciding whether to rely on this:

- **It was made by one person with a lot of help from AI**, said plainly rather than left to be discovered, and part of any support it receives is intended to go towards environmental work to offset some of what that help costs.

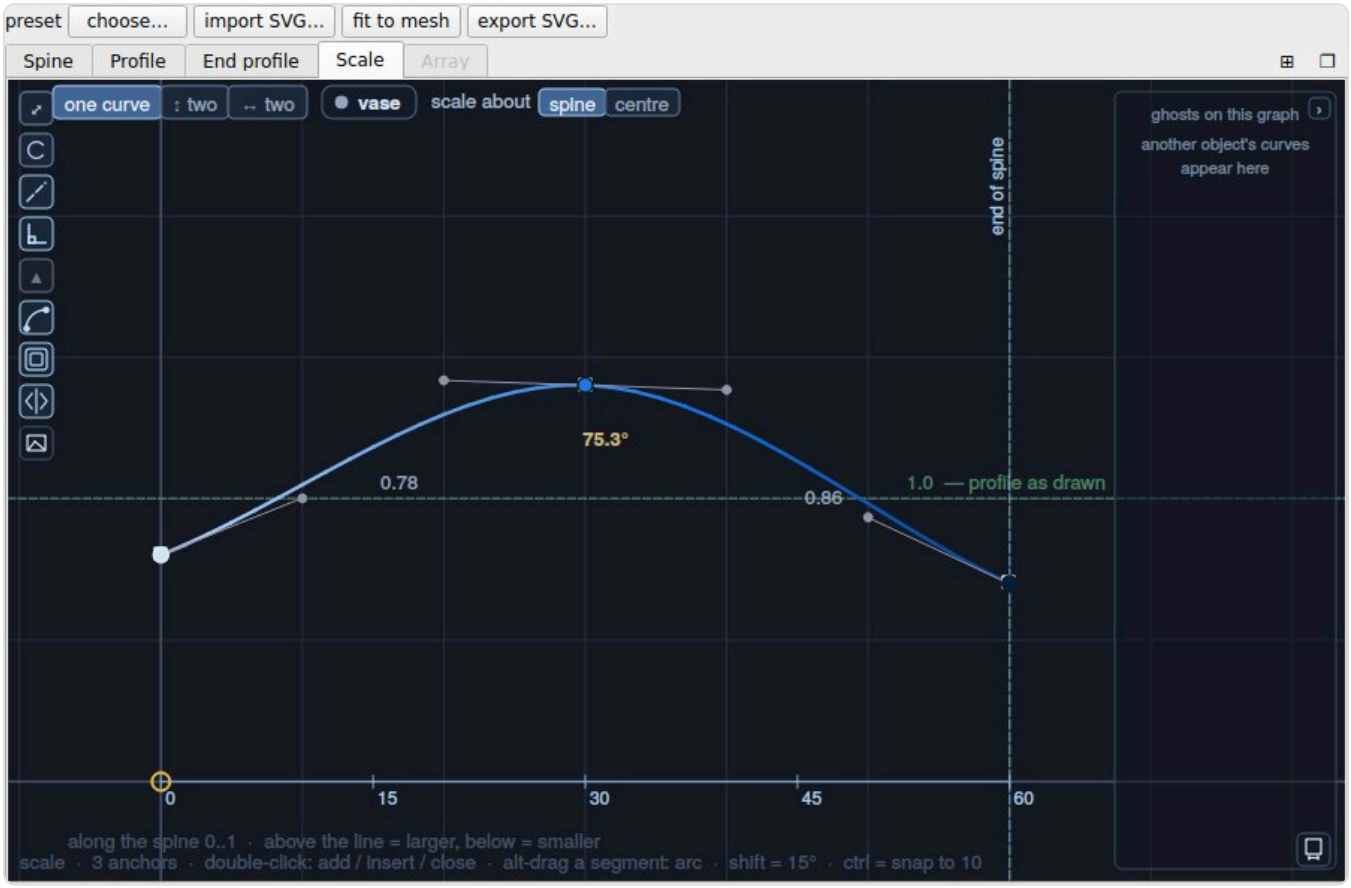
- **If the author stops, the app is not meant to go dark.** That may mean opening the source, or handing it on — no promise about which yet. What is certain is that your projects are saved as plain, readable JSON, so nothing you make here is locked inside one program.

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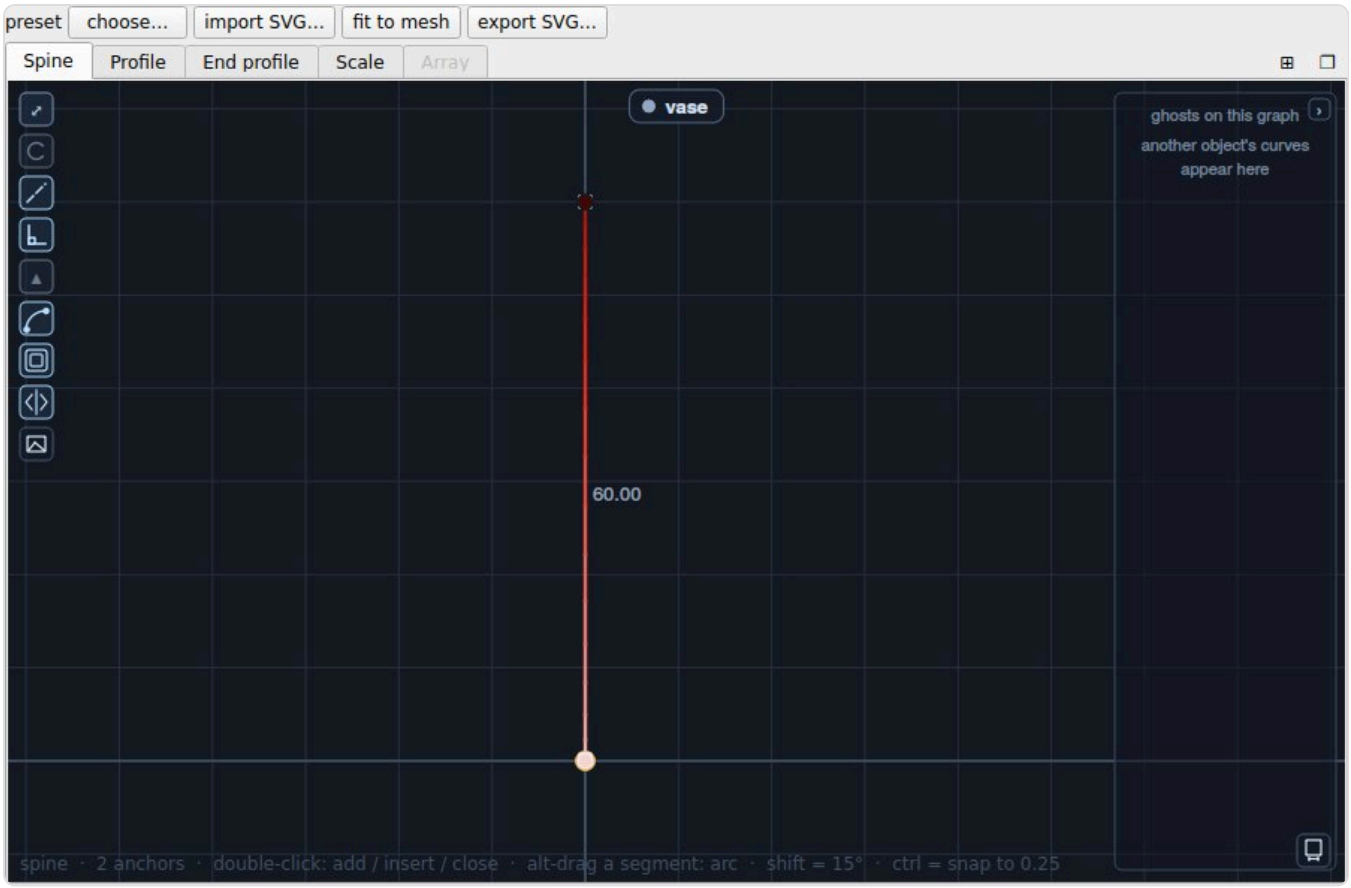
2. Drawing and shaping

The four graphs

Tabs under the 3D view. Each is a drawing you can edit at any time.



The Scale graph: how wide the profile is along the spine (here a vase: 0.8, 1.4, 0.7).



The Spine graph: the path the profile follows.

graph	what it is
Spine	the path the shape follows
Profiles	the closed cross-section that rides the spine - and, when it changes along the way, every other profile it morphs through: the start, the end and any number between, picked on the strip of ticks under the graph
Scale	how big the section is at each point (x = along the spine, y = size)

Profiles along the spine. Besides the start and the end, a sweep can carry any number of profiles in between, each at its own place along the spine; the sweep runs through them all. They all live in ONE graph, **Profiles**: under it a strip shows the spine with a tick for every profile, the start's and the end's included - **click a tick** to edit that profile in the graph above (the profiles either side of it show faintly behind, so you draw each against the ones it runs between), **click an empty spot** to insert one there, **drag a tick** to move it along the spine (it stays between its neighbours), or type its place in **at** (mm from the start, with the share of the spine beside it). **add** puts a new profile halfway between the one picked and the one before it, shaped as the sweep already is there - nothing changes until you edit it (on a sweep without an end profile it switches the morph on, the end a copy of the start). **remove** takes the picked one out; the sweep then runs through its neighbours. The tick at the start edits the start profile, with its presets and shape list; the **morph** box on the graph switches more than one

profile on and off. A converted mesh's sweeps are made of these - a finger has ten to forty - so every one of them can be reached and reshaped. Scripts (and the AI) have the same: `model.profiles`, `add_profile`, `remove_profile`, `move_profile`, `set_profile`.

The scale graph is drawn **in proportion**: its x axis is ticked in the spine's real length, and there is a dashed *end of spine* line. So the graph reads as the shape's silhouette rather than an abstract curve. Its handles have a **hard border**: a handle can never go past the points its segment runs between, however it got there (a drag, an inserted point, a typed value, a file) - a handle beyond the first or last point made the graph double back and the sweep fail. Doubling back on purpose is still done by placing the points themselves backwards.

All the graphs at once (⊞). The ⊞ beside the graphs' ❐ shows every graph in use side by side, up to four - for an ordinary object the spine, the profile, the end profile when it is on, and the scale. The tabs stay above as the list of graphs: the lit one, also lit in the grid, is the one *preset*, *import SVG* and the rest act on, and **a click inside any graph makes it that one**. Drag the lines between them to share out the room. Click ⊞ again for the tabs. It is remembered, and works in the graphs' own window too (❐) - on a second monitor all four get a good size.

export... beside a graph writes its curve out: as **SVG**, or as **DXF** in millimetres (a polyline within 0.01 mm of the curve) for a plasma cutter, waterjet, laser, CNC router or another CAD program - pick the type when saving.

On a narrow graph the words along the top shorten, then the controls wrap to a second row. Under about 360 px the **ghost list folds to a tab** (‹ *ghosts*, with how many are shown); click it to open the list, and the **›** at the top of the list folds it again at any width.

Drawing

Spine presets. The spine graph's **preset** menu offers a straight **line**, a **circle** (sweeps a ring) and an **n-gon** (a frame with mitred corners). Each starts at the origin, so the object stays where it is.

Every preset in the **preset** menus - profiles, spines and scale graphs - has a small picture of its curve beside its name: a profile as you look down the spine, a spine with its start marked, a scale graph as the side outline it gives the part.

Profile presets. The profile graph's **preset** menu has a circle, an n-gon, a star, a splined ring, an involute gear (and a meshing pair), text, and the shapes that become something when swept: an **ellipse**, a **rounded rectangle**, a **slot**, a **D shaft** and a **keyway shaft**; a **fluted** circle (a column - give it twist and it is a barley twist), a **lobed** ring (twisted, a rope - one turn per lay length), an **auger blade** (a thin radial blade on the axis; twisted, a screw flight - add a shaft as a second + object), a curve of **constant width**; a **rack** that runs on the gears (same module and pressure angle; its pitch line is y = 0, so a gear's centre sits module x teeth / 2 above it) - and a **thread**. A thread is the shank seen in cross section: the outline sits on the major diameter along the crest, runs down a flank, sits on the

minor diameter along the root and climbs back - one turn per pitch, so the four arcs share the circle the way crest, flank, root and flank share the pitch (an ISO V-thread: 1/8, 5/16, 1/4, 5/16, depth 0.54 x pitch). Give it the major diameter, the pitch and the number of starts; with a straight spine it also sets the **twist** that makes it a thread, 360° per pitch of length (divided by the starts). A bent spine leaves the twist to you. Threads are long twists, so the object is marked *allow slow*; a long bolt is a heavy build - model the threaded length only.

The same gestures on every graph. Click a point to select it (and drag it). **Double-click** on the curve to insert a point there, or in empty space on an open curve to add one at the end - a new point matches the one before it, smooth after smooth, a corner after a corner. **Alt-click** a point switches it between a corner and a smooth point; **Alt-drag** a segment bends it into a true arc.

- **Double-click** on the curve to insert a point; on empty space to add one at the end (open curves)
- **Drag** a point to move it. Near-misses will not create a point
- **Lining up**: dragged within a few pixels of another point's height or position across, a point (or a corner, or a handle) lands exactly in line with it and a dashed guide shows which. A handle also lines up with its own point, which makes a level or upright tangent. Holding **Ctrl** (the grid) or **Shift** (the angle) takes over from it
- **Snapping and guides** have their settings on the graph: the grid button at the bottom right, beside the numbers switch. *Snap to the grid* makes points, corners and handles land on the grid while you drag (off: hold **Ctrl** when you want it). *Cell size* is the grid's cell - automatic (a round step that follows the zoom) or your own size; the grid is drawn at it. *Angle* is what **Shift** locks to. *Line up with other parts* lets a dragged point also line up with the points of the other curves shown on the graph. *Midpoints and equal spacing* lets it catch halfway between two of your points and one equal step past them - the two equal gaps are marked "=". The settings are the same on every graph and are kept
- **Double-click a point** to switch it between a sharp corner and a smooth one
- **Select a point, type a number, press Enter** — the segment reaching that point takes that length. Escape cancels

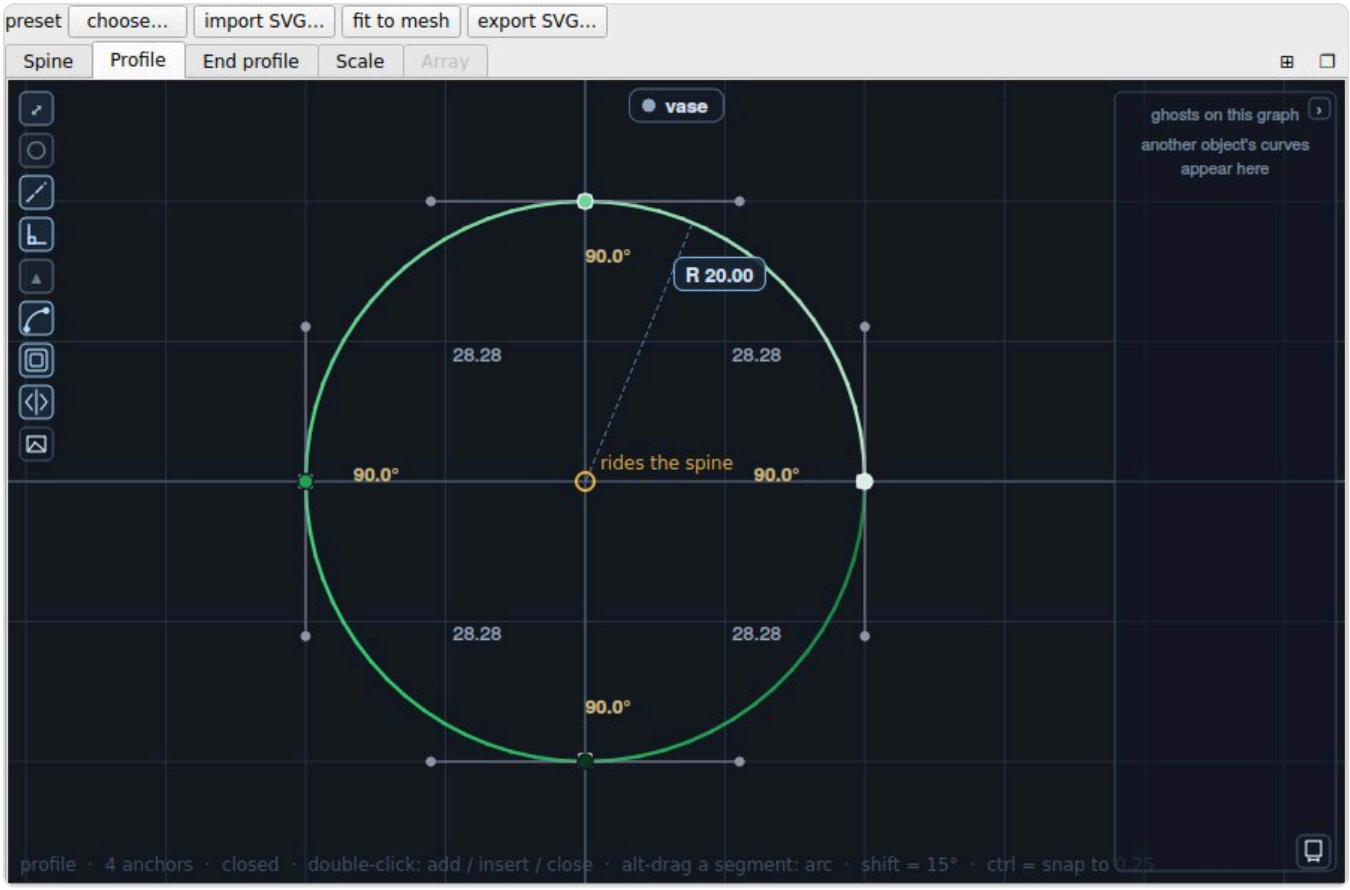
Any number field takes arithmetic. Type `360/7` for a seventh of a turn, `12+3`, `2.5*4` or `(10+2)/3` — the relationship stays visible while you are typing it instead of being rounded away in a calculator.

When you type a length, **which end moves** is yours: the darker point, the lighter one, or **both** — both grows or shrinks the segment about its middle, so it stays centred where it was. You can also set how many points travel together.

Rounding a corner: every sharp corner shows an `R` label. **Drag the label** to set the radius and watch the shape change, or **double-click it** to type an exact value. A point with handles is a curve point, not a corner, so it has no radius — remove its handles first.

Drawing a spine or a profile

+ and **–** do two things. **Click** in the 3D view and a ready-made object is placed, as always. **Drag** and you are **drawing** - on the face you started on (its plane), or on the floor in empty space.



The Profile graph: the cross-section, with its radius and angles shown.

- **Drag** draws freehand; it is smoothed into a few curve points, and where your stroke turns sharply the point stays a **corner**.
- **Click** adds a point. **Drag from the last point** pulls out its handles (like a pen tool).
- **Backspace** (or Delete) takes the last point back.
- **Right-click** (or double-click, or Enter) finishes. **Escape** cancels.

Open is a spine, closed is a profile - the shape decides:

- An **open** drawing becomes a new object swept along it, with the profile, size and scale preset from the placement settings. A **+** one lies on the face or floor it was drawn on; a **–** one is centred on it, so it cuts a groove half its profile deep.
- A **closed** drawing - a stroke that comes back to where it started, or a click on the first point - is finished at once and becomes a new object with that profile, pushed out of the face by the placement settings' spine length. Drawn with **–** on a face, it cuts a pocket of that shape.

The switches at the top left while drawing: **fine / normal / smooth** - how closely the curve follows your hand (fewer points the smoother) - and, when an object is selected, **closed: its profile**: a closed shape then replaces the **selected object's profile** instead of making a new object. The object being drawn shows as a wireframe as you go, and afterwards its points are edited in the graphs and in 3D like any other.

Circular arcs

A bezier can only approximate a circle, and drifts the moment a handle is dragged. So an arc here is kept AS an arc — a radius and a side — and its handles are worked out from that. They are read-only, and they are rebuilt whenever you move a point. Arc segments are drawn in amber with their radius beside them.

Click the **arc** icon (below the others on the graph) to turn arc mode on. The segment under the cursor lights up, so you can see which one you are about to change. Then:

- **Drag a segment** to bend it into the arc that passes through your cursor. Radius, which side it bulges to, and whether it goes the long way round are all read off where you drag, so there is nothing to choose. Near the arc that would leave its neighbour with no corner it **snaps to tangent** (the readout says so).
- **Click a segment** (no drag) for the **tangent arc**: the one arc that carries on smoothly from the segment before it (on the first segment, the one that runs smoothly into the next). It needs no radius, and it **stays tangent** when you move points — the radius re-solves itself. Its label shows **T**.
- **Click an arc** to make it straight again.
- **Click the amber R label** to type an exact radius. The side and the long-way-round choice stay as you bent them, so bend it roughly, then type the number. It has to be at least half the distance between the two points. A typed radius is yours: the arc is no longer kept tangent.

Points still drag as usual while arc mode is on. Click the icon again, or press **Esc**, to leave it. Holding **Alt** while you drag a segment does the same as arc mode, without the icon.

Turning a hand-drawn curve into an arc replaces its handles; undo brings them back. For a whole circle, the *circle* profile preset is four true arcs.

The profile's radius

Every closed profile shows its radius on its graph (the **R** label): the distance from its centre to its farthest point - a circle's radius, an n-gon's corner radius. Click it and type a value: the whole shape scales about its centre to that radius, corners, roundings and arcs together, so it keeps its form.

Offset and mirror a drawing — how to hollow something

(For mirroring an OBJECT in 3D, see *Mirroring an object* further down. This is about mirroring one drawing on its own graph.)

In the object panel: **offset / mirror**, with a dropdown for which drawing it applies to.

- **Negative offset** goes inward, positive outward
- Your drawing stays exactly as you drew it and remains editable
- The **amber** curve on the graph is what actually gets swept

To hollow a shape into a tube:

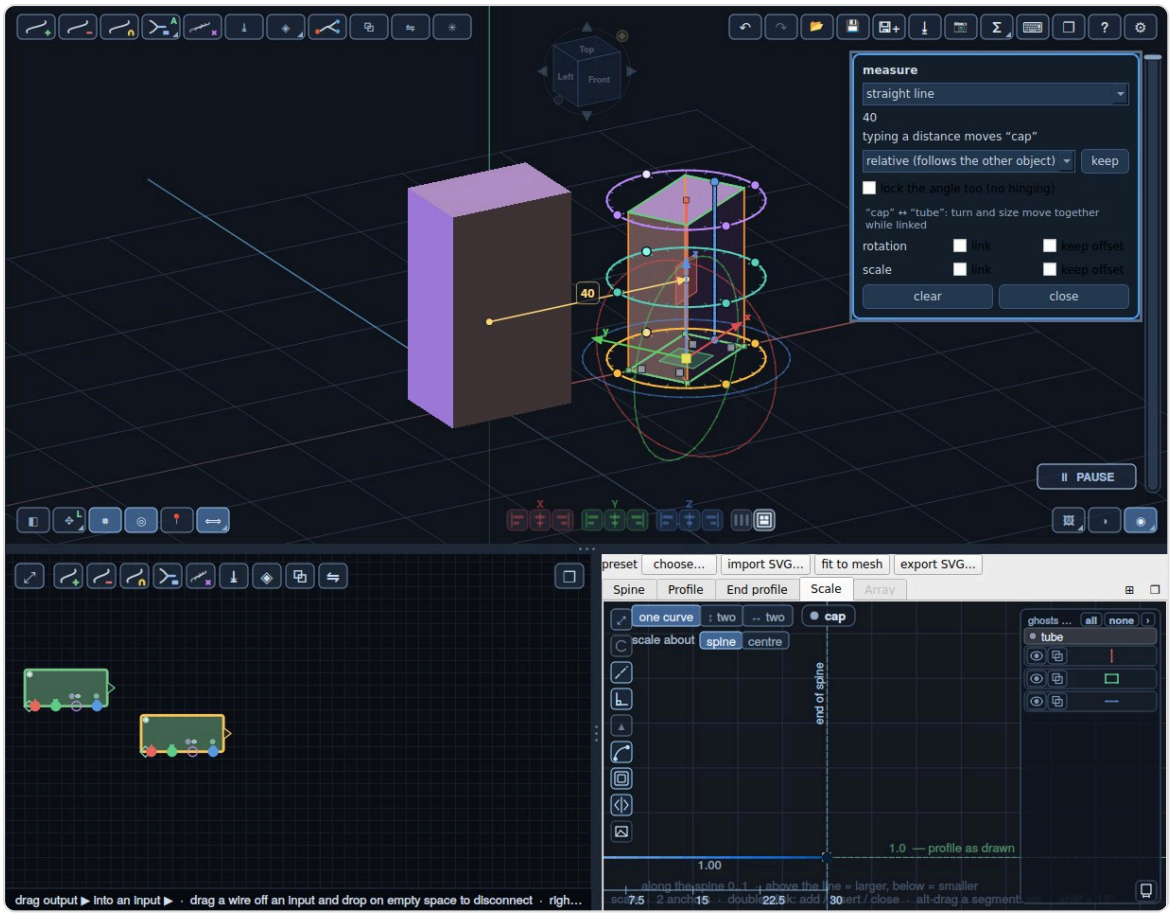
1. Duplicate the object (toolbar, or right-click → Duplicate)
2. On the copy: `ghosts…` → *LINK a curve here (live)* → the original → `profile`
3. Set the copy's **offset** to minus your wall thickness
4. Set the copy's **sign** to negative
5. Combine the two

The wall thickness stays a number you can change, and editing the original drives the wall with it.

If an offset is impossible — too far inward, so the shape would collapse or split in two — it says so instead of building something wrong.

Guides — measuring to another object

This is the dimension tool: pick two things, type the distance you want, and your geometry moves to match.



The measure window between two parts: keep the distance, link rotation and scale.

Click the **guide tool** on the left of a graph, then:

1. Click **your** point — or **your edge**, by clicking along it rather than near a corner
2. Click a point or edge on a **ghost** to measure *to*

Ghosts are other objects' curves, shown dashed. Arming the tool shows the ghosts it can measure to, so there is nothing to set up first. While it is armed every point you can aim at is drawn as a dot, and whatever is under the cursor highlights before you click it.

Edge to edge gives the shortest gap between the two edges — not a point-to-line distance, and not a crossing somewhere off the ends.

Then type the number. The guide shows its value; type a new one and your side moves (the ghost never moves — it belongs to another object, which is what makes the number unambiguous). As with segment lengths you choose **how many points** move and **which way**, so a whole run can travel together.

Locking a guide makes it hold. A locked distance is kept by a live solver, whatever moves the curve:

- **Dragging.** The point you drag stays under the cursor as far as its locks allow. With one lock it **slides** along that distance; with two (say, so far from this edge and so far from that one) it is **pinned**.

- **Other edits** — typed lengths, moving the whole shape, undo — are followed by the same solve.
- **A live reference that moves** takes the locked point with it, on any object, selected or not. A **frozen** guide keeps its target where it was, so nothing follows.

Only points that carry a lock are moved; the rest of the curve stays where it is.

No hinging. A distance alone lets your point swing round a target point, or slide along a target edge, at that distance. Tick **lock the angle too** in the guide's window and the direction is kept as well. The point then stays exactly where it is relative to the target, or at its spot along the edge. Such guides show ∠ beside the lock.

The arrow says which drives which. Every guide has an arrowhead pointing at *your* point. The ghost drives; your curve follows.

The overconstraint check. Locks that cannot all be true at once turn **red** and read *conflict*. The solver then moves **nothing** in that group: it never quietly picks a winner, so you decide which lock to change. A lock that only repeats what the others already force turns **amber** and reads *redundant*; it does no harm. When you lock a guide, the status bar says whether its point is now fully placed, can still slide, or conflicts, and with which guide.

Constraints on your own edges. The button beside the guide tool (an L with a small square) constrains edges of the curve itself. Click one of your straight edges and choose:

- **Horizontal** or **Vertical**
- **Parallel to…**, **Perpendicular (90°) to…** or **Equal length to…**, then click the second edge
- **Tangent to a circle or arc…**, then click a ghost's circle or a round stretch of another object's curve. The edge slides across until its line just touches it, and it follows when that object moves. A stretch that isn't round is refused.

The edge turns to the direction straight away, and the same live solver holds it from then on. Drag one end of a horizontal edge up and the other end comes with it. Each constraint is marked on its edge with a small symbol (H, V, ∥, ⊥, =, T, numbered when it pairs two edges; a tangent also shows the circle it touches, dotted). **Click the symbol to remove it.** Esc or the button again stops the tool.

Constraints and locked guides are checked together: ones that cannot all hold turn **red** and nothing in that group moves; one that repeats what the others already force turns **amber**. An edge cannot be horizontal and vertical at once (only a zero-length edge is both), so that pair is refused.

Which ghosts you see: `ghosts…` → tick objects individually, and pick which of *their* curves to show (spine, profile, and so on) per object.

Placing a section as a profile

In the section panel, **use as profile** makes the chosen loop the selected object's profile - and with **move the object there** ticked (the default) it also moves and turns the object, so its spine starts exactly at the cut and heads out along the cut plane's normal. Undo takes the whole step back. (If the object is scaled unevenly, the section still sits exactly on the cut, but the sweep leans: an uneven stretch shears an object's sections against its path.)

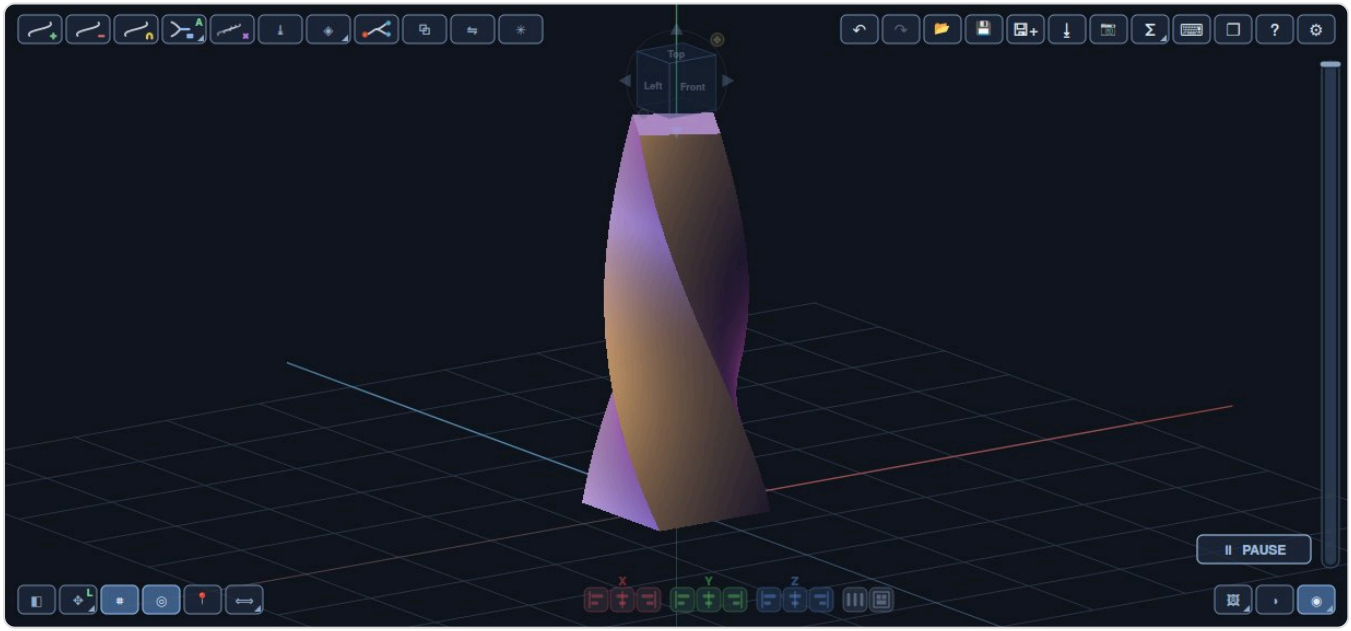
Where the scale scales from

The scale graph means **distance from the spine**. So a profile drawn away from its origin — 30mm out, say — moves *in* as it shrinks: at half scale it sits 15mm out. That is what you want for a taper.

When the offset is a feature's **position** rather than part of its size, pick **scale about: centre** on the scale graph (in the row along its top), or tick **scale about the profile, not the spine** on the object - the same setting. The profile then shrinks where it stands: 30mm out at full size, still 30mm out at half. It makes no difference to a profile drawn around its own origin.

Twist and roll

Twist turns the profile progressively along the spine: 90° means the far end is a quarter turn round from the near end.



A square profile, twisted 120° and tapered along the spine.

Roll turns the whole sweep by one fixed amount — the profile starts rotated and stays that way. Use twist for a spiral, roll to sit a square on its corner.

Both are in the object's panel, and both in the small dialog the node's parameter chip opens, so you can set them from the node tree too.

Spin turns the whole object about the start of its spine - the spine swings round, while the first section stays exactly where it was (roll takes the opposite turn to hold it). Spin is an **action**, not a stored number: type an angle in "spin: turn by" (panel or node dialog) and it is applied once, into the object's rotation; the field goes back to 0. That is how flat 2D spines make 3D shapes: spin one piece against the next. An **attached** piece turns about its joint by roll alone.

Roll is always one turn (-180 to 180). Twist is not wrapped - 500° really is more than 140°.

The dials

Three small rings, drawn on the selected object's own sections:

- **spin** (amber) round the **first** section,
- **roll** (teal) round the **middle** of the spine,
- **twist** (violet) round the **last** section.

Grab a dial by one of its **four dots** - only the dots take a click, so the rest of the ring never steals one from what lies under it. Tick marks every 15° (longer every 90°) turn with it. While you drag, nine light **ghost sections** show where the shape is going; the model rebuilds when you let go. Dials work **relative**: a twist of 500° is tweaked from 500, not reset.

Switch all three off with the bottom-left switch (on ⌗ splines the dials are hidden, and in direct editing; right-click it for the checklist), or each one in Settings. The manipulator's **off** setting hides the gizmo only - the dials and curves keep their own switches.

Curve points beat the gizmo. Near a visible spine or profile point, a press grabs the point (and turns curve editing on) instead of turning the object.

Camera moves glide. Clicking a plane square or a face of the view cube turns the camera smoothly rather than jumping, so you keep your bearings; any click stops it where it is.

Asymmetric scale

Along the top of the scale graph, pick ↕ **two** or ↔ **two** (instead of **one curve**). You then get two scale curves — one above the axis, one below — and the profile's two halves are scaled independently. The ▲/▼ button switches which one you are editing; the other is shown as a ghost. Each half follows its own curve in both directions - it grows or shrinks in width as well as in height - and the two blend smoothly across the split line, so where they meet the profile stays smooth.

Both curves can close to a point at an end (a tip, a teardrop) and the surface stays smooth. If only one of them reaches 0 there while the other does not, the object is built with straight facets along its length.

Which way it splits is that same choice: ↕ **two** splits top / bottom, ↔ **two** left / right. The graphs look the same either way; what changes is which axis of the object the second curve drives. Since a spine only curves in one plane, this is what lets you put the asymmetry across the bend or along it. (The same switch is on the left panel, under **asymmetry**.)

The scale graph starts as a flat line at 1 - the profile as drawn, all the way along. Choose **box** in the placement settings - (0,0) (0,1) (1,1) (1,0) - for a graph whose corners you can round: it sweeps exactly like the flat line until you do, and rounding them is how you give a sweep a rounded start or end.

The scale graph in 3D. With *curves in 3D* on, the selected object carries its scale graph as a blue line on its surface, running along it like a seam. A symmetric graph runs on the side of the profile that lies in the spine's plane, so clicking the spine's square shows it beside the spine, flat, the way the graph draws it. An asymmetric one is two lines, the second dashed: top and bottom, or right and left when it splits left / right. Its points are there too: drag one along the object to move it along the spine, or in and out to change the scale there. Press anywhere on the line and pull to make a point there and drag it in one go; double-click the line to add a point without moving it, double-click a point to make it sharp or smooth, and right-click a point to delete it. The graph follows as you go; one undo puts it back. The first and last points stay at the ends, and the points at scale 0 (where an end cap closes) are shown small and are not grabbed. When a bend was eased, the profiles and this line sit on the eased spine - the one that was built - which is drawn dashed in red. The line always runs from the start of the spine to its end, however wide the graph is drawn (the graph is read on its own width). A smooth point shows its two handles on the object too - drag one and its partner mirrors. Where the graph is drawn from rounded or chamfered corners, you move the corner itself, and its radius goes with it. All of this works with point editing on as well.

Reality check. The stats panel's last section checks a part you made against the design. Import a scan (OBJ or STL, in millimetres), select the object and press *compare with the selected object*: the scan is lined up with the design wherever it was scanned from, every point is measured to the design's surface, and the scan is coloured - green within your tolerance, red where it stands out, blue where it falls in - while the design turns into a ghost around it. You get the match (the share within the tolerance), the average, the RMS, the worst either way and how much of the design the scan covers. Tick *allow a uniform scale* to fit the size too: a casting's shrinkage, or a print's compensation, is then reported as a percentage instead of counted as error. *Which object is it?* tries every part in the scene and ranks them by match. *Put the scan back, uncoloured* returns it to where it was imported.

Match photos of the print does the same with photos instead of a scan. Pick one or more photos and click on the part in each; the part's outline is cut out of the photo (a plain background behind it helps), and for each photo the camera is found - where it looked from, its roll and its field of view (taken from the photo's lens data when it has it, otherwise fitted roughly). You get each photo's outline match

(how much the two outlines overlap) and how far apart they lie, in mm; the result window shows the design's outline over every photo, green where they agree and red where they part, and can turn the 3D view to that photo's camera. *Photos check* chooses whether they are compared with the selected object or with every object in the scene, ranked. A photo shows the outline only, and a symmetric part can fit from more than one side: more photos from other sides check more.

Layer jumps. Under the cross-section graph: the height to measure over and the change (in % of the section's area) that is too much. The graph gets a strip from cool to red, and the stretches past the threshold are drawn red; *show the jumps on the part* colours the part the same way.

Closing the stats window clears everything it drew in the 3D view.

Cutting and joining sweeps

All on screen: on the **profile strip** under the Profiles graph, and behind the **scissors icon** - in the node editor's row and in the 3D view's toolbar beside the stick figure - which opens cut, join and connect (the last used shows on the icon; the node menu has them too). On the strip, **click an empty spot** to insert a profile there, shaped as the sweep already is at that point - nothing changes until you edit it; press **cut**, then click the strip where to cut - a profile's tick or anywhere between. The scissors on the node editor's row cut the selected sweep at a distance you type. A cut leaves an **=** node of the two halves standing exactly where the sweep stood - nothing moves. The section at the cut is the profile the halves share; the twist reached there turns the second half's profile and the rest of the twist stays with it; the scale graph is shared out in millimetres. A sweep that morphs between profiles is cut too: each half runs through the whole's sections, within a quarter of a percent of it.

From the mesh. With a reference mesh loaded, **mesh** on the strip replaces the picked profile by the mesh's own section where that profile sits - the plane across the spine there, the loop round the spine's point, brought into the profile's frame with the scale and twist undone - and the node menu's **All its profiles from the reference mesh** does every profile at once. A drafted figure's leg becomes the mesh's leg, still a sweep with editable profiles. One undo step.

Connect. Two sweeps selected and connect (the scissors' row): ONE new sweep runs from the nearer end of one to the nearer end of the other, their sections as its start and end profile - nothing else is added (no = nodes; combine it with the two yourself if you want one solid). Its spine lies in the plane holding both ends' directions; an end that stands a little off that plane (up to about its own size) is met by setting the end profile off by that much, so it is still one piece. Only ends much further out of one plane get two planar pieces meeting smoothly at a midpoint. Each end reaches a hair (4 % of the outline's size) on into its body, getting a little smaller there, so an = with the bodies is an ordinary overlap rather than two faces laid on each other - the case booleans get wrong. Between the ends it morphs from one outline to the other - or, when both are the same outline turned against each other (a square meeting a square at 30°), it turns, with a copy of the outline every 15° of the turn, and the square stays square.

Select two or more sweeps (shift-click) and press **join** on the row (or in the menu, **Join the selected spines**): one sweep, standing where the first stood, their profiles along its spine and their scale graphs in a row. The spines must lie in one plane; a piece drawn the other way is turned round. Where two ends stand apart - a little or a lot - a smooth stretch runs from the one along its own tangent into the other, a bend rather than a corner, and the profile morphs across it, so the pieces' profiles may differ there. Pieces that **overlap** - one running on past where the next one starts, like two committed figure sticks sharing a ball (the first runs on to the ball's far side, the second starts at its centre) - have the overlapping part cut away first; where they then meet at an angle, each is cut back by the joint's size and a smooth bend carries the joint round the corner at about its full size. The same outline at another size (a thin stick's circle and a thick one's) goes into the scale graph, so the joined sweep keeps one profile. **Join them into a closed loop** also runs a stretch from the chain's end back to its start - a ring with smooth corners; a chain that already lands on its own start closes by itself. Different rolls, headings or twist rates, profiles that differ where two ends touch, or a twist on a loop, are refused with the reason. A combine left with one input collapses into it, so the graph stays tidy.

Coils — springs, threads, shells and horns

Every sweep - adding or cutting - has a **coil** box with a tick: tick it and that sweep's own profile winds round its own spine; untick it for a plain sweep again (the winding is kept). A cutting sweep wound round makes a thread's groove. The ≀ icon (in the stick figure's corner group, beside the route tool; also on the node editor's row) is the quick way to a new one - a **coil**: an object whose wire winds round its spine. It is the one object whose path is not drawn in a plane — everything else about it is a sweep's: the profile and the scale graph are drawn as usual, it combines, cuts and exports like any object, and its node carries the ≀ mark in its sign's colour.

- **The spine is the axis.** A new coil starts on a straight spine exactly as long as it climbs: a plain spring. Bend the spine and the coil follows it — a phone cord, a ram's horn. *Straighten the spine to the climb* puts it back.
- **Preset**: spring, thread, conical spring, shell, ammonite, golden, horn. Every number stays editable afterwards.
- **Radius**: constant; logarithmic (multiplied by *grow per turn* every turn — 6.854 is the golden spiral, shells and horns live here); or Archimedean (a steady step outward, a clock spring).
- **Climb**: a pitch in mm per turn (springs, threads), or a share of the radius (shells and horns: the bigger the turn, the further it drops). A pitch of 0 lays it flat (an ammonite).
- **Turns**, **left-handed**, and **the profile grows with the radius** (shells: the opening widens with the whorl).
- The object's **twist** turns the profile along the wire (a flat ribbon, a twisted thread).
- **Morph**: give the coil an end profile (and mid profiles, on the strip) and the wire changes shape along its length - a round wire growing into a star, a horn's mouth flaring.

- **Heading**, how the profile sits on the wire: *upright* (in the plane through the coil's axis - threads and shells), *square to the wire* (x pointing outward - a spring's true section, a ribbon lying on the coil), or *rolling with the wire* (square to it, but rotation-minimising: a flat profile slowly rolls round the wire, as a free ribbon would).
- The **scale graph** reads along the spine. *Scale about the spine* (the default) scales the whole winding: the coil narrows toward the spine as the wire thins — a tapering cord, a horn's tip. *Scale about the profile* only thickens or thins the wire, the coil keeps its radius.

The climb is spread along the spine's length, so a longer spine stretches the coil. A loft through the exact profile every 15° makes one closed solid.

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3. The 3D view

The 3D view's own controls

Top left - making things: add a positive (+), negative (−) or keep-inside (/) object, combine (=), auto-combine (A=), array (∗), **import** a CAD part or a mesh (⤓), a **lattice** (◈), a **route** (⤳), and **copy** (⧉) and **mirror** (⇋), which act on the selected object.

Grouped icons. A few icons hold more than one tool, to keep the rows short. A small ◢ **in the corner** shows it: click the corner (or right-click the icon) and a row with the others opens; pick one and the row shuts. The icon then shows the tool you used last. - **=** holds **A=** (auto-combine). The = stays the icon, and a green **A** on it says auto-combine is on. - ◈ **lattice** holds ✎ **drawing on the surface** - both make implicit shapes. - ◫ **extract** (bottom right) holds the **section** (extract a plane), the **line** extractor and the ⋒ **border** where two parts meet. 🖼 **references** (show or hide the reference mesh) is with the display switches. - ✥ **the manipulator** (bottom left; move / turn / scale / direct / off) holds the **pivot** and **world / local**; an **L** on ✥ says local, a dot says a temporary pivot is set. - ⟺ **measure** holds **dimensions**. - ◖ **what is shown** (bottom right) holds ◎ **isolate**, the **axes**, the **grid**, **ortho / perspective** and **light / dark**. **Isolate** (or **I**) hides everything but the selection - a combine with its inputs, a part with the combine it is in - with one click, and one more click brings it all back (an eye you shut yourself stays shut). No key to hold.

The switch groups (=, ✥, ⟺, ◖) keep their own icon; a state you could not see otherwise is marked on it.

Top right - the document: undo, redo, **open** (📂 - open, open recent, new), save, save as, export, a **picture** of the model (📷), the **stats** (Σ), the script editor (⌨), **own windows** (❐), **help** (? - how to use, the scripting reference, terms, a newer version, supporting development, about) and the **gear** (⚙).

Own windows - for a second monitor (❐). Put the **3D view**, the **nodes**, the **graphs** or the **panel** in a window of its own and drag it to any screen; it is the same app, so everything stays in step - one model, one undo, the same selection, and the shortcuts work in every window. The first time, a window opens big on your other monitor (if you have one); after that it opens where and as large as you left it. **Close the window** to put the pane back exactly where it was. The graphs have their own ❐ at the right of their tabs, the nodes theirs in their top right corner, and the View menu has all four. Panes that were in their own windows when you quit come back out the next time you start.

The gear holds the Settings; everything else has its own icon, and an imported part's right-click reloads it from its file - so the menu bar is not needed. On Windows and Linux the gear's last entry, **Hide the menu bar**, puts the menu bar away (remembered, and every keyboard shortcut keeps working); tick it again to bring it back. On a Mac the menu bar belongs to the system and stays.

Icon size (Settings > Icon size, also under the gear): *Normal*, *Large* or *Extra large* makes the 3D view's icon rows and the node editor's 1.25 or 1.5 times bigger, glyphs included - handy on a large or high-resolution screen. Remembered. When a row no longer fits beside another one it moves to a line of its own instead of overlapping.

Bottom left - handling the model: the manipulator chooser - each click the next of move and turn together, move, turn, scale, ⇕ **direct editing** or **off** (no gizmo at all, clicks just select and orbit) - then the **what is on the object** switch - ⌗ **splines** (the plane squares and the curves in 3D), ◎ **dials** (spin, roll and twist) or ○ **none** (a clean view, nothing on the object); its picture shows which is on, and a right-click gives the checklist (splines and dials together) - then what a click selects, reposition, the temporary pivot, measure, and world/local.

Bottom middle: align and distribute, for two or more selected objects.

Bottom right - how it is shown: reference images and meshes, with the **section** tool and the **line** extractor in the same icon, the world axes, the **grid**, perspective or ortho, the **view** button (see *How the model is drawn*), what is displayed (result, components or both), and the **sun** (☀): light or dark for the WHOLE interface - the 3D view and its background, the node editor, the curve editor and the panels switch together. Dark is the default; your choice is remembered. Your objects' own colours stay as they are.

Hidden dials, squares and curves cannot be grabbed either - that is the point: switch them off and nothing gets in the way of the model.

The lights ride the camera in every mode, so there is never a dark side turned away from you. The colour view lights warm against cool, which makes curvature read as a shift in colour as well as in brightness; the clay view lights it neutrally and softly, leaving the form alone.

Keys

key	does
Ctrl+Z / **Ctrl+Shift+Z**	undo / redo
Ctrl+N / **Ctrl+O** / **Ctrl+S**	new, open, save (**Ctrl+Shift+O** adds a file to the scene, **Ctrl+Shift+S** saves as)
Ctrl+D	duplicate the selected node
A / **S** / **C**	add a positive object / a negative object / a combine
Delete (or Backspace)	delete the selection - with the pointer on a curve point in 3D, that point
F	fit the model in view (while editing a curve in 3D: face its plane)
O	perspective or flat (orthographic)
Shift+W	wireframe

key	does
P	pause rebuilding
B	full build now (in draft mode)
L	the lens: the full model in a frame
Q / **W** / **E** / **R** (in the 3D view)	the gizmo: move and turn / move / turn / scale
X	in the 3D view: world or local axes; in the node editor: disconnect the selected node
D	direct editing on, and back to the manipulator
I	isolate: everything but the selection hidden; again, all back
B	the sculpting brush on / off
Esc	cancel or stop the current tool
Enter	finish a drawing; apply a typed value
Tab / **S** while placing	the next distance to type / snapping to points on or off
Ctrl+R	script editor
F1	this help

Keys can be changed in the settings (the gear). A key a menu already uses stays with the menu, so no key ever does nothing.

Every edit shows at once

Whatever the scene, an edit shows in the view within about half a second: the **quick look** - every unchanged part as it was, the edited part rebuilt, parts that are only added shown side by side, roundings and bridges left for the moment. The real build (every boolean and blend) follows in the background with the view usable; the status bar counts while it runs, and the quick look is replaced when it lands. A new edit during that build asks it to stop - whatever it had already built is kept, so nothing is built twice - and starts again from the new state. Pause (P) stops it the same way.

A part that will not build (an outline that cannot sweep, a shape the kernel refuses) is said in the status bar, by name; the rest of the scene is built and shown as usual. A sweep that would leave thousands of faces is shown but not combined, with a note on what asks for them.

Editing in the 3D view

Select an object and two small squares appear, one lying in each graph's plane. **Click a square** and the camera turns to face that plane - from the side you are already looking from, so it never swings round the back - and its points become editable. **Double-clicking the object** switches editing on or off too. The curves' points are already drawn before that, small, beside the radius squares; grabbing one opens editing too, and then the handles and their tethers appear. Everything that can be dragged lands exactly under the cursor:

- **spine points** and **profile points**
- **bezier handles**, so a curve can be shaped and not merely moved
- **rounded corners** — the corner a radius was cut from, which the curve itself does not pass through
- **an array's own curve**

A curve is flat, so its points can only move within its plane. Seen edge-on, that plane offers a single direction and a drag cannot mean anything — so it is refused, with a note saying so.

Each square wears its own graph's colour — red for the spine, green for the profile, purple for an array — so the square and the curve it belongs to always match. The camera is yours again immediately after a click — the square moves it once, it does not hold it. A square also *shows* its plane: a sliver means you are edge-on to it, and edge-on is the one angle from which a drag is refused.

Points are squares in their graph's colour; handles are small grey discs on a line back to the point they belong to — the same as on the graphs, so the same thing looks the same in both places.

On a **scale** graph, a handle stays between its own point and the next one. A curve running backwards there would no longer give one scale per position along the spine — it would double back, which the sweep then has to walk in several passes, and that is easy to cause by accident and slow when it happens. Doubling back on PURPOSE is still available: place the points themselves out of order.

Where a corner has a **radius**, the point you drag is the corner itself, not the two points where the arc meets the edges (nor their handles — none of them are offered). Those are consequences of the radius, so they are not offered — exactly as on the graphs — and moving the corner keeps the radius.

Drawing new curves in 3D is deliberately not offered: the graphs are where drawing belongs.

Direct editing — push and pull the model itself

The manipulator chooser (bottom left: move and turn, move, turn, scale, ⇕ direct, off - each click the next, and Q, W, E, R pick the first four) has a step for it: **⇕ direct**. The next manipulator leaves it again, and brings back the splines or dials that were shown. **The quick way both ways: the D key**, or **direct editing** in the right-click menu - direct editing on, and again back to the manipulator and what was

shown before. With it on, the model's own faces and edges answer the pointer - there is nothing to select first. **The app opens in direct editing** the first time, and after that in whatever you left it in - splines, dials, none or direct - when you closed it:

- **Hover** a face or an edge: it lights up and a label says what it is (*block: side 2*, *block: end cap*, *c: seam*) and its number now.
- **Drag a side face** to push it out or in. It moves its profile segment, all along the object. The neighbouring sides keep their lines, so a trapezoid stays a trapezoid; a round side changes its circle's radius; a smooth face (an oval, a curved side) moves out evenly and keeps its shape - or, with *View > Direct editing: smooth faces bulge where grabbed*, the curve's point nearest where you grab it moves and its neighbours ease along.
- **Drag an end cap** to pull it - the spine gets longer or shorter, its last piece running on. The cap has two more grips on its rim: the ● **dot** tilts the end (the spine's direction there, in its plane) and the ■ **square** scales the end (the scale graph's value at that end). Tilting one end leaves the other end exactly as it was; with *View > Direct editing: tilting an end keeps the other end in place* off, the whole spine bends with it.
- **Past zero, a cut.** Keep pulling an end back through the other end and the part carries on out of it the other way, and turns from adding to cutting: a block pushed down into what it stands on becomes a pocket that deep, with the same outline; a hole pulled up out of it becomes a boss. A taper is mirrored across that end. A part placed on a face sits a hair into it, so the cut also reaches a hair back out past the surface - the pocket opens at the top, and a boss sinks a hair in. The label says *past zero: it cuts now*. Drag back above zero and it adds again. This works on a straight part with one profile and no twist; anything else says why it can't.
- **A straight side** has a ● **dot** at its end: drag it to turn the side about its middle (its neighbours keep their lines).
- **A side facing along the spine's plane** has a ◆ **diamond**: drag it to bend the part there - a smooth point is put in the spine where you grabbed it (or the one already near is used) and pulled sideways.
- **Drag an edge** to round it - between two sides it is the profile's corner radius, round a cap it is the rim rounding, between two parts it is the seam's fillet (a fillet already there can be dragged on its own face). Dragged past zero it becomes a chamfer, as everywhere. While you drag, the rounding is drawn on the edge at once - the two lines where it meets each face, and arcs across it (a chamfer as its flat) - and the part itself is rebuilt when you let go.
- **Every drag shows what it makes** before you let go: pushing a side, the turn dot, the corner square, the length, the tilt dot, the end size and a bend draw the new shape as rings across the part and lines along it, over the part as it was. A cage point bends the part along with it, and a net point lifts the face around it. An edge between two sides also has a ■ **square**: drag it to move the corner itself, out or in along its middle line; both sides follow.
- **A hole** is pushed the same way: its wall's arrow points into the material, and the wall follows the pointer.

- **Type a number** instead - digits while hovering or dragging, then **Enter**. **Esc** cancels; the face goes back.

Everything is written into the graphs the object is made of - there is no separate history of direct edits - so the curves, the fields and the view always agree, and the result edits the usual way afterwards. Each gesture is one undo step. While dragging, a quick picture shows where the face goes; the exact shape is built when you let go. An edit that would break the shape is refused and the label says why (*the corner has room for 4.00 mm at most*, *pushed through the other side*, *the profile follows another object's - unlink it first*); a drag stops at the last value that worked.

The numbers are true millimetres in the view: on a part that is scaled, or tapered by its scale graph, a 5 mm push moves the face 5 mm where you dragged it (the profile itself moves less or more). On an object that morphs - an end profile, profiles along the spine - a push, a turn, a rounding or a moved corner is made in every one of its profiles, when they have the same corners; otherwise it is refused and said.

A profile built from several shapes is edited through its shapes: a side pushed moves the side of the shape it came from, and the profile is derived again; a shape's own corner rounds as that shape's, a corner where two shapes cross as the group's. Turning, moving corners and smooth faces are for ordinary profiles.

An imported CAD part (STEP, IGES, BREP) has no curves to write into, so its faces themselves move: drag a face and the kernel moves it, its neighbours running on to meet it; drag an edge between two of its faces to round it (past zero: a chamfer). A mesh part (STL, OBJ) has no faces to push and says so.

On a bent or twisted part, which way a side faces is read where you grab it, so the side and the rounded corner next to it are told apart right along the part.

Face nets — shape a face by a net of points

In direct editing a face has a **#** beside it: click it to put a net of points on the face (3 × 3), click it again to take the net off. Drag a net point out of the face to lift the face there, or in to dent it; a number follows the drag, or type one. ⊕ and ⊖ beside the **#** make the net finer or coarser (2 × 2 to 9 × 9), the face keeping its shape as near as the new net can hold it.

The face's borders never move - only its inside - so the solid always stays closed and the faces next to it are untouched; the edges are still moved by pushing and pulling. The face stays a real CAD surface.

A net belongs to its face by name, so it rides along when the part is edited underneath: made longer, tapered, combined with others (the net is made on the part itself, before any combine). A net on a side remembers where its side is in the profile, so it follows its side when points are put in or taken

out, or a joint turns smooth or sharp; a net whose side is gone is kept, asleep, and wakes if the side comes back. Nets work on swept parts, imported CAD parts and the fillets a combine makes.

The box cage — bend a whole object

For an array, a big combine, or whenever a whole object should bend, bulge, taper or lean, hover it and click the ⬚: a cage of 3 × 3 × 3 points goes round it. Drag a point - it moves in the view's plane - and everything inside follows smoothly; where the point stands, the object moves exactly as far. ⊕ and ⊖ beside the ⬚ make the cage finer or coarser (2 to 6 points a side), the shape it makes kept. With *View > Direct editing: cage points move in mirror pairs* a point and its twin across the cage's middle (along the object's own X) move together, mirrored, so a symmetric part stays symmetric. Click the ⬚ again to take the cage off. The result stays a CAD solid and every face keeps its name, so pushing, pulling, rounding and face nets still find their faces. Each drag is one undo step, and the cage is saved with the file. The switch's last step, ⬚ **cage**, puts a cage round the selected object too.

The sculpting brush

✎ beside ⌗ in the bottom-left row (or **B**) switches the **brush** on, in direct editing. Drag over a face and the surface rises under the brush, most in the middle and less toward the brush's edge: the face gets a **net** by itself the first time the brush touches it, as fine as the brush asks (about one net step per brush radius; or a fixed size), and every stroke moves that net's points - so a sculpted part is still a real solid with every face named, saved like any net, and goes on being edited the usual way. The part is rebuilt when the stroke ends. A stroke that crosses onto another face carries on there with that face's net; a fillet's own face is left alone.

The brush is a share of the model: its **radius** is a percentage of the model's span (the box round everything drawn, corner to corner; 5 % to start), and a **dab** a percentage of the radius (8 %) - so it is the same brush on a pendant and on a gate, and the net a face gets follows it. While the brush is on, **a bar at the bottom of the 3D view** has both as sliders, with the millimetres beside them; the panel's sculpt row shows the same numbers and either moves the other. A small brush still moves the nearest net point with every dab. Leaving direct editing switches the brush off. A drag in empty space beside the part turns the view, as it does without the brush. The bar also has **pull / push / smooth**, and a press **held still** keeps sculpting there (a small dab every 60 ms) until you let go. The net's resolution: a face's size divided by the brush radius, plus one point, kept between 3 and 9 per side (or a fixed count in the sculpt row). When a press on a part does nothing the status line says why: on a fillet's face (left alone - sculpt the face beside it), or on a part that cannot take a net.

The **sculpt** row in the node box has the brush: **pull** (out), **push** (in) or **smooth** (the points under the brush drawn toward their neighbours), the **radius** on the surface, the **strength** (how far one dab moves the surface at the brush's middle - a slow stroke lays more dabs) and the **net** a face gets.

Each stroke is one undo step. It is a brush on nets, not a polygon sculptor: a face's net is at most 9 × 9, so the brush makes smooth hills and dents, not sharp creases - draw those in the graphs.

Where you see them. Nets and a cage are not nodes of their own: they belong to the part (or combine, or array) they shape. Its card says so - *# 2 nets*, *⬚ cage 3* - and its panel has a **shaped** row with **clear nets** and **clear cage** - and **soft**, the falloff.

Soft falloff. Dragging one point of a net or a cage normally moves that point alone. **Soft** (in the shaped row; remembered) makes its neighbours follow it, less and less with distance, out to that many net or cage steps: 0 moves only the point, 1.5 gives a smooth dent from one drag, 3 a broad swell across a fine net. It applies to every drag, nets and cages alike, and a drag never piles up. From a script, each call has its own: `model.net_point(face, a, b, mm, soft=2)`, `model.cage_point(part, i, j, k, z=4, soft=1.5)`.

Editing points in the 3D view

With an object selected and its curves shown (⌗), its spine and profile can be edited in the 3D view much as in the graphs: - **drag a point** to move it (in the curve's own plane); - **double-click the curve** to add a point there - only while you look straight at that curve's plane (click its square first), so a point is never added by accident while turning around the model; - **double-click a point** to switch it between **sharp** and **smooth** (bezier handles, which you can then drag); - **right-click a point** for its menu (sharp / smooth, delete), or point at it and press **Delete**; - every **corner** has a small square on a stem along its bisector: **drag it into the corner** to round it, **back past the corner** to chamfer it (a diamond), onto the corner to make it sharp again - the same control as in the graphs and on the fillet rings. Click the square to type the radius.

What a click selects

The first button of the bottom-left row chooses what clicking in the 3D view selects: the **part** you click (◧), or the **whole assembly** it ends up in (⛁) - the combine at the top of its tree, however deep the part sits. The node editor still selects anything directly.

Drag with the right mouse button to select everything inside a box - no keys needed (Shift or Ctrl with a left drag does the same). In ⛁ mode the box selects whole assemblies. A right click that does not move still opens the menu. The same right-drag box works in the **node editor** (it selects the cards it touches) and in the **curve graphs** - spine, profile, end profile and scale (it selects the points inside); Shift adds to what is selected, Ctrl takes away.

Arrange the nodes. The second button on the node editor's row (beside fit-all), or *Arrange nodes* in its right-click menu, tidies the whole graph: every tree flows left to right into its result, each input one column left of what it feeds and input **a** above **b**, so no wires cross; a long chain of combines

becomes one straight line with what joins it just below. Loose nodes go underneath. Undo puts everything back where it was.

Deleting. Delete removes what is selected - all of it, not just the last thing clicked. In ⛁ mode, or with **Shift+Delete**, everything feeding the selection goes too: select a placed library model and press Delete, and all its parts go with it. Deleting a combine on its own (the part mode, or the node editor's plain Delete) frees its inputs instead. The right-click menus have both: **delete** and **delete with its parts**. Undo brings it all back.

Moving the gizmo for a moment

The ⌖ button (bottom left, right after the manipulator chooser) snaps the gizmo somewhere else: click a point on the model and turning and scaling then happen about **that** point rather than the object's origin. It **snaps** to what you probably meant - one of the curve's own points, a corner, the middle of an edge, or the centre of a round edge (the axis of a hole or a boss) - and the status line says which it took. The gizmo's **axes turn to match**: along the edge it landed on, square to the face it was clicked on, so dragging and turning follow the part rather than the world. Press **Escape** to put both the pivot and the axes back; the world/local button also returns the axes. Nothing moves when you set it, a ring-and-cross marker shows where it is, and pressing the button again puts it back.

It is temporary on purpose: it is never saved, and an autosave or an undo taken while it is on records the object exactly as it is. For a permanent version, use **origin** in the object's panel: that makes the object turn about another object's origin and follows it if that object moves.

How the model is drawn

The **view** button (bottom right) cycles through nine ways of drawing the model. SweeperCAD starts in **clay + edges**; the view you pick is the one it opens with next time.

- **colour** - shaded and glossy, lit warm against cool;
- **grey** - matte and soft, like fired clay;
- **clay + edges** - the clay grey with the solid's own edges (the CAD edges between faces, not the mesh) drawn dark on top, only where you can see them: edges behind the part stay hidden. A round surface's seam, and the line where one surface is only stored in two halves, are left out;
- **wireframe** - every mesh triangle;
- **edges** - the solid's real face outlines and nothing else: a line along the sweep for every corner of the profile, a ring only where one loft piece meets the next, the caps, fillet edges and combine seams. The selected object also shows, thinly, the rings it is lofted through;
- **glass** - shaded in colour but see-through, so a hole, a hidden part or the spine shows through;

- **zebra** - black and white light bars in reflection, for checking how smoothly surfaces meet: the stripes **jump** where surfaces meet at an edge (G0), **kink** where they meet smoothly but the curvature changes (G1), and **flow on unbroken** where the curvature carries on too (G2). Judge a fillet, a chain of sweeps or a smooth imported surface by eye;
- **direction** - faces judged against a direction you choose in the **stats** panel (Σ): **red** where a face turns away from it more than the angle past square, **amber** within 10° of that, **blue** where it lies flat at the far end. With **+Z** it shows overhangs when printing upright; with another axis a mould's pull direction, the side a coating is sprayed from, and so on;
- **chrome** - polished metal in a lit studio, **in the object's own colour** (a red part is red metal, a gold one gold), with the colour view's warm and cool lights. A plain grey part reads warm and cool as it does in colour.

The bottom-left row and the right-click menu

The row, left to right: what a click selects (part or whole combine), the manipulator, its pivot, the curves and plane squares, the dials, the placement pin, the ruler and the kept dimensions, stats, and world / local axes. **Right-click in the 3D view** and the menu lists the same things in the same order (up to the dimensions), then the creation icons of the top row: **+** positive, **−** negative, **/** keep inside, **=** combine, **A=** auto-combine, **✱** array, then **copy** and **mirror** of the selection. +, − and / start placing a new object - click where it goes.

Right-click

In the 3D view, a right-click (without dragging) opens the **quick menu** - as a **pie** round the cursor or an ordinary **list**, chosen in Settings > Right-click menu. Whatever is under the cursor is selected first. It offers, in this order: - **manipulator** (move / turn / scale / off), **dials**, **curves & planes**, **dimensions**, **view** (every view mode, direction and curvature included), **grid & axes** (grid, axes, local axes, fit and reset the view), - **the selected part**: duplicate, mirror, delete, measure, reposition, pivot, draw on it, route from here, - **stats**.

In the pie, move toward a slice and click, or press its number; a slice with ▸ opens its own ring; the middle (or Esc) goes back or closes.

While a tool is running, a right-click means **done**: it ends a click-point curve on the surface (or stops drawing), finishes a route with the points picked so far, and ends placing.

On the app's small windows (a value to type, a choice to make), a right-click presses **OK** - but not inside a text field, where it is still copy and paste, and never on a question (a delete stays a deliberate click).

Draft mode — when the scene gets heavy

When two full builds in a row take more than ten seconds (one slow build is often one seam's rounding after a move, and does not count), the scene switches to **draft mode**, and the app tells you so. From then on every edit shows a **draft**: the most expensive parts are simplified (fewer sections, a simpler outline) or left unjoined (their booleans skipped, the cutters shown as the usual ghosts), until the draft takes about a second. Edits stay instant, however heavy the scene.

Be in no doubt about what you are looking at:

- an orange **DRAFT** banner sits across the top of the view while a draft is on screen — *not what would export*;
- every node the draft simplified or skipped wears a **DRAFT** tag on its card;
- the status bar says *DRAFT* and how long the last full build took.

Press **FULL** (next to PAUSE, or the B key) to build the real model, with every boolean and blend, exactly as it would export. It runs in the background with the view usable, and CANCEL stops it. After it lands you are looking at the real thing until your next edit.

The lens - full in one spot. When FULL is too much and you only want to check one blend or one cut, press **LENS** (left of FULL, or L) and drag a frame over that part of the view. Every object the frame touches is built in full from then on - cutters and blends included - while the rest stays the draft; the frame stays on screen, labelled, and the status line says *full in the lens frame, draft elsewhere*. Edits inside the frame keep it full. The frame is a choice of objects, not of pixels: orbiting the view does not re-pick. Press LENS again (or Esc) to take it off. A frame whose objects alone take over a minute comes off by itself and says so.

Exports, pictures and STEP/STL always come from the full model. If you export while a draft is on screen, the full build runs first and the export follows when it lands — you can never export a draft by accident.

A heavy file opens in draft mode straight away, using the build times it was saved with, so the first thing you see is a quick draft rather than a long wait. A scene that becomes light again (fewer parts, simpler outlines, or empty) leaves draft mode by itself, and so does FULL. The threshold is ten seconds; the `draft_seconds` preference changes it, and *Settings* can switch draft mode off altogether. The first build after inserting an asset does not count - every part is built once then, which is always slower.

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4. Building parts

Placing objects exactly

Click **+** and the next click in the 3D view places a new object on whatever it lands on — a face, or the view plane in empty space. The placed object faces *out* of the surface it was put on.

While you aim, the new object's **wireframe** follows the cursor, posed exactly as it will land (green for +, red for −), with the stick showing which way it faces. Tick **"line it up with the nearest edge"** in the placement settings and it also turns so its sides run along the edge of the face you point at - the edge the distances are measured from - by the smallest turn that does it. **"show the object while placing"** switches the wireframe off if you'd rather see only the stick.

Turn it with the wheel. While placing, the mouse wheel turns the object about the face it sits on, 15° a notch (the angle shows under the dot); the view does not zoom until it is placed. With the edge line-up on, the turn counts from the edge - two notches is square across it. Assets show their wireframe too, and turn the same way.

The placement settings window (the small icon beside +) holds what a new object starts as and how it lands. It opens at the right edge of the 3D view, out of the way of where you place things (drag it by its title to put it elsewhere; double-click the title to fold it). At the top, a picture of what the next click makes: the profile as you look down the spine, with its width and height, and the side view - its length, shaped by the scale graph, the segments marked. Under it, what it is:

- **new profile** — every profile preset as a picture: an n-gon by default (a square; its sides beside the pictures), a circle, a star, a gear, an I-beam, a heart... The one picked takes the **profile radius** as its size.
- **spine length** — how far it is swept (default 10)
- **profile radius** — how big it starts (default 7.07: the square's corners, so a new object is a 10 mm cube)
- **scale** — which scale graph it starts with (the default flat line, the *box*, or a preset such as *to zero*), each with its picture

And how it lands:

- **depth** — on the surface, or sunk in by an amount
- **keep** — a distance from the nearest edge, and a second, different distance from the edge the cross arm meets (see below)
- **snap** — how close the pointer has to be before a guide takes hold, in pixels; 0 turns snapping off
- **also snap to the middle** — offers the midpoint of each guide

The profile and spine-length fields grey out when you are repositioning or placing an asset, because they only apply to a *new* object.

Guides. While you hover a face, a dashed line runs from its nearest edge through the pointer. Set a *keep* distance and one point on that line is exactly that far from the edge; a second arm runs across the face at 90° so the far end of a narrow face is reachable too. The dot appears only when your click will land on it — no dot, no snap. Where the two arms cross is a point the set distance from *both* edges.

Two different distances. Fill in both *keep* fields — say 5 and 10 — and the point is 5 from one edge and 10 from the other. Each distance is measured square to *its own* edge, so it stays exact on a face that is not rectangular. Which two edges? The nearest one, and the one the cross arm runs into — so the pair follows your pointer as you move around the face. With both fields set, that point is the only thing the tool snaps to.

Both midlines at once. With *also snap to the middle* on, moving near where the two midlines cross snaps to the middle both ways rather than to one of them.

Readout. Beside the pointer, four distances — left, right, front, back — from the pointer to the edges of the face under it, along the face, in screen directions. It is a readout, not an input.

Repositioning (the pin icon) and **placing an asset** (double-click it in the shelf, or its *place* button) use the same hover, guides and snapping.

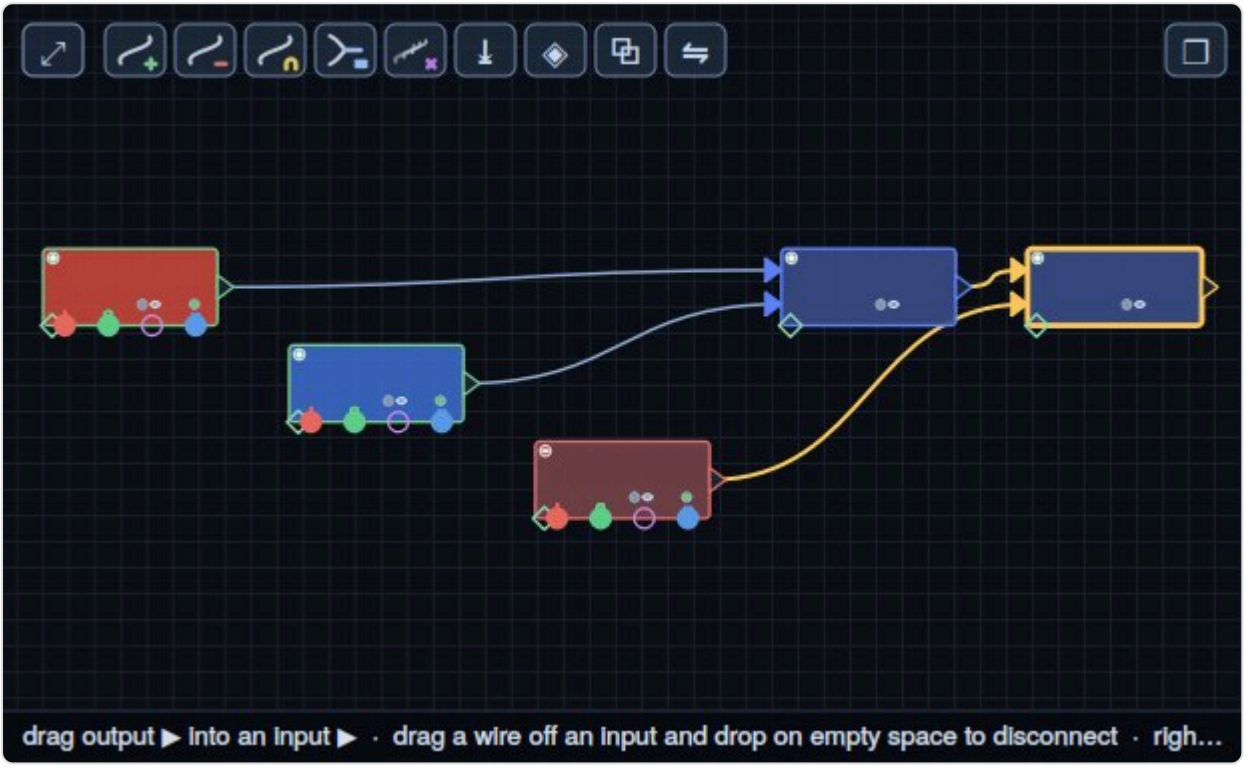
Carrying a spine on. While placing, repositioning or dropping an asset, point at the end (or the start) of any object's spine: a cyan ring shows there. Click it and a small menu asks — *Continue "..." from its end* attaches the object so its spine starts right there and carries on the same way, matching that spine's twist and scale at the joint (it sinks a hair into it, as any placement does, so the joint can be rounded); *Just place it here* places it normally. *Continue "...", profile linked* also links the new object's profile to the face it continues from - that object's end profile, or its profile while it has none - so a row of lofts stays joined: give a segment an end profile and the next one starts with it. Continued pieces are combined into one solid. Runs and stacks have the same choice: *link each segment's profile to the one before*. Moving the first object later takes the attached one with it (the same attachment the node editor's attach port makes). Choose *Don't offer this again* to switch it off; the placement settings window has the switch (*offer to continue a spine when clicking its end*) to turn it back on.

The camera keeps working while you place. Zoom and pan (middle-drag, or shift-drag) move the view mid-placement; orbit does not, because the left button is placing.

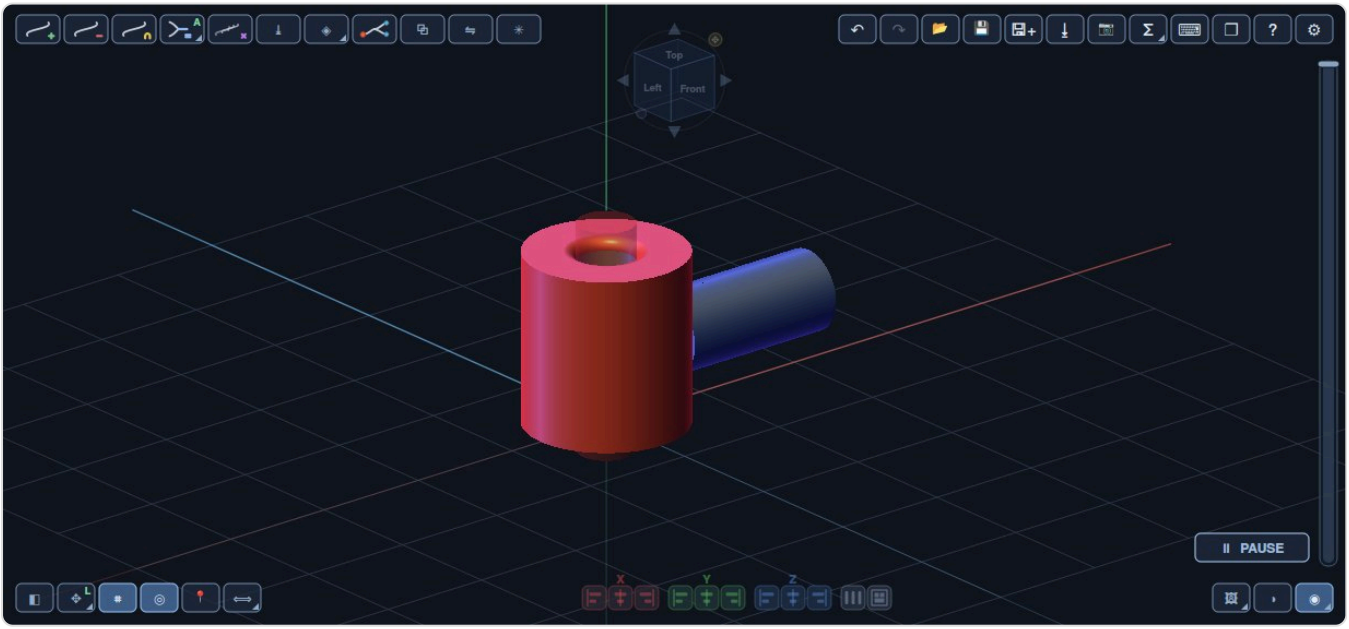
Twist and roll. *Twist* turns the profile progressively along the spine — 90° means the far end is a quarter turn from the near one. *Roll* turns the whole sweep at once, about the spine, including its first section. Use roll to orient a square or an n-gon; use twist to make it spiral.

Nodes

Each card is the colour of what it is: green a positive object, red a negative, yellow a keep-inside one, blue a combine, purple an array, orange a bridge, cyan an imported part (a mesh with a dashed border), magenta a lattice (a colour you give a node yourself wins). **Any node takes +, – or /:** the badge at its top left goes + → – → / (+) → / (–). A combine or an array normally takes its sign from what goes in; set from its badge it keeps its own - a combine of two parts set to – cuts as one piece - until the badge comes back round to the sign its inputs give it.



The node editor: two objects combined, then a negative one (the hole) taken out.



The result: each part keeps its own colour when the combine has none.

- **+**, **−**, **/** add a positive, negative or keep-inside object
- **=** combine two selected objects
- ✱ array
- ⭳ import a CAD part or a mesh
- ◈ lattice
- ⧉ **/** ⚇ duplicate the selected node, or duplicate it *with everything feeding it*

On a node: double-click the name to rename, click the colour chip to recolour. The chips along the bottom right are lock, visibility, blend mode (on combines), **curve or surface** (on arrays), colour, and the node's key number - twist on an object, the radius on a combine, the copies on an array. Clicking that number opens the node's own dialog: twist, roll and spin for an object; everything about the array for an array.

Imported parts look different on purpose, since they have no graphs: a CAD part is a cyan card with "CAD part"; a polygon part is cyan too, with a **dashed** border and "polygon · no CAD" - a reminder that it is a mesh. A lattice node has one input and no output: it is an end point.

The boolean is never chosen — it follows the signs. `(+)(+)` fuses, `(+)(−)` cuts.

The stick figure — balls and sticks, like ZSpheres

The ⦿ icon (beside the route tool, in the same corner group) opens the stick figure: a quick way to block in a figure, a creature, a branching part - anything made of limbs - before refining it with the graphs, the nets and the brush.

- Click the floor or the model: a ball. With a ball selected, a click links the new ball to it; with none selected it starts a **new branch** - click the selected ball again (or *New branch* in the panel) to drop the selection. Separate branches commit together, and combined they make one shape of several bodies (never bridged across the gap).
- Drag out of a ball's rim: a new ball grows under the cursor, linked to the one you dragged from, its size inherited. Release to place it.
- Drag a ball's centre dot to move it. Scroll over a ball to size it (the panel shows the size too). Drag the middle of a stick: its **bend point** - the spine between the two balls runs through it from then on; drag it (or the stick again) to shape the stick. One bend per stick: ball, bend and ball always lie in one plane, so the spine stays an exact flat curve and nothing twists; for an S-curve press **B** to turn the bend into a ball and bend each half. Delete removes the selected ball (with its sticks) or bend point; Ctrl+Z undoes inside the tool (Ctrl+Shift+Z redoes). Esc closes the tool; the icon brings the figure back.
- The selected ball shows two small buttons beside it: **−/+** flips its branch's polarity, **×** removes it - no key needed.

- **Mirror** (in the panel): a ball drawn on one side gets its twin on the other, linked the same way, moving and sizing with it - a character in half the clicks. A ball on the mirror plane is its own twin.
- **Commit as sweeps** turns the figure into ordinary objects, two ways (*Commit as* in the panel): **one sweep per stick** - each stick a capsule: its spine runs from the far side of one ball to the far side of the other, its scale graph a quarter circle up to the first ball's size, straight (or tapering) to the second's, a quarter circle down - each its own object, nothing fused: a plain collection of sweeps to edit and combine as you like (*Stick ends* chooses whether a stick runs to the far side of its balls, as described, or stops flat at their centres so cylinders and blocks simply meet at the joints) - or, with *and combine them into one* ticked, already chained through = nodes into one solid (where sticks share a ball, one carries it whole and the others at 99 %, so the "=" meets no coincident faces); or **fused** - a ball at every joint (a sweep with a half-circle scale graph, an exact sphere), a cone between joints, a half ball at a free end, smooth runs where a chain bends gently, all combined into one. Every graph is editable afterwards; the figure itself stays on the result ("Its stick figure..." in the node's menu draws it again).
- *Rounding* adds a fillet where a stick meets a ball. Off (0) a figure of fourteen balls builds in seconds; a rounding at every joint takes minutes.
- **The section on screen.** The selected ball shows its section: its outline (blue), a square handle at each end of one axis and one on the other, and a dashed **dial** round it with a yellow **knob**. Drag an axis end to squash the section; drag one end of the first axis further than the other and it becomes an **egg** - fuller on that side. Drag the knob round the dial to turn (roll) the section. The panel's squash, roll and **egg** fields show the same numbers. A stick between egg balls carries the egg as its section (exact: two half ellipses); a fused joint ball takes it too.
- **Handles on a bend.** Click a stick's bend point: its two **handles** appear (blue dots on white lines). Drag one to turn and stretch the curve through the bend - the other turns with it, opposite, keeping its length, as on any graph. They stay in the stick's own plane (ball, bend, ball), like every curve's handles. The committed stick is exactly this curve; a stick whose handles were never touched keeps the automatic curve. Ctrl+Z takes a drag back; the mirror twin follows.
- **Web** (panel, or the ◢ button beside the selected ball): skin between the selected ball's sticks - a wing's membrane. A ball with two sticks gets one web between them; with more, one between each two that lie next to each other going round it (a wrist's fingers), leaving out a gap wider than 170°. **sag** pulls the free edge from tip to tip back toward the ball (a bat's scalloped trailing edge), **billow** bows the skin out of the bones' plane (negative: the other way). Click **Web** again to take them off. **One gap at a time:** the selected ball shows a dot in each gap between neighbouring sticks - filled where a web is, hollow where one can go; click a dot to web (or unweb) just that gap. Committed, each web is ONE ordinary sweep - a fan: its spine an arc round the ball from one stick to the other, every profile a thin strip from the ball out to the edge in that direction (more of them where the edge runs steeply up to a tip), the skin half as thick as the thinner tip's ball. A web at a cutting ball cuts.

- A ball can **cut** instead of add: select it and click the **−** button that appears beside it (the − / + keys work too) - the ball turns red and every stick touching it cuts: committed, those sweeps are negative, taken away from the rest after every positive part is in. So a hole is one ball grown out of the body and flipped; the body stays as it is. **+** turns it back; a ball grown off a cutting ball cuts too. *Whole branch cuts* in the panel flips everything connected to the selected ball at once.
- A ball's **shape** (in the panel) is a circle or an **n-gon** with a side count - 4 a square, 6 a hexagon. A stick between two square balls is a block (flat ends at the balls' far sides, an exact prism); between a polygon and a circle the section morphs. Polygon joints are blocks too.
- **Tangent to the balls** (panel, on by default, with far-side ends): each round stick is the cone that just **touches** both its balls, so a half ball turns into the cone without a crease. Its scale graph follows each ball's circle up to the touching circle - a little behind the centre, toward the smaller ball - then runs straight. Off: the cone runs centre to centre at the balls' sizes and meets each ball at an angle.
- **Stick ends** (panel): to the **far side** of each ball (the default: half balls, blocks past the centre), or flat at each ball's **centre** (meeting at the joints); **and combine them into one** (on) joins them.
- A ball can be **squashed** across its stick (two factors) and **rolled** about it, in the panel: a flat palm, an oval forearm, a squashed head. Committed, the sticks' sections are the balls' ellipses and a joint's ball an ellipsoid.
- A chain of balls that bends gently commits as ONE smooth sweep along a curve through them, their sizes its scale graph (*smooth runs*, on by default); a turn over 60 degrees, or a bend tighter than the ball, is a joint and gets a ball instead.

Keep only what's inside (/)

The third sign, next to + (adds) and − (takes away): a **/** object **keeps only what lies inside it** of whatever it is combined with. It is drawn **yellow and see-through**, like a negative is red.

- **Placed or drawn on a face**, a / reaches from just below everything under that face to just above it - a cookie cutter: draw an outline on the top of a part and the part is cut to that outline, all the way through. With nothing selected it joins the object it was put on.
- **What is left counts as + or −.** Its badge on the node (and the sign box in the panel: *keep inside, adds* or *keep inside, cuts*) decides: kept as −, the trimmed piece then cuts whatever it is combined with next - a lens-shaped hole from two arcs, say. Clicking the badge goes
- → − → / (+) → / (−) → +.
- **Two / together** make one smaller keep-inside shape (inside both).
- **The trim blends** like a join: fillets, chamfers and smooth (G2) blends round the edge where the cut runs.

- A / trims only inside its own combine - put the parts you want trimmed and the / in a combine of their own, the same way you scope a negative.
- Nothing inside the /? The combine says so instead of leaving you guessing.

While a /'s combine - or anything in it - is selected, the parts it trims show **faint grey** at their full size, so you see what was cut away.

Uses: cutting a part to an envelope; keeping the overlap of two sweeps (two silhouettes, side and top); clipping a lattice or an array to an outline.

Bridging under = — parts apart, joined by one smooth piece

The = decides by what it is given. Parts that **meet** are joined by the boolean and blended as the **mode** says: *none* (sharp), *fillet* or *chamfer*. Parts that stand **apart** are **bridged**: one smooth piece grows between them, tangent to both, and the result is one solid with no seam to round. Apart means **ends facing each other** (a sweep's start or end caps), or one part being a **body** the others grow onto, or three or more meeting; parts that merely stand side by side - two boxes upright next to each other - stay a group, as they always did (the status line says so). Move the parts together and the same = joins them again.

Bridge as the mode: the = has a fourth mode, **bridge**, next to none, fillet and chamfer (in the panel, in the dialog that makes an =, and on the node's blend chip, which steps through all four). Under *bridge* parts that stand apart are **always** bridged, side by side too: each part grows from its side facing the other. Parts that meet have nothing to bridge, so they are joined as they are, sharp, and the status line says so. Under the other three modes the = still decides by itself, as above. While the parts are apart the bridge's dials show in the panel: **reach** (how far each part carries on before it turns), **bulge** (how full the middle is) and **flare** (below). Only solid parts of one sign are bridged: two positives make one positive, two negatives one negative (two cutters apart become one channel), two keep-insides one keep-inside. A negative pulled clear of its positive is left alone, and so are an array's copies.

Which end joins: each part's flat end that faces the other (a sweep's start or end), or what the bridge box's row for that part says: *auto*, *start*, *end*, *the side facing the other* (a flat side, so two boxes side by side are bridged side to side) or *body*. **A part with no such end is the body**: the other part's end outline is **projected along its own direction onto the body's skin**, that patch is cut away, and the piece grows out of the opening — leaving the body along its skin, the way a finger leaves a palm or a branch a trunk. A plate beside a thin part (a peg over a block's top) is taken as a body too, so the peg grows out of the plate instead of a funnel running from the whole top. **Flare** is how far the projected outline is grown before the opening is cut - the radius of the root, in mm: 0 means half the part's own radius, about the part's radius is the soft web a real finger has, more is a buttress. The piece leaves the skin tangent and turns in towards the part, as a fillet would. The part's end must be clear of the body (not inside it); move it out a little if the status line says so.

One part alone under = passes through as ever; choose **its start** or **its end** in the bridge box (the row per input, below) and that end is closed with a smooth **dome**, tangent all round, **bulge** setting how far it reaches.

More than two: while its parts are apart a combine has a **free port** below a and b - wire more parts in, and all of them go into one bridge (a palm and its fingers; several pegs grown onto one block, one after the other). A web between three or more wants **round ends**, so an end with corners (a box's, a star's - inward corners too) is **eased round** over its own last bit first: the part keeps its length, its corners soften into a rounded end, and the web joins that. Untick *round cornered ends in a web* in the bridge box to refuse such a web instead (said in words) and join cornered parts two at a time. When the parts meet, the = joins two at a time and sets the extra inputs aside (said in the status line) - chain another = for the rest.

The bridge node

A **bridge** joins any number of objects with **one smooth piece where they meet** - a palm and its fingers, a Y, a fork, a knot of tubes. Each input joins with its **flat end that faces the others** (a sweep's start or end), and the bridge leaves every one of them **tangent**: the result is one solid with no seam to round.

The = does this by itself for parts that stand apart (see *Bridging under =*), so the bridge node is no longer in the toolbars; a file that has one keeps it, with the same dials and ports. A bridge has a port for every input and **one free port** below them - wire another object into it to add it, or drag a wire off a port to take that input out again.

One input closes its end with a smooth **dome**, tangent all round - a fingertip, a rounded tube end, a bullet nose, on any profile; **Bulge** sets how far it reaches (1 round, more pointed, less flatter). **Two inputs** are joined by a **tube** from end to end, leaving each along its own direction, its section turning from the one end's outline into the other's.

Which end joins: each input joins with its flat end that faces the others (*auto*), or - in the node's dialog, one row per input - a sweep's **start** or its **end**. Needed when both ends face the middle, or neither does: two parallel tubes joined at their tops make an arch.

How it is made, for any profile and any number of inputs:

- Between each pair of **neighbouring** ends, a **web**: from the side of one end to the side of the next, its lowest line the **crotch** (between two parallel fingers, the U at the bottom of the gap). Neighbours are the ends next to each other round the joint; ends that come from all sides are joined to the ones beside them over the joint.
- What the webs leave is a **hole** on each side (the back of the hand and the palm), filled over all of them.

Reach - how far each input carries on into the bridge before it turns. **Bulge** - how full the middle is: 1 is as it falls, more is fuller, less is flatter. Both are on the node (click its number) and in the left panel.

The new pieces take the **colour** of the input they grow from - a web the colour of the larger of its two ends, a hole the colour of the biggest end (usually the palm); colour the bridge itself and the whole result takes it. A bridge takes **+, – or /** like any node: its inputs are always joined as solid parts, and its own sign says what the result does to what it is combined with next - a **negative bridge** cuts a smooth branching channel.

Each bridge is made in a helper process, trying a few layouts (side by side when the first is not good enough) and keeping the smoothest; a layout the kernel cannot make costs only that try, and a bridge never takes more than 30 seconds. While you move or edit, the quick look shows the inputs as they are; the bridge follows when you stop, and one made before comes back at once (undo, redo). An input without a flat end facing the others (an end closed round or to a point), or a joint that would fold, is said in words, and the inputs are shown as they are.

Sharp corners: an end with corners (a box's) keeps them - a tube from box to box, or from a square end to a round one, and a dome on a box's top carry the corners on as edges, smooth into each part. In a web of three or more, a cornered end is eased round just before the web (see *More than two*).

Mirroring an object

Mirror... sits in a node's right-click menu and on the ⇋ button, both in the node editor's toolbar and in the 3D view. The ⧉ beside it duplicates: for a combine it asks whether to copy just that node or everything feeding it, as the node menu does. The pop-up asks:

- **across which plane** - the object's own left/right, front/back or top/bottom plane (mirrored in place, so the image sits where the object is), or a world plane, YZ, XZ or XY;
- **through** the world origin, the object's own origin, or the centre of what is mirrored (world planes only);
- **keep the original**, which makes a mirrored copy rather than flipping the object itself;
- **include everything feeding it**, so a combine is mirrored with all its parts.

Only the placement changes: the graphs stay exactly as drawn, and a combine mirrors with everything inside it. A mirrored copy is an independent copy - it does not follow the original afterwards. Mirrored objects carry a ⇋ badge on the node, and their graphs say "mirrored in 3D", because dragging a point to the right there moves the mirrored shape to the left.

A stack of objects

The placement settings carry a **segments** field: 1 places a single object, more drops a run whose spines join end to end, with a **combine** tick to union them into one solid. It greys out when the click will not create - repositioning, or placing an asset.

Add a stack... in the node editor's right-click menu asks how many, then makes a run of objects whose spines join end to end. Each one after the first is attached to the one before it, so they stay joined when you move the first.

By default the stack is **placed with a click** in the 3D view, the same as placing a single object - it stands out of the surface under the cursor, and the whole run comes with it.

By default the segments are **combined into one solid**, so the stack is a single body - ready to fillet at the joins and to export as one part. Untick that and they stay separate objects that touch, which is lighter while you are still shaping them.

They are ordinary objects afterwards: edit any of them, fit any of them to a scan, delete one and the rest keep their chain. Their profiles are independent - changing one does not change its neighbours.

A stack of sweeps from cuts

In the section panel, **stack from cuts...** rebuilds what the plane is cutting - a mesh reference, or your own parts - as a chain of sweeps in one go. Say how many cuts: they are taken evenly along the plane's direction, from just inside one end to just inside the other, and each pair becomes a piece - the lower cut its profile, placed at the cut with the spine heading up to the next, the upper cut its end profile. Every piece's start is **linked** to the end of the one below, so editing one cut moves both pieces that share it and the chain stays continuous. More cuts follow a curved body more closely; six is a good start. Undo takes the whole stack back. (Point the plane along the part's length first - the cuts follow the plane's direction, whatever it is.)

Close the ends (ticked by default, beside the button): a cap at each end of the stack instead of stopping in a flat end just inside the part. The outer cuts move half a step inward, and each cap runs from its cut on to the part's very end, the cut as its profile and its scale graph measured from the part - exactly 1 at the cut, down to 0 at the tip - so a rounded end comes out rounded and a pointed one pointed. A cap's spine follows the end: the cuts are carried on past the outermost one, and the spine runs through the middle of each, leaving the cut square to it - so a tip that leans or bends away is followed to its point. (A spine bends in one plane: the cap is turned so the lean lies in it, and what leans across that is left to the scale.)

A bent body. Select an object whose spine follows the part's curve - a thin helper drawn along a hose, a horn, a handle - and *stack from cuts* asks whether the cuts should **follow that spine**: planes square to it at equal steps, and each piece carries its own stretch of the curve. Piece two is placed

exactly on the end of piece one, and so on, so the chain is continuous by construction; the helper can be hidden or deleted afterwards.

Arrays - along a curve or on a surface

An array repeats whatever is wired into it. It **takes that object's sign**: an array of a negative is negative (drawn red and see-through), so an array of holes cuts.

Where the copies go is chosen on the node itself - the **curve / surface chip** (a wave, or copies on a line) and the copies chip both open the array's dialog - and in the panel, "place on":

- **along the curve** - the Array graph, as always: count, one per point, follow the curve, spin and scale per copy.
- **on the surface of an object** - pick the object; the copies stand on it, each turned to face out of it (a negative one faces in, so a hole reaches into the part). **Scatter** spreads them evenly over anything (with a minimum spacing and a random seed); **grid** puts rows along the spine and columns round the profile of a plain sweep. **Depth** sinks each copy into the surface (negative lifts it off); **random twist** turns each about its own axis.

Each copy is placed by **its own origin** - the point the object was drawn around - not by the world's origin, so it does not matter where the object sits before you array it. The copies follow when the surface changes. Untick a copy in "copies shown" to leave it out.

Where the curve sits. An **open** curve starts at the object's origin: the first copy is where the object was. A **closed** curve goes **round** the object: its middle (the middle of its extent) is on the object's origin, and the Array graph marks that point. Untick **"closed curve: round the object"** to place a closed curve exactly as drawn instead. Copies round a closed curve are spaced evenly all the way round - six copies, six equal gaps - and each turns about its own origin to follow the curve. While you drag a point the middle stays where it was; it settles when you let go.

Editing the curve in the 3D view. The array's purple curve is drawn where the copies are - it moves and turns with the array - and is edited there like a spine: drag a point, **double-click a point** to switch it between sharp and smooth, drag a smooth point's **handles** (the other side follows), **double-click the curve** to add a point, and right-click a point (or point at it and press Delete) to remove it.

The object being repeated shows as a see-through **purple ghost** where it was modelled - the array graph's colour - whenever no copy sits there (round a closed curve, say). It is not part of the result: placing, snapping, measuring and exports pass it by. Click it to select it and edit it like any object; the copies follow.

The original as well. The copies replace the object. Tick **"keep the original too"** to keep one more exactly where the object is - a hub inside a ring of copies. It is not added twice when a copy already sits there.

Wrapping a part onto a surface

A part modelled flat - standing on a flat floor - can be bent to lie on the part under it. Tick **wrap** in the placement settings and place it, or select a part already in the scene and set the left panel's **wrap** row: the way, and the part to wrap onto. A wrapped part's node has a wrap chip with the same choices. The wrap is live: move the part, or change the part below, and it wraps again.

- **Straight down** - each bit drops straight onto the surface below it and keeps its height above it. Right for a part small against the curve under it; on steep sides it stretches, and says by how much.
- **Along the surface** - each bit is carried along the surface at its own distance: over a rounded edge and down the side, round a tube.
- It refuses, and says why, where it cannot work: over a sharp edge, past the edge of what is below, on a surface too steep, or a part too tall for an inside curve.

Hollow

Every object and combine has a **hollow** field (under "result"; 0 = solid). It makes the part a closed shell of that wall thickness, as **exact CAD**. On a combine the cavity is built from the parts, so the wall is never thinner than asked (thicker where a union joins). On some **filleted** combines the CAD kernel cannot do it; the app refuses rather than return a wrong part, and points to the lattice node's hollow instead.

The same setting sits **on the node** as a small chip: a solid dot when the part is solid, a lit ring when it is hollow. Click it to type the wall thickness (0 = solid).

Draft — sides that lean, so it comes out of a mould

A part comes out of a mould only if its sides lean away from the way it is pulled. **Draft** in the node box (under round rims) leans them by that many degrees: on an **object** the sides lean in toward the **end** of its spine - the start keeps its size, the end shrinks - exactly, every side a straight wall at that angle (a profile of straight sides is leaned by the kernel as planes; a curved one through its profiles). **Below zero** they lean out. On a **combine**, a bridge or an **imported part** the draft leans every flat and round side that stands steeper than the angle, going up from the part's lowest point; a curved side the kernel cannot lean is left straight and said. The card says ⟋ *draft 2°*; the field takes names and maths, and `part.draft` is a name other fields can use. From a script: `model.draft(part, 2)`.

Does it come out? The **casting** window has a **draft** check: pick the way the mould comes apart (up and down, front and back, left and right) and the least lean a side needs (about 1° for a smooth mould, 2 to 3° for sand or a textured surface), press **check draft**: the view colours every bit of the surface - **green** leans enough, **yellow** too little, **red** a straight wall, **purple** an **undercut**, a spot the mould cannot let go of because part of the part sits over it in the way it is pulled (a groove round a neck; a cap on a stem is fine, the mould splits at the cap). The worst spots are ringed, and the numbers say the smallest draft and how much of the surface falls short. A two-part mould split at the widest section, from the shape alone - a first look. From a script: `model.draft_check(part, 1.0)`; for an AI app, `sweepercad_check` with `draft`.

Screw holes. The profile menu's **screw hole (negative part)...** adds a ready hole as a negative object: **M2 to M12**, **clearance** (the screw passes through), **countersunk** (a flat head sits flush, 90°), **counterbored** (a socket head sits below the surface) or a **tap drill** (to cut a thread in), at the depth you give. Its top sits at the surface (it reaches a hair above, so the cut opens cleanly) and it goes down; combine it with the part, move it, or place it on a face with + while it is selected. Typical workshop values - check them against your screws. From a script: `model.hole("M4", "countersunk", 8, at=(x, y, z))`.

Machine parts — nozzles, gears, pulleys, a snap fit

The profile menu's **machine parts** make a whole part from one form, as ordinary objects (a sweep, or a few in a combine): afterwards edit their graphs, move them, cut them, give them a draft like anything else. Each form says the numbers worth knowing as you type. From a script: `model.machine_part("timing", teeth=20, belt_width=6)`.

- **Rocket nozzle**: throat and exit diameter, wall, a **cone** (its half angle) or **Rao's bell** (a parabola from the throat angle to the exit angle, the length a share of the 15° cone's). A tube open at both ends: the chamber, an arc of 1.5 Rt into the throat, 0.382 Rt out of it, the contour. The contour is the sweep's **scale graph**, so you can reshape it there; the form says the area ratio and the angles.
- **Ring gear (internal)**: teeth, module, face width, rim - a disc with the involute space cut out, the hole a gear of the same module runs in. For a planetary set: sun + planets + ring (ring teeth = sun + 2 × planet).
- **Helical gear pair**: like the meshing pair, twisted to a **helix angle** at the pitch circle (the twist is in each gear's twist field), opposite hands, the centre distance apart; a small pinion gets the profile shift that saves it from undercut.
- **Chain sprocket**: teeth and the chain (06B, 08B, 10B, 12B, ANSI 25 to 60, bicycle): a seating arc for each roller on the pitch circle (p / sin(π/z)), flanks out to the tip circle, a bore.
- **Timing pulley**: GT2, GT3, HTD 3M/5M, T2.5, T5 - the belt's pitch line sits the pitch-line distance outside the pulley, each groove a round pocket; **flanges**, a **hub** and the **bore** as ticks. The form says the outside diameter and the belt widths.

- **V-belt pulley**: Z, A, B, C, SPZ, SPA, SPB; pitch diameter, the number of grooves, a bore. The groove section (36°) swept round, a web, the bore. Too small a diameter for the belt is refused.
- **Flat-belt pulley**: diameter, width, a **crown** that keeps the belt centred.
- **Snap fit**: a cantilever **hook** - beam length and thickness, width, **undercut** (how far the head stands out), the **lead-in** angle to push it in and the **return** angle (90° locks, 45° can be pulled out again) - standing up with its head on top, and beside it the **slot** it snaps into, a negative part through a wall, the head plus clearance. The form says the **strain at the root** when it snaps (1.5 × undercut × thickness / length²) and which materials take it (PLA about 1.5 %, PETG 2, ABS 3, nylon 4, PP and POM 6).

A slot round a curve you drew

Draw an **open** curve on the profile canvas - a line, an arc, a bend - and pick **slot round this open curve…** from the preset menu: the curve becomes the centreline of a slot of the width you give, its sides following the curve, its ends round. A straight line gives the stadium exactly; a bent curve a curved slot - the outline a pin of that width leaves following the curve. (The slot replaces the drawn curve; redo it after changing the centreline.)

The lattice node

◈ makes a lattice. With an object selected it **fills that object**; with nothing selected it fills a box. It has one input and no output - it is always the last step - and it is drawn and exported as a mesh.

Types, grouped: surfaces (gyroid, Schwarz-P, diamond, Neovius, IWP, lidinoid, split-P), 3D struts (cubic, BCC, BCC+cubic, FCC, octet, diamond, star), 2D fills extruded along an axis (square, diamond, triangle, kagome, honeycomb), foams (Kelvin, rhombic, random Voronoi), and **your own**: a custom cell typed as struts, or **any object of yours** as the cell, repeated every cell size.

Surface cages - hollow wireframes on the object's surface, nothing inside: a **Voronoi cage** (the borders between cells spread evenly over the surface) or a **strut cage** (points spread evenly over the surface, neighbours joined by straight round struts - an even, mostly triangular net). **Cell size** is the spacing, **wall / strut** the strut's thickness, **seed** another random spread. A **tile cage** lays **your own 2D cell** (edit custom cell) into every quad of an even net over the surface - each quad turned the same way, so a tile with a direction keeps it; with no 2D cell yet it is a plain quad cage. On sharp edges the net comes out bevelled at about the cell size. A **rind** is a shell just inside the surface, **depth** thick, with a smooth round hole in every cell - the carved-gourd look; **wall / strut** is the rim left between two neighbouring holes, and the rims are rounded where hole meets shell. "Pattern in" and skin do not apply to cages (nor depth, except for the rind).

Form: sheet (a wall either side of the surface) or strut (the solid network). **Cell size** and **wall / strut** thickness in mm; a **skin** closes the outside with a solid wall.

Pattern in decides where the pattern goes: the whole volume, a plain **hollow** wall, a **patterned shell**, a **perforated shell**, **emboss** (standing out of the surface) or **engrave** (cut into it), with a **depth**. "Follow the sweep's surface" wraps 2D fills round a sweep, bends and all, with no seam.

Decal: text (any font), an image or an SVG, projected onto one side of the part and embossed or engraved there.

Place - a repeating pattern (gyroid, honeycomb, struts, foams, grid, Voronoi) can be slid **across / up / along** the object and **turned**, as numbers in the panel or by hand: with the lattice selected an amber handle with three spokes sits on the part - drag it to slide the pattern, Shift-drag to turn it; the numbers follow and the model rebuilds when you let go.

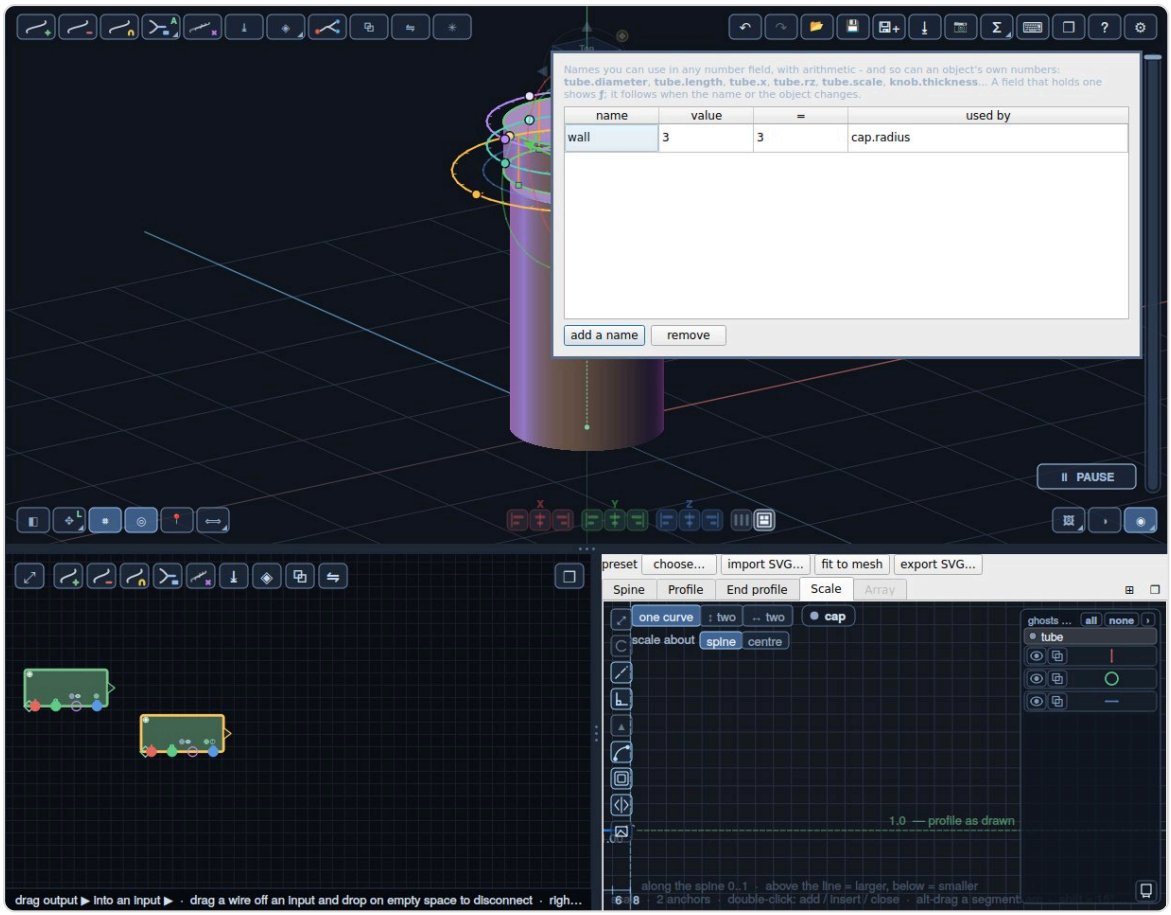
Grading - "vary" thickness or cell size **by** a direction, radially, by a **grayscale image**, or by **your own graph** (the Grade tab, x along the part, y in mm). **Attractors**: other objects that thicken (or thin) the pattern near them, fading over a distance you set.

Cut: holes - other objects cut right through wall, pattern and skin: drain holes for a hollow part, or bolt holes.

The view shows a lighter **preview**; tick "full detail in the view" to see exactly what will be exported. Exports are always full detail.

Parameters

Any number field takes a **name** instead of a number, with arithmetic: `tube.diameter + 2`, `wall / 3`, `tube.length`. Every object offers its own numbers under its node name - `diameter` and `radius` (a circle profile), `width`, `height`, `length` (the spine), `twist`, `roll`, `x`, `y`, `z`, the rotation `rx`, `ry`, `rz` (degrees), the scale `sx`, `sy`, `sz` and `scale` (the uniform size), and a lattice's `thickness` and `cell` - and the name completes as you type (Tab takes it). A field that holds one shows **ƒ** before the value; hover it for the expression; type a plain number to let go. When what it refers to changes - typed, dragged on a graph, dragged in 3D - the field follows and the model rebuilds, down a chain (a cap that follows a sleeve that follows a tube). It works in twist, roll, position, rotation, scale, a lattice's wall and cell, the circle-radius chip on the profile graph, and the length of a straight spine (click its label). **Edit → Parameters...** is the table of your own names (`wall = 3`), each with who uses it. A typo keeps the last value and says what is wrong; a deleted object does the same.



Edit → Parameters...: your own names and who uses them; the cap follows the tube.

Types - one design in several sizes. Below the names, the same window holds **types**: each row a type (M6, M8, M10; small, medium, large), each column one of your names, each cell that type's value for it. **New type from the values now** keeps the model as it is under a name; edit its cells for the next size; **use this type** writes its values into the names, and everything that follows them - sizes, positions, a lattice's wall - follows. The type in use is marked ●, and an edit to its cells goes into the model at once; an empty cell leaves that name as it is. Types are saved with the project, and using one is one undo.

Levels - named heights. Further down the same window, **levels**: a floor, a platform, a roof - each a name with a height, drawn in the 3D view as a dashed outline round the model with its name and height. Select parts and **stand the selected parts on it**: each part's bottom goes onto the level (it keeps its place across), and from then on it moves up and down with the level - change a deck's height and everything on it follows. A level is an ordinary name too, so it can follow others (`rail = deck + 900`) and be used in any field; **take them off** leaves the selected parts where they are, free again. Renaming a level takes its parts along.

Pick instead of typing. While a number field has your cursor, click a part in the 3D view or its node in the graph: the part's number for that field (`tube.rz` in a rotation field, `tube.x` in a position field) drops into the field and the field keeps the focus - the part is not selected. Type the rest (`+ 90`) and press Enter. Clicking the selected object's own node is an ordinary click.

The ƒ chip. A node with a linked field shows a small **ƒ** among its chips. Hover it to see the links and dotted wires to the objects it follows (they show only while you hover, so they never tangle with the curve links); click it to select the node, with the links in the status bar.

Turn and size from the measure tool. Position is the kept dimension above (relative); rotation and scale have their own row in the measure window. Select one object, measure from it to a point on the other, and tick **link** on the **rotation** row and/or the **scale** row - the two are linked from then on, both ways: turn or scale either one with the gizmo or a field and the other does the same (a **ƒ** in the transform fields). With **keep offset** ticked when you link, both keep what they have now and move together from there: 5° more on one is 5° more on the other, half its size stays half its size. Without it the values are shared - the selected object takes the other's rotation or scale now, and they stay equal. Untick **link** to free them again (a plain number typed into one field frees that axis only). Each object turns and scales about its own origin (or pivot), and the kept offset is a real composed turn in the other object's frame, so it holds whichever way that object is tipped - not three angles added up, which would only be right about a single axis. This is separate from a kept dimension's **lock the angle too**, which fixes the measured bar's direction. A linked pair shows a dotted line between the two origins in the 3D view with what is linked at its middle (↻ rotation, ⤢ scale, ↔ a linked position); click the badge to unlink one or all of them. When a kept dimension already joins the two objects, the badge sits on that bar instead and no second line is drawn. It comes and goes with the kept dimensions (the ⊢ switch).

`rotx(obj, ox, oy, oz)`, `roty(...)`, `rotz(...)` can be typed into any rotation field yourself: the object's rotation composed with a turn of (ox, oy, oz) degrees in its frame, one axis of the result per field.

Assemblies — joints between parts

A part can be **held by another** and move about an axis that moves with its holder: the lid on a box, an arm on a base, a wheel on an axle, a slide in its rail. The **joint** box in the panel (for a top-level part) has it:

- **held by** — the part that holds this one. Pick it and the joint is made: where the part is **now** is its pose at the joint's zero.
- **kind** — a **hinge** turns about the axis, a **slider** moves along it, a **ball** turns about the point (its angle, plus a tilt and a swing), **fixed** just rides along.
- **axis** — **pick the axis in 3D...** and click a face of the holder: a round face (a pin, a hole) gives its own axis, a flat face its normal through the click. Or world **X / Y / Z** through the part's origin. The axis is kept in the holder's own frame, so moving or editing the holder moves the axis with it.
- **value** — the angle or distance, held inside the **range**; the **slider** under it runs through the range, **play** runs it back and forth until stopped.

- **check collisions** — steps through the range, moves this part (and everything jointed to it) as the joint moves it, and says where it runs into another part, with the deepest spots ringed in the view.
- **take the joint off** — the part stays where it is.

Chains: a joint on a jointed part (hand on arm on base) — they resolve holder first before every build. **Moved by hand** while jointed, the part keeps the move as its new rest. The card says *⟲ hinge 30°*. Export writes the parts where the joints hold them.

Joint values are names. `lid.angle`, `slide.slide`, a ball's `tilt` and `swing` work in any number field and can be bound: a wheel's `angle` set to `crank.angle * 2`, two lids that open together. From a script: `model.joint(lid, box, "hinge", origin=(0, 20, 30), axis=(1, 0, 0), min=0, max=110)`, `model.set_joint(lid, 45)`, `model.unjoint(lid)`.

What it does not do: **closed loops** (a four-bar linkage, a scissor lift - those constrain each other in a ring and want a solver) and **contact** (a cam pushing a follower). Each joint has one holder and the value you give it.

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5. Fillets and chamfers

Rounding and chamfering a corner

Drag the radius on a corner and it rounds. **Drag back through zero and it chamfers** - the same control, the other direction. The corner is trimmed identically either way; a round joins the two trim points with an arc, a chamfer with a straight cut.

Zero has a small landing place in the drag, so you can put a corner back to sharp without sliding past it - about two per cent of the corner's own edge, so it feels the same however far you are zoomed out.

Double-click the radius to type it and the dialog asks which you want, since a typed number cannot say it the way the drag does. Zero is still a sharp corner and comes back exactly - the corner point is never moved.

A fillet that changes along the seam

Select a combine and every **ring** where its parts meet shows in the 3D view (with the curves switch ⌗ on - the same switch that shows the curves in 3D), with **one point** carrying the ring's radius: **0** when the seam is still sharp. Each point has a **stem** with a small square on its end - the longer the stem, the bigger the radius. Everything is done with the mouse alone: - **drag the square out** of the ring: a **fillet**, wider the further out; **drag it in**, past the ring: a **chamfer** (the square becomes a diamond) - the same control, the other direction, as a corner in the graphs; back on the ring it is **sharp** again. With one point the ring is the same all round; - **click a +** between two points to add a point there, then set its square: the blend now changes from one point to the next, smoothly, all the way round (a ring has no start or end). Chamfers are constant: a ring's points are all fillets or all chamfers; - **drag a point** to slide it along the ring; - **pull a point clear off the ring** (it turns red with a cross) and let go to remove it; the ring always keeps at least one point; - **click a point** to type its radius exactly (a negative number is a chamfer).

The combine's **mode** follows the rings: none while every ring is sharp, fillet otherwise. A negative radius in the seam table is a chamfer.

Where two round parts meet, the ring is made of several seams end to end (one per face). The points belong to the whole ring, not to one seam. Every change rebuilds the blend and is one undo step.

Quick shapes: under the seam table choose a **shape** (the same all along, one swell, thin in the middle, two swells...) between the **thin** and **full** radius, and press **apply** - for the rings of the seams selected in the table, or every ring when none is. A ring with a shape shows "ring 0.3–3 (drag in 3D)"

in the table. For one seam on its own you can still type points as `t,r; t,r` (t from 0 to 1 along the seam).

An ordinary fillet is round (G1): the surfaces meet with the same direction but the bend jumps - from flat straight to the fillet's radius - and the zebra view shows a kink in the stripes where the blend starts. On polished metal that kink is a break in every reflection.

Rounding the rims of an end

A sweep's flat end meets its sides at a sharp rim. **Round rims** in the **node** box (under hollow) gives the **start** and the **end** rim each a radius - and so does the **rims chip** on the node itself, beside the hollow chip, or dragging the rim in direct editing. It is built into the part itself: a boss with a rounded top, a block with its edges eased, a peg whose end will not catch. **Below zero it is a chamfer** of that size - the same control, the other direction, as a corner's drag back through zero. 0 is sharp and comes back exactly. While a rim is rounded or chamfered, the node's card says so (◠ *rims 2 · 0*) and its chip is lit.

The rounding is part of the object, so it stays through every join: a part rounded this way and then combined with another keeps its rounded rim, and the seams the join makes on it are listed like any other (the new faces are *round0* and *round1*). A rim that will not take the radius is tried at 70, 50 and 30 % and the status line says so; one that takes nothing is left sharp and said. An end with no flat cap - a closed spine, an end scaled to a point - has nothing to round and says so once.

To round where two parts MEET, use the combine's blend; to close an end with a full dome, use a bridge with one input. The rims are for the edge in between: a flat end that is no longer sharp.

Smooth (curvature-continuous) fillets

Tick **smooth (curvature-continuous)** in the combine's blend box. Each fillet is then rebuilt so it matches not only the direction of the faces it joins but also how they bend (G2): the stripes in the zebra view flow round the corner without a kink, and so do reflections on the cast part.

- **fullness** (0.5 - 2, default 1): how far the blend reaches into the corner. More fills it out, less pulls it in toward the corner. The blend starts and ends where the ordinary fillet of the same radius would.
- Radius, variable radius and rings work as before - the smooth blend is built on top of the ordinary fillet, between the same two lines.
- A seam with corners - a square peg, a rib, a boss - is smoothed as one loop: round each corner the blend turns on the peg's edge the way a ball rolling over it would - the peg's edge stays sharp down into the blend, as it does on a real filleted boss - where the ordinary fillet runs each

side on past the corner and cuts them against each other with a hard crease. A very narrow tip (a star's point with a blend nearly as wide as the face) may keep the ordinary fillet.

- A seam that runs into a flat end cap (a sweep's end) is smoothed too. Where it cannot be done safely - a seam ending somewhere else, a chamfer - that seam keeps the ordinary fillet, and the status line says how many were smoothed and how many kept.
- **Which is which**: with the smooth switch on, the seam table's status reads **✓ 2 · G2** for a seam that came out curvature-continuous and **✓ 2 · G1** for one that kept the ordinary fillet (hover it for why), and the ring's label in 3D says **G2** or **G1** after the radius.

Fullness that changes along the seam. With the smooth switch on, each point on a fillet ring gets a second, **hollow square** a little beyond the stem's square. Drag it **out** for a fuller blend there, **in** for a flatter one (0.5 - 2; the label reads `r 2 · f 1.4`); with two or more points the fullness eases from one point to the next, like the radius. Click the hollow square for a menu to type it or go back to the default.

An uneven chamfer

A chamfer's point has a **hollow diamond** beyond its stem's diamond: the chamfer's **second distance**. Untouched, both distances are equal. Drag the diamond out or in and the chamfer becomes uneven - `c 3/1.5` cuts 3 mm into the seam's first object and 1.5 mm into the other. **flip the chamfer's sides** in the diamond's menu swaps them. With several points the two distances each ease round the ring.

When a fillet will not take

A seam that refuses to blend is asking for a radius that does not fit. Which way to move depends on the shape: on a finely traced or sliced outline the radius is usually too LARGE for the room between facets - a 16-sided profile blends at 0.5 mm where it fails at 1 mm - while on smooth surfaces meeting at a shallow angle it can be too SMALL — measured on a box and cylinder union, 0.25 to 1.0 all failed and 1.5 blended every seam. When one fails the app tries a few larger radii on that seam and tells you the first that works, so the message names a number you can use rather than a dead end.

A fillet the kernel makes by running off the face it should join - landing on the far edge of a narrow strip with a crease - is never kept, however "valid" the kernel calls it: the search goes on to the radius that stays on both faces, tangent.

Rounded another way. Some joins the kernel's fillet cannot make at any radius - a handle whose outline turns tighter than the radius where it meets a curved body is the usual case. Then the app rounds the seam another way: it trims a narrow strip off both faces along the seam and fills the gap with a surface that meets both faces smoothly. Where seams meet at corners - three edges of a box at one point, a square peg's loop turning at the peg's edges - each run of seam between corners is

done that way, the runs meet in a mitre where a loop turns, and what is left open at a corner where three meet is filled with a patch that meets everything round it smoothly. The result is checked like any fillet (one closed, valid solid), and the combine says *rounded by trim and fill*. If the full radius does not fit, it tries half the radius before leaving the seam sharp. A seam that stops on a face, with no other seam carrying on from its end, is still left to the kernel. It is a switch under **Settings > Geometry kernel** (*round seams the fillet cannot, by trim and fill*), on by default.

What the app does first. A seam around a boss or a hole is one LOOP, even though it is made of several edges. If the full radius does not fit, each loop is blended at the largest radius IT can take, the same all the way round — so a loop comes out even, never half round and half sharp, and one tight spot does not shrink the other loops.

Each spot as big as it fits. Even is not always what you want: a boss standing close to one edge of a plate gets the small radius its cramped side allows all the way round. Tick **each spot as big as it fits (uneven)** in the combine's blend box and every stretch of the loop gets as much of the radius as there is room for there — measured from the seam to the nearest edge of the faces beside it — easing smoothly between, so it is large where there is room and small where it is cramped, never a step. The status line says how far it eased down, and where.

Two options for the hard cases, in the combine's blend box - the first on unless you untick it, the second off until you tick it:

- **where it will not close: round face by face.** A seam that runs over one part's sharp edge - a bar ending right on a box's corner, a cap cutting two faces - sometimes will not round at all. Ticked, it is rounded against each flat face in turn and trimmed by the next face, so the part's edge carries on across the fillet as a crease. Tried only when the ordinary fillet has failed; two solid parts, one radius.
- **too big for the face: roll over the edge.** Normally a fillet too big for its face is made smaller until it stays on the two faces it joins. Ticked, it keeps (nearly) the radius you asked for and runs over the face's edge onto the next face; the edge it rolls over shows as a ridge.

When a rounding crashes the geometry kernel. Rare, but it happens on awkward shapes. The app then looks for the combine at fault and tries it at half and at a quarter of its radius, and only if those crash too shows that one combine without its rounding — every other rounding stays. The status line names the combine and what was done; your own radius is not changed.

Where it happened. In the 3D view, a loop that got a smaller radius than you asked for is marked in **yellow** while its combine is selected, and a seam that was left sharp is marked in **red** all the time — so an edge you did not mean to leave sharp is noticed without opening anything. The combine panel lists the same seams, with the reason for each.

The **section switch**, next to the reference switch, cuts a loaded reference mesh. Click a surface and the plane is taken from it, the same way placing works; the sections found are drawn over the model, and clicking one makes it the selected object's profile and keeps a copy in your curve library so you

can use it again. Cut across a cylinder and you get its circle; cut down the side and you get the rectangle through the middle — the click decides which.

fit to mesh, beside the SVG buttons, samples whatever else is there — a loaded reference mesh, or another object in the scene — at this object's own ends. When there is more than one thing it could sample it asks which: everything, the imported mesh, everything you modelled, or one object by name — combines included, since a combine is what you actually see. Picking "everything" skips the parts that feed a combine, so a tube is counted once rather than three times. The section panel offers the same choice, in the same words and uses what it finds as the profile. It asks which: the start, the end, or both — fitting one leaves the other exactly as you drew it. Fitting an end profile turns on "morph to an end profile" for you; you do not have to tick it first. Both ends get the same number of points so the morph between them does not twist, and the result is a normal editable curve. The object's ends have to be inside the mesh.

The scale graph is applied on top. The sweep's section anywhere is the profile (morphing into the end profile) times the scale there, so what is measured at an end is divided by the scale graph's value at that end: with a graph at 0.5 the fitted profile comes out twice as big, and the built object still meets the reference. (Using a section from the section panel as a profile does the same.) **fit the end profile's shape, its size into the scale** is the other way round: the end profile is kept at the start profile's size, so it only changes the shape, and how much bigger or smaller the end is goes into the scale graph's end, where you can see and edit it.

fit everything: spine, profile, scale does the three below in the order they depend on each other: the spine first (the other two are measured from it), then the profile - the reference's section a little inside the start, cut square to the spine there, and an end profile likewise inside the end unless the reference narrows to a point there - and the scale last, measured against that (morphing) section. Measured: a straight line and a plain circle became an S-shaped oval horn at 99.3% of its volume, on average 0.4 mm from its surface. An object is never measured against itself: the combines it is part of are left out of what is sampled.

Two more ways to fit everything, for **stacks** (pieces joined end to start by *create stacked objects* or *connect spines*):

- **fit everything, keep its ends** - one piece inside a stack: its two ends stay where they are, in place and in direction, and only its inside follows the reference.
- **fit the whole stack** - the stack's spines are joined into one, centred as one curve, and split again at the joints, so each piece ends in exactly the direction the next one starts in (an attached piece is turned to match its neighbour's end; pieces fitted one by one cannot meet that). Sections at a joint are cut exactly there, so both neighbours share the seam. Measured: three straight pieces became an S-shaped horn, on average 0.6 mm from it, the joints closed. The stack has to lie in one plane; one that turns in 3D is fitted piece by piece with *keep its ends*. The other pieces of the stack are never sampled as reference.

The same menu has **centre the spine in it**: draw the spine roughly through the reference - a straight line is enough, and it may stick out at the ends - and every point of it moves to the middle of the reference's surface on its two sides, square to the spine in the graph's plane, round after round until it settles. An end that stuck out is cut back and put on the reference's surface; an end inside stays where you drew it. The result is a normal editable spine with as few points as hold it (measured: four rough starts on an S-shaped tube all came within 0.2 mm of its true centreline, with 3-5 points). A spine is a curve in its object's plane, so how far the reference's centre lies off that plane cannot go into it - the status line says how far to move the object instead.

fit the scale graph: the spine is walked from its start to its end and, at every step, the reference's surface is found on both sides of it - in the plane the graph is drawn in, or, on an asymmetric object split front/back, across it (the split switch decides). Its distance from the spine over the profile's own half-size becomes the scale there, so the sweep takes on the reference's changing thickness along the whole spine, not only at the ends. A symmetric object gets one curve (the mean of the two sides); an asymmetric one gets its two curves, one per side. Only the stretch between the start and the end profile is read. Where the reference stops before the spine does, the scale runs to **0** there, and the sweep ends in a point at that spot - the rest of the spine is left out, and the status line says at what fraction. So: put the spine through the reference (a line taken off its surface, or drawn), give it roughly the right profile, fit the ends, then fit the scale. With the spine centred first, a straight line drawn through an S-shaped tube came back as the tube at 100.3% of its volume.

It is the exact opposite of applying a scale: measured on an S-shaped spine, a scale graph applied to a sweep and fitted back comes back within 0.4%. That holds while the profile turns with the spine (*follow the spine*, the default, or *up*). With a **fixed** or **yaw** heading on a curved spine the sections lean into each other and the fit is approximate - the status line says so.

A scale graph that reaches 0 and stays there - fitted or drawn by hand - ends the sweep where it first reaches 0, with a point; before this a zero in the middle of the spine was clamped to a hair and ran as a needle to the end.

The panel chooses how a section is sampled:

- **mixed** fits curves along the smooth runs and keeps creases sharp. Best for scans and most real parts, and it passes exactly through the corners.
- **smooth** fits one curve all the way round - fewest anchors on anything round (a 48-sided section goes from 145 points to 3) - but it rounds corners off.
- **cornered** keeps straight segments and sharp anchors.

Fewer anchors matter: they are what every later fillet and boolean pays for.

A section can be a **spine** as well as a profile: "use as spine" sweeps the selected object ALONG the line you cut instead of across it - trace a rim or a seal on a scan and follow it. A closed section gives a closed sweep.

The **section switch**, beside the reference button, cuts a loaded reference mesh. Switch it on and a plane appears in the middle of the model, already cutting; move and turn it with the ordinary gizmo and the curves follow it. The panel also takes a height and a tilt as numbers, if you know the cut you want. The curves it finds are drawn in place - the chosen one bright, the others dim. **Tab** picks between them, **Enter** makes the chosen one the selected object's profile, and **S** saves it to your curve library, where it is offered like any other saved curve.

The measure panel offers a **lock**: the straight line between the two points, or only the distance along X, Y or Z, or the distance ignoring one axis ("ignore Z" is the horizontal distance between two things at different heights). The line on screen always shows what the number counts.

Typing a distance moves the object along whatever the measurement counts, so the number you type is the number you then read. The window names the object that will move, and says so plainly when nothing is selected — measuring still works, but there is nothing for a typed number to act on. Switching the tool off closes the window with it.

Keep a measurement as a dimension. Select the object that should hold the distance, measure, and press **keep**. The measurement becomes a lasting dimension, drawn in the view with its value and a lock. It keeps the measurement's lock too: along X only stays along X only.

- **Relative** (the default) measures to a point on the other object. When that object moves, yours follows.
- **Absolute** measures to that spot in the world, which never moves, for example "6 above the floor". If the far end isn't on another object, the dimension is absolute anyway, and the status bar says so.

The solver moves **positions only**, never rotation or size. While you drag with the gizmo, the object slides only where its dimensions leave it free: one along X stops it moving in X, and three pin it. Dimensions that cannot all hold turn **red** and nothing in their group moves; redundant ones turn **amber**; one whose object was deleted turns **purple** and keeps its last position. Click a dimension's label to retype it, make it absolute or remove it. An attached piece cannot keep a dimension, because its attachment already places it.

Placing can keep its distance too. In the placement settings (right-click the + or − icon), **keep distance** can be off, relative or absolute. With it on, an object placed by snapping to "N from an edge" keeps that distance as a dimension. It is measured square to the edge, across the face, and turns with the object it was placed on when relative. With two distances (from two edges), it keeps both.

The ⊢ icon (bottom left, beside the measuring ruler ⟺) shows or hides the kept dimensions. It turns itself on when you measure or place, and it is remembered between sessions. Hidden dimensions still hold.

No hinging in 3D either. Tick **lock the angle too** before you keep a measurement, or in a dimension's window later. The bar then keeps its direction as well as its length: the object cannot swing round the other end, and when the dimension is relative it turns *with* the other object. These dimensions show

∠.

Which drives which. A kept dimension's arrow points at the object that moves. In its window, **swap** hands the moving over to the other object. The measure line also shows an arrow at the end a typed number would move.

The **what-is-shown** button carries its state: a filled circle for the result, a hollow one for the components, a ringed dot for both.

Each of these tools refuses to switch on when it has nothing to work on, and says which thing is missing: the **pin** needs an object selected to reposition, the **section** switch needs something to cut - an object or a reference mesh to cut, and **measure** needs something in the scene to click.

The **measure switch** lets you click two points and see the distance between them. The points snap to edges and features the same way placing does, and a third click starts a new measurement.

The **✛ switch** at the bottom right draws the world axes through the origin — red for X, blue for Y, green for Z, the same colours the gizmo uses. The grid gives you a floor; the axes give you a direction on it. Switch them off on a dense model, where three long lines through the middle are clutter.

While the **pin** is carrying something it is drawn faint, so you can see the surface you are aiming at through it. Nothing has moved yet: cancel and it is exactly where it was.

When you move something with the **pin**, it ignores the object being moved, so a click lands on whatever is behind it rather than on the thing in your hand.

A click on **empty space puts the object on the ground**, where the grid is - not floating at whatever depth the camera implied. Click a surface instead and it goes there, as before.

An object placed ON a surface sinks a hair into it — one per cent of its own size — so the two genuinely overlap and there is a seam to fillet. Without that they would touch and share no volume, which looks identical on screen and blends nothing.

If the app says **nothing to blend**, the two objects are not actually overlapping — they may look as though they touch while sharing no volume at all. Move one into the other a little.

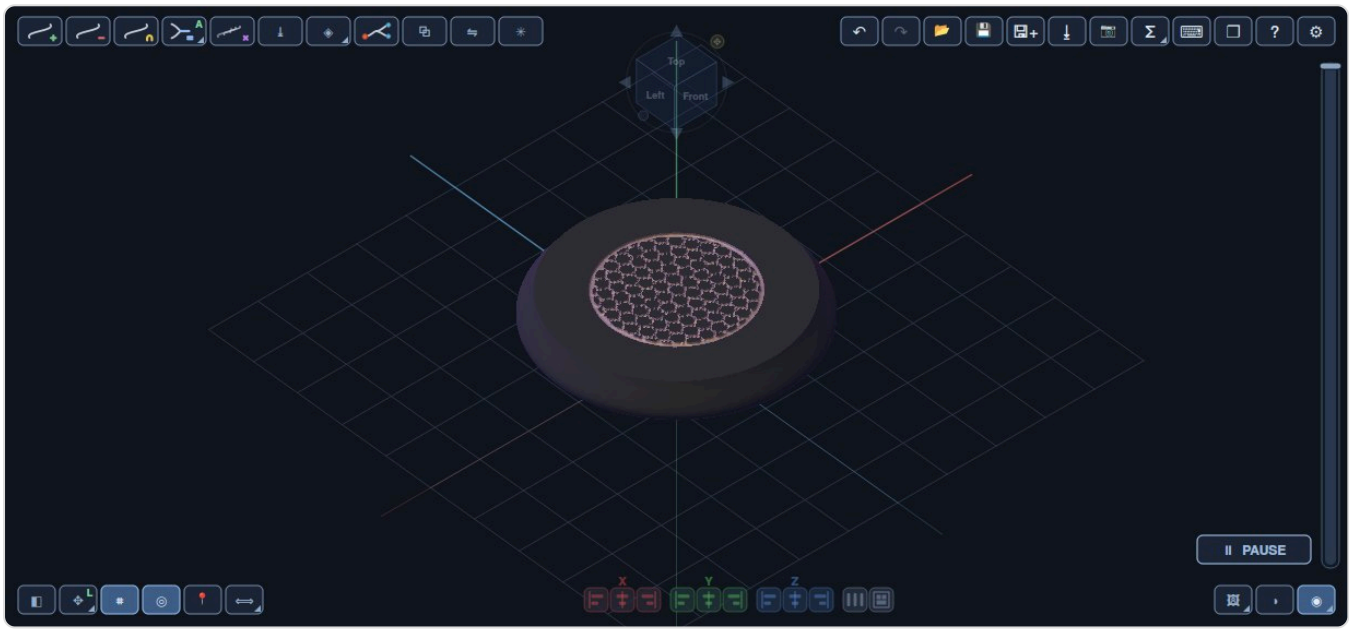
Other things that stop a fillet: - **stacking blends** — a radius that reaches an existing fillet surface - **objects that barely touch** — move them into each other a little - **forcing it** — the 'force' tick makes it try anyway, which can fail loudly instead of leaving the edge sharp

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6. Surfaces, routes and more

Drawing on the surface

✎ in the 3D view draws on a part like a pencil, and the strokes become **wires**. Select the part, click ✎, and: - **drag** across it: a freehand stroke that sticks to the surface (the stroke is evened out and the hand's jitter smoothed away); - **click points**: a smooth curve through them, hugging the surface; **Enter** (or a double-click) ends it, Esc drops it; - **Shift-drag** erases the strokes it touches; **Ctrl+Z** undoes; - a press **off** the part still turns the view, so you can turn round it and draw on the other side; **Esc** (or ✎ again) stops.



A closed loop drawn on the surface, filled with a honeycomb that follows the curve.

With a drawing tablet, **pen pressure** sets the thickness as you draw (firm is thick, light is thin). The strokes are 3D, on the surface.

The drawing is a **lattice node** of type "drawn", with the part as its input: a light, closed **mesh**, made from the strokes' distance field - no CAD booleans. The part stays on screen, see-through, to draw on. The **drawing** panel: - **result**: **wires only** (the part is just the guide - a cage for casting), **wires, the part cut out** (their inner side follows the part exactly: a **tight fit**, with a **gap** for a sliding fit or shrinkage), **raised on the part**, or **cut into the part**; - **fillet**: a rounded blend where strokes meet - good for the metal's flow; - **tapered stroke ends**, **pen pressure**, and **fill closed loops**: a loop drawn back to its start becomes a sheet on the surface inside it (for patches, not for loops that wrap all the way round a part).

The wire thickness is the lattice's **wall / strut**. Select the drawing and click ✎ (or **draw** in its panel) to add to it later.

The border where two parts meet

The extract group in the bottom-right row (◫ plane, ⌇ line, ⋒ border) has its own icon now; the reference switch moved to the display group. **⋒ border**: select the two parts that meet (Ctrl-click the second) and click it. The curves where their surfaces cross come out as **spines** — a thin round sweep on each, placed where the border is: swap the profile, or use the spine elsewhere.

- A border that lies in a plane (a pin through a slab, a ball through a plate) is **one spine**, closed when it closes.
- A border curved in 3D (two pipes crossing: a saddle) is cut where it leaves its plane into **stacked planar spines** meeting end to end — every piece an ordinary spine you can edit.

Extracting a line from a surface

The **line** button (bottom right, grouped with the references and the section) takes a curve off any surface - a reference scan or one of your own objects. Click two points on the surface: the curve between them is previewed in the 3D view, and a small panel sets:

- **plane** - the curve lies in one plane through your two points, and this picks which: **along the view** (the stretch that looks straight from here - aim it with the camera), **upright on the surface**, or **shortest** (of all the planes through both points, the one giving the shortest curve).
- **points** - how many, ends included.
- **curve** - straight between the points, or smooth.

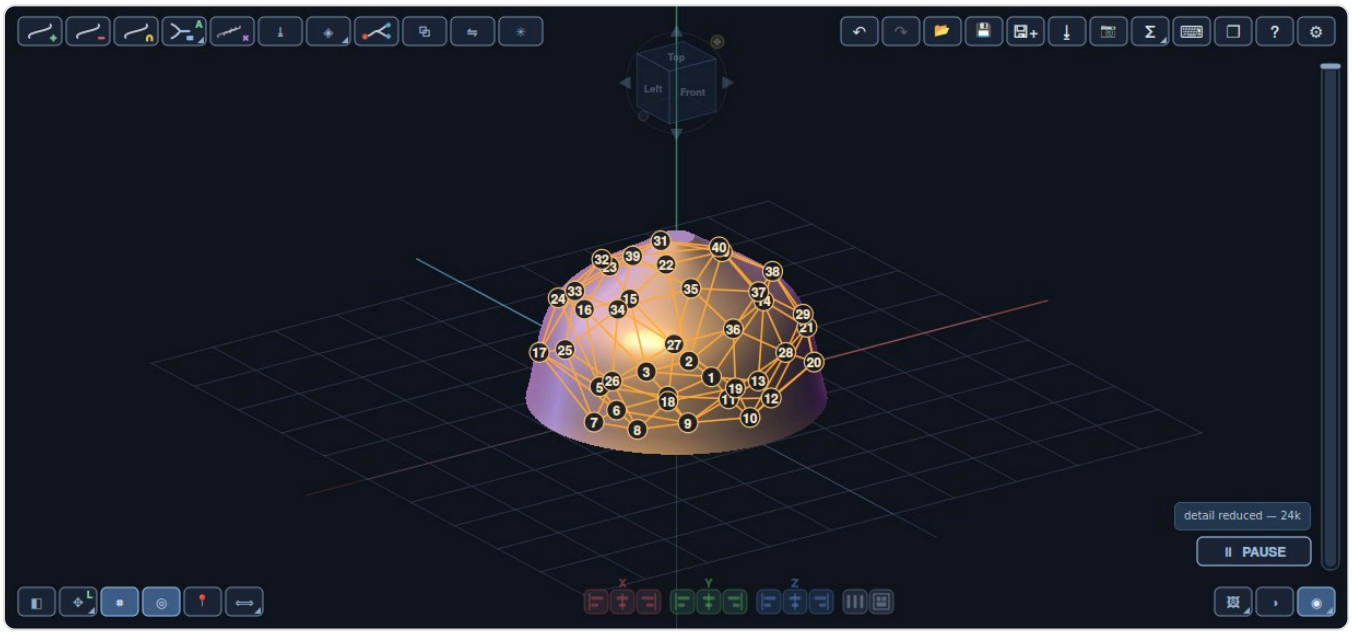
Save to library keeps it as an SVG with your other curves; **Use as spine** makes it the selected object's spine - and, with **move the object there** ticked (the default), moves and turns the object so its new spine lies exactly where the line was taken, whatever its scale and wherever it sits in a combine.

Use as scale makes the line the selected object's **scale graph** instead. Take it down the side of a scan - a vase, a handle - with the object's spine already in place: each point's distance from the spine, over the profile's own size in that direction, becomes the scale there, so the sweep takes on the scan's changing thickness. If the line stops short of an end, the scale is held level to it; points past the spine's ends are left out. (The line is cut by a plane: on a straight-sided cone, use *upright on the surface* - the view plane can just touch the surface there instead of cutting it.)

On an **asymmetric** object (two scale curves, one per half), each point goes to the curve of the half it lies on - and one cut fills both: the line you picked runs down one side, and the other way round the cut runs down the far side, so with the plane *upright on the surface* a single extraction gives both halves. The cut should run along the object's split axis (left/right or top/bottom); if it is well off, the status line warns, and if it is nearly square to it nothing is changed. A spine gets a loft piece per point, so a handful of points is usually plenty. A third click starts again.

Building it from struts

✳ **build as struts** (in the make-it group at the top right, with Σ, casting, machining and unfold - or **File > Build as struts…**) turns the model into a net of struts and connectors you can build for real - straws, tubes, sticks - with a connector that joins many struts at a point. Say how many struts; the app lays a net of near-equilateral triangles over the model, pulls every strut towards one length while the net stays close to the surface, and shows it over the model: nodes numbered from the bottom up, struts as lines (tick *label every strut* to see A-B on each). The panel says how equal the struts came out (a closed curved shape cannot be tiled with exactly equal triangles - the number of struts meeting at a node, 5, 6 or 7, carries the curvature; expect a few percent), the object's size, the tube to buy and the connectors by how many struts meet. **Real strut length** scales the whole plan so every strut is the length you actually have. **Two or three lengths** snaps the struts to classes instead, for a connector with no play. **Save the plan** writes three files: the plan as text (nodes with positions and how many struts meet, struts as pairs of node numbers with lengths, and for every node the struts that land on it), the same as CSV, and the net as an OBJ of lines.



A dome as a near-equal strut net, its nodes numbered.

Routes

A **route** is a tube found between two points, around the parts. Click ⇝ (in the 3D view, or right-click in the node editor), click the **start** and then the **end**. Either can sit on a part or in free space, for example down to the floor. **Shift-click** points in between to make it pass through them. Esc cancels.

Its settings window opens with the tool, like the placement window (right-click ↝ to open just the window). Set the tube radius, the plane, the clearances and the rules there *before* you pick the points; with no route selected the window shows what the **next** route starts with, and that is remembered. With a route selected it edits that route, live. The same settings sit in the inspector's **route** panel when the window is closed. Drag the title to move it, double-click to fold it.

Editing its points. Select the route and its points show in the 3D view: the **start** green, the points it passes through blue, the **end** orange, and a small **+** halfway between each two. - **Drag** a point to move it. It stays in the route's plane, and a point that sat on a part stays with that part. Hold **Ctrl** while dragging to drop it onto the surface under the cursor instead. - **Click a +** to add a point there, then drag it where you want it. - **Click a point** for more: type its position, remove it (points in between only), let go of its part, or pick all the points again.

It stays in **one plane**, so its spine is an ordinary spine you can edit in the spine graph. By default the plane is vertical, through both ends; it can be horizontal instead, or **best**: a few tilts about the line between the ends are tried and the shortest wins - that can look into a hole or round a tall wall from the side.

A plane per leg (a chain). Pick the route with points in between, then choose **a plane per leg** under plane: the route splits into one leg per stretch, each in its own plane (vertical through its ends, or horizontal), joined into one piece in a combine. At every corner both legs run along the line their planes share, so the tube carries on round the corner smoothly - no kink, no ball. With vertical legs that line is vertical, so the tube turns while rising or falling. Drag a corner's dot to move it (both legs follow); a setting changed on one leg goes to them all. To add or remove a corner, pick the points again.

In the **route** panel: - **live**: it is found again whenever its ends or the parts around it change. Its ends remember the part they were clicked on and follow it. Off, it is an ordinary object you edit by hand, and **re-route** finds it again once. - **tube radius**: the profile circle. - **style**: **smooth** takes any angle; **square** keeps to straight runs along the plane's two axes with as few turns as it can, like pipework. - **wander**: meanders instead of taking the tightest way, still around the parts and under the rules. **another** gives a different meander. - **keep away**: how far to stay from other geometry *where possible*. It squeezes closer where it has to; **0 creeps** along the surface. - **never closer**: a hard limit. The ends are exempt, since that is where it touches. - **min bend radius**: straight runs are fine, but every bend is at least this round. - **straight between bends**: to bend the route from real tube or bar, set the bend radius to the bending die's and this to the straight the bender's clamp needs: runs too short to hold both bends and the clamp are flagged under the route. - **Lay it on the surface…**: the route's points joined OVER the part they sit on - the shortest way across its surface, a tube's radius off it (plus the "never closer" gap) - made of straights and bends at the bend radius, a gentle curve as several small bends. It is built as STACKED SPINES: one ordinary sweep per plane, joined in a combine, so every piece stays editable, and the pieces meet in the middle of a straight. The route itself is hidden. A tube can only approximate a surface curved two ways; the bending tool (⌒) reads the pieces as one tube and tells you how far it strays. - **direction** with its rules: - **one way**: never go back against the direction. - **max lean**: never lean more than this from the direction. - **min slope**: e.g. a drain that is

never flat. - **ends**: **at the point** you clicked, or only on the plane it sits on: **anywhere on its surface's plane**, or **anywhere at its height**. The route then arrives wherever on that plane is easiest. - **cling**: within this distance of a surface the route may break the direction rules, creeping along it like a vine - under a lip, along the floor, into a crevasse. In free air the rules hold. - **inside the part - a channel**: runs *through* the parts instead of around them and is cut out of them: the route turns negative and joins the part it starts on. **keep away** is then the wall it leaves to the outside. Both ends run on a little past the surface so the channel opens cleanly. (Holes already cut into the part are not avoided yet.) - **leaves** / **arrives**: free, square to the surface, vertical, horizontal, or along or against the direction. - **stay above the floor**.

If no route exists under the rules, it says so and keeps the last one; relax a rule or add a point to pass through.

Drop onto the floor

↧ at the end of the align row, after distribute (or right-click > the part > drop onto the floor...): each selected object falls from where it is, as it is held, onto the floor or the objects below it. When it touches, it tips over that point or edge the way its weight takes it, and keeps going until it stands - on a face, or leaning on another object. You see it fall. It doesn't bounce, and it doesn't slide (it grips where it touches). Set **turn** to *never* and it falls straight down without turning. An object that starts too low - in the floor, or inside another object - comes straight up out of it first, and the panel says how many did. **What you didn't select stays put**, and the falling objects land on it; hidden objects are ignored. **Gap** keeps them that far apart.

Importing CAD parts and meshes

⭳ (or File > Import CAD or mesh) brings in another program's work as a part of your model - not just something to look at.

- **STEP, IGES, BREP** become exact **CAD parts**. A file with several solids asks: one object, or one per part.
- A **closed STL or OBJ** becomes a **polygon part**: an exact solid made of its flat triangles, so it booleans like anything else. Open meshes, overlapping pieces or more than 100,000 triangles are refused with the reason - import those as a reference instead.
- An **OBJ or STL** also offers **Smooth CAD surface**: it is read as a subdivision cage (as in SubD modellers) and turned into the smooth surface it stands for, as real CAD patches - it then goes into STEP like any CAD part. Add edge loops in the cage to keep an edge sharp. Any other mesh is **prepared** first: corners that coincide are welded (most exports store each face with its

own corners), doubled faces dropped, holes closed, and a dense mesh (a scan, a sculpt, most STLs) is rebuilt as a light cage whose smooth surface is fitted back onto it. The status bar says what was done. Sharp edges come out rounded - it is a smooth surface.

- A **STEP or IGES with surfaces but no solid** (exported as loose faces, often with hairline gaps) is **healed into a solid**: the faces are sewn together across their gaps and closed, at the smallest tolerance that closes them - the status bar says how far it had to reach. Reload heals it the same way.

An imported part has a transform and a sign and nothing else: move it, turn it, scale it, add it or cut with it. **File > Reload imported part from source** re-reads the file and keeps where you put it.

A mesh as editable sweeps (beta)

The third answer for an STL or OBJ, **Editable sweeps** - and **Convert to sweeps…** for a mesh already loaded as a reference (in the references list, and at the bottom of the *fit to mesh* menu) - rebuilds the mesh as ordinary SweeperCAD objects: sweeps and combines you edit like anything you drew.

Method: sections along the spines (the default). Every part of the mesh - the trunk and each limb or branch - becomes one sweep whose profiles are the mesh's own cross-sections, cut square to the part's centreline at stations along it and carried along the spine, so the sweep IS the mesh between stations. A part ends where its section meets another's: walking a finger from its tip, the first section that also holds the palm's spine is where the finger merges - its root; the trunk (an arm: forearm, wrist and palm are one sweep, it never splits) runs to its last section that holds no other spine - through a junction on the way (a pipe with a branch, the thumb's spine inside the palm), and no further once it has become one finger's. The trunk is the skeleton's path that carries the most of the part, or the part's own axis when that carries more (a screw, a shaft, a part with a wide head). A free end runs on to the last of the mesh round the axis, so a slanted cut is sectioned down to its last sliver. Stations sit where the section changes size or shape, few where it does not, and both sides of a step are kept; where the section changes faster than the stations are spaced (over a dome), a section is cut halfway and kept when the straight loft would miss it. A part without branches is laid a second time, through the centres of its own sections, so a dome is cut square to its own axis. **Negative space**: the outer shape is read with every hole filled - any space a cut through the part encloses all round, a bore, a cross-hole, a socket, a navel - so its spine ignores them (a block with a bore has a straight spine). Each hole is then a negative sweep on its own centreline, sectioned square to it, run on out through an open face, a closed bottom left where it is. Where a hole breaks through the outside, the outer sweep runs straight across it. Sharp corners (a block's edges, a hex socket) get a sample of their own in every section of that sweep. **Hard surfaces as CAD**: each section is first read as what a maker draws - a circle, a regular polygon, a rectangle - and otherwise fitted as lines and arcs with real corners where the sides meet. Where a part's section stays the same along a straight stretch, the stretch becomes one sweep with that one profile on a straight spine, built exactly - planes and true cylinders, radii you can read and type; a step between two such stretches (a screw's head on its shank) makes two pieces that overlap at the step. A part whose section really changes keeps the sections-along-the-spine sweep. **A machined part** is tried first as layers: looked at along the

directions its flat faces face (and its main axes), its cuts stay the same for a while, step, stay again - a plate and its bosses, a flange and its hub, a bracket's foot and upright. Each layer's outlines become exact extrusions of their fitted sketches, an outline that carries on through the next layer stays one piece, and the direction with the fewest layers wins. A hole with a step in it (a counterbore over its bolt hole) becomes two exact pieces. **A bolt circle is one hole, arrayed**: round holes of one size, parallel and evenly spaced on a circle become one hole and an **array** of it round the circle's centre - change the count, the circle or the hole, and they all follow; places on the circle with no hole are left out of the array (its skip). A counterbored hole still counts: its shorter bore lies inside its counterbore. A step between layers sits at the level of the mesh's own flat face there, and a scan's noise and a simplified (decimated) mesh are read too - the hexagon of a socket, the rims of a block, the step of a plate. A rounded or chamfered rim at a part's end (the cuts there the layer's outline evenly inset, by an amount that grows toward the face) is the extrusion's own rim rounding or chamfer, fitted to its size - not sections. A step's **inside corner** rounded or bevelled - a fillet under a screw head, where a shank meets its collar - is read the same way from the smaller layer's cuts, which grow evenly toward the step: the combine joining the two pieces carries that fillet or chamfer, to its size. **A draft** - walls leaning at one angle, as a casting pattern's or a moulding's do to come out of the mould - is read too: where every cut is the wide end's outline evenly inset by an amount growing straight with the height, that stretch is one layer with its object's **draft** set to the angle. A smooth taper is not read as steps: across a real step the cuts on either side differ, across a drift they do not. **A turned part** (every cut one circle, their centres on one line) is one circle swept along its axis, with its half section as the scale graph - up the start face from the axis, along the outline as lines and arcs, down the end face - edited like any scale graph. Tips are closed by a dome (a bridge node); limbs from the trunk's end run on into it and are rounded where they meet; a limb out of its side is rounded on. The profiles' points are numbered by angle round each section's own centre, so the surface between two stations runs straight. On a scanned arm: about 0.6 mm from the scan on average, as six sweeps; a socket-head screw comes out as one sweep and one negative sweep, 0.08 mm; a shaft with an off-axis cross-hole and a blind bore as three, 0.07 mm. **Classic** is the earlier way below - body and limbs by thickness, two profiles and a fitted scale - still there for parts it suits.

It is a **beta**: parts made of tubes and branches, turned parts and blocks with bores come out well; a soft, sculpted shape (an arm, a figure) comes out recognisable but rough - each sweep has one profile at each end, and a long part that changes its shape along the way is simplified. For a sculpted mesh, the import's **Smooth CAD surface** answer follows it far better for now.

- **A body and its limbs** (a hand, a figure): the thick body is found first - everything thinner than about half of it is taken off - and every piece that is long and reaches out is a **limb**. The body is one sweep along the axis its sections change least along, measured against itself; each limb is a sweep along its own centreline, its root carried on into the body.
- **A branching part** (a Y, a tree, a bracket of tubes): every branch is a sweep that ends flat a little short of the fork, and one **bridge node** joins the branches at each fork - the fork comes out as one smooth piece, as it is in the mesh, with no seam to round. A body's end and the limbs that

grow from it (a palm and its fingers) are bridged the same way; a limb out of a body's side joins it with a rounding. With *forks joined by a bridge* off, the branches run into each other and are rounded where they meet instead.

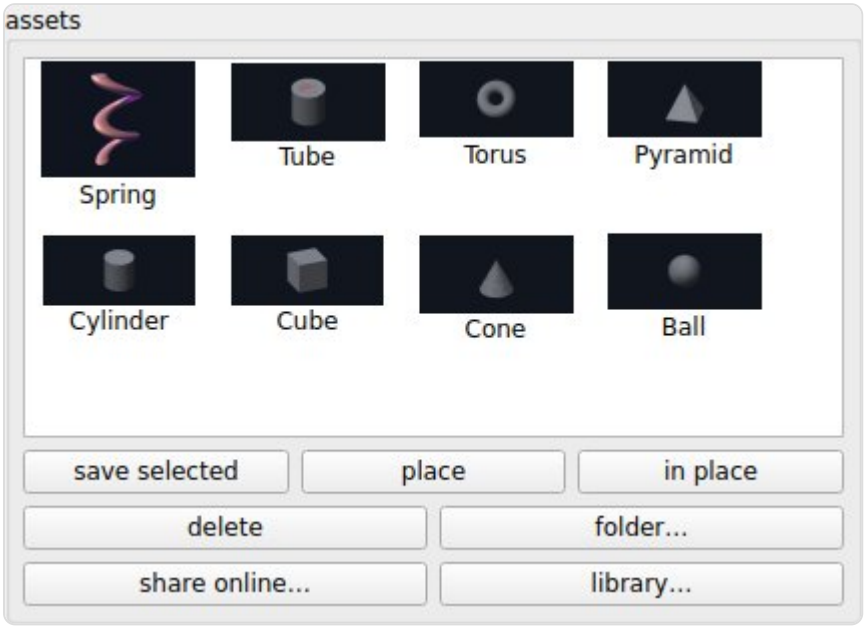
- **One body** (a block, a stone): when its section is the same all along, it is an **exact extrusion** of the part's own section, corners and all.
- **Any units**: a mesh file does not say what its units are. The panel shows the size and offers a **Scale** - a very small or very large mesh is suggested a power of ten that brings it to a size the modeller works well at (1 keeps it as saved). The grid it is read on, and the rounding where the parts meet, follow the part's own size.
- **A mesh as it comes**: corners are welded, loose bits left out, small openings closed on the mesh itself, and a mesh that is still not a clean solid - holes in it, overlapping or doubled pieces, flipped triangles - is first made solid from its surface (the outside flooded in, what it cannot reach is the part); everything after is measured on that. A clean CAD mesh is read exactly, its corners kept.
- **Holes**: holes, bores and pockets are filled first to find the main shape (just enough that no tunnel is left through the part), then each is cut back as a **negative sweep**; a hole that opens out runs well past the surface.

Each sweep's profile and end profile are cut from the mesh, its scale is measured along its spine, and its tips close round. The forks are bridges; whatever else meets is joined in combines and **rounded where they meet**; each rounding is tried in the build's helper, and one the kernel cannot make is left sharp - nothing hangs.

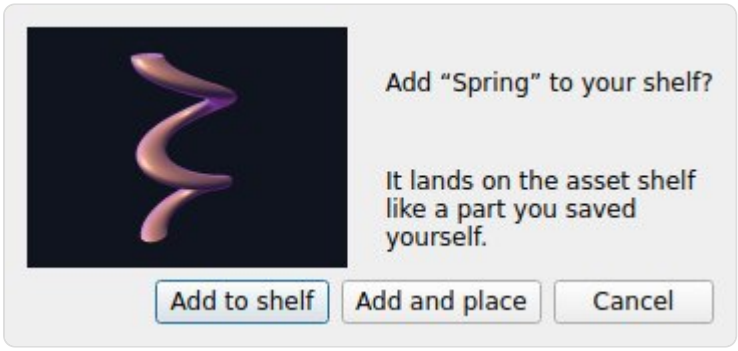
The panel shows what it found before anything is built: the body, each limb in its own colour and the holes in red, in place of the mesh. **Detail**, **Fill holes up to** (too much and two fingers close up into one), **Limbs thinner than**, **Round the joints**, **tips closed round** and **forks joined by a bridge** change it; **refine the surface with nets** (on by default, for the sections method - the fine tuning on top of the sweeps and blends) finishes it: where a sweep runs smooth past the mesh - round a fork, where sections had to be dropped (the ball of a thumb) - its side faces are reshaped by a face net fitted to the mesh, and kept only where the face then stands closer. Flat faces are never netted, a face whose lift would reach through another part of the same sweep is left as it was, and a face buried inside another part is not touched. A face net keeps its border, so a bump that sits on the seam between two faces stays. The nets are ordinary face nets afterwards: edit them by hand or clear them. It makes the conversion slower (about a minute on a whole hand). **Preview** shows the change. **Build** puts everything in the node editor as one group, laid out - one undo takes it back. A spine lies in one plane, so a limb that twists through 3D is flattened into its best plane (the status bar says which, and by how much).

Assets

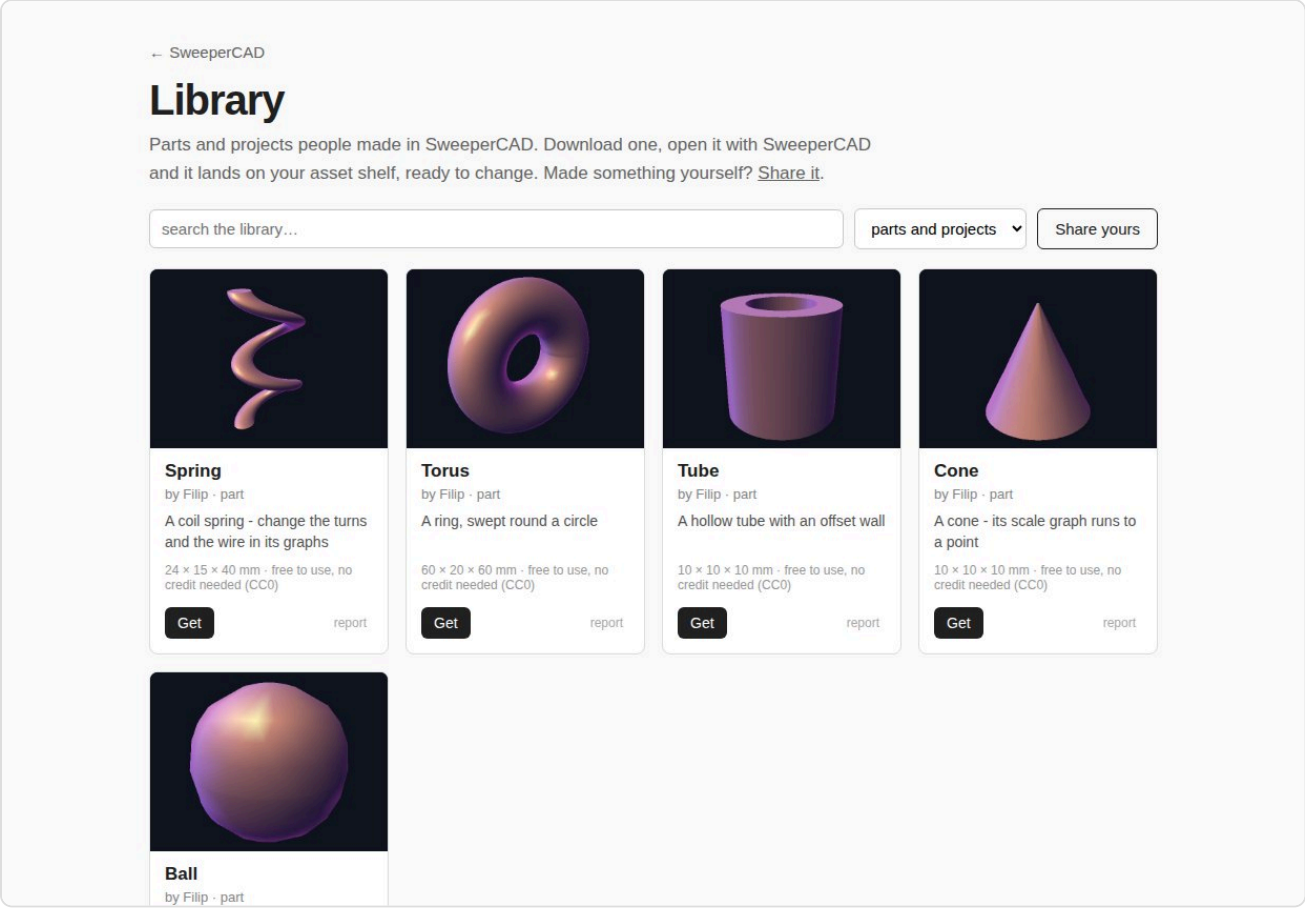
A starter shelf comes with the app: a ball, cone, cube, cylinder, pyramid, spring, torus and tube, each made of a profile, a spine and a scale graph you can open and change. They are copied into your library the first time the app starts; after that they are yours - rename, change or delete them and they stay that way.



The asset shelf: the starter parts, and a Spring just downloaded from the online library.



Opening a part downloaded from the library: add it to the shelf, or add and place it.



The online library at sweepercad.com/library: free parts, each checked and drawn by the robot.

Save selected in the shelf saves the selected node with everything feeding it, with a thumbnail of *that* asset. The name you give it is the name the shelf shows AND the name the node gets when you place it — a second copy is numbered. Place one with a double-click or the *place* button; it lands with the same guides and snapping as a new object.

Assets are ordinary files. **folder…** opens the folder they live in, or points the shelf at a different one — a synced folder, a stick, or one beside a project. They live in `~/.sweepercad/assets/` unless you say otherwise, and a folder that has gone away falls back to that.

A shared shelf stays up to date. The shelf watches its folder: when a part is added, changed or removed there - by you, or by someone else through Dropbox, iCloud or a network drive - the shelf redraws by itself. A file that is still syncing appears once it is complete. Only the shelf follows the folder: parts you already placed in a scene are your own copies and never change under you.

The online library. Parts other people made, free to download, live at sweepercad.com/library. *library…* on the shelf (or *Assets → Online library…*) opens it in your browser. Click **Get** on a part and open the downloaded `.sweeperasset` file: double-click it, drop it on the 3D view, or use *Assets → Add an asset file to the shelf…*. SweeperCAD shows its picture and asks: **Add to shelf**, or **Add and place** to drop it straight into the scene. A file that holds anything other than shapes is refused.

Sharing a part. Select it and click *share online…* on the shelf (or *Assets → Share selection online…*). SweeperCAD saves it as an asset in your Downloads folder, shows it to you, and opens the library's share page in your browser: pick the file there, give it a name and a line about it, and send. A checking robot opens it, builds it and draws its own picture before it goes up - nothing you did not mean to share travels along. Anyone can report a part; one reported several times is taken down until it is looked at.

SweeperCAD itself never goes online. The library is a web page: your browser does the downloading and the uploading, the app only reads and writes files on your computer.

How smooth a blend came out shows on the combine's card as a small badge at its top right after each build: **G0** when the seams are sharp (or chamfered) as asked, **G1** when they carry ordinary tangent fillets, **G2** when every blended seam was rebuilt curvature-continuous - and an orange **G0** when a blend was asked for but a seam had to be left sharp or reduced; the seam table says which.

Adding a whole file to the scene. *File → Add file to scene…* (Ctrl+Shift+O), or drop a `.sweeper` file from your desktop onto the 3D view: its objects join what is open, alongside, as they were saved - the file you are working in stays the current file. A name already taken gets a number. Its reference images come along; so do the saved build times, so a heavy addition can put the scene in draft mode.

Placing or inserting an asset. *Place* (or a double-click on the shelf) puts the asset where you click in the 3D view, with a light preview on the cursor and the same guides and snapping as a new object. *Insert* puts it straight into the scene, beside what is there, without the placing tool. Every asset keeps its finished shape and a light preview in its file, so even heavy ones (the robot, the potted plant) appear quickly; edit anything inside one and it rebuilds the ordinary way. The starter models come with theirs; an asset you save, or add from the online library, gets its own a moment later - built in the background, by this computer, never taken from someone else's file.

Putting an asset back in place. *In place* (beside *place* on the asset shelf) inserts the selected asset exactly where it stood when it was saved - turned and moved as it was, the combine it sat in included, so a nested part comes back to its spot. Assets saved before this remember no spot and are inserted as saved.

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7. Checking and making

Stats and checks

The **Σ** icon in the 3D view opens the **stats** panel. It scrolls, and each section folds away by the tick in its title (and stays folded): - **selected**: the selected part's extent, volume, surface, weight and material cost (material and price per kg are chosen here). - **direction view**: the direction and angle for the **direction** view, and a switch to show it.

- **schedule**: every visible part in one table - like parts (the same sizes and volume) counted together, with the box round each however it is turned, its volume, and its weight in the material above (and the material's cost). Sort by any column; click a row to select those parts; **save as CSV** for a spreadsheet. **Fill the schedule** reads the model again after edits.
- **between parts**: separate parts closer than the distance you set are flagged **yellow**, parts that **touch orange**, and parts that cut into each other **red** (with how deep). The spots are ringed in the 3D view. It checks again after every rebuild while the panel is open, or on **check now**. The pieces inside one combine belong together and are not checked against each other. Parts that are **interlocked** - that cannot be slid apart in any straight direction (linked rings, one caught in the other's hollow) - are flagged **purple**. A part that only comes out with a twist counts as locked.
- **along the direction**: **find islands** rings every piece that starts in mid-air - nothing below it to grow from (with +Z: when printed or cast upright) - with its height; parts that touch hold each other up. **Cross-section graph**: the section's area over height; hover to read it, click to cut the view there (along Z). **Curvature view**: the surface coloured by how it bends, full colour at the radius you set or tighter. Three kinds: **bulging / hollow (mean)** - convex warm, concave cool, flat grey; **dome / saddle (Gaussian)** - curved both ways like a dome warm, like a saddle cool, and grey where it is flat or only rolled (a cylinder, a cone): those parts unroll flat; **tighter than (radius)** - red where the surface bends tighter than the radius: a cutter of that radius will not reach in there, a printer will not hold it. The **1.2** button in each graph's bottom-right corner (or **View > Numbers in the graphs**, Ctrl+Shift+N) shows or hides the lengths, angles and radii written on the curves - with many points they crowd the drawing. Hidden, a corner's radius still shows on the corner you point at or select (so you can click it and type a new one), and the selected point's x, y always shows. The choice is remembered. **View > Curvature combs on curves** draws a comb along each curve in the graph editors (profile, spine...): the longer a tooth, the tighter the curve bends there; a smooth curve has a smooth comb, and a jump in the comb is a kink in the curvature. Not on graphs whose two axes have different units (the scale graph).
- **strength**: quick checks of the selected part (or everything) in the material chosen at the top, along the direction view's axis - where it would be pulled apart (and at what force), how far it moves when pushed sideways at the top while standing on its far end (and the push that bends

it for good), and the load on top that buckles it. The graph shows the bending stiffness along the height; the weak spot is ringed. Beam estimates: rough for stubby parts, good for slender ones. Each material has a typical Young's modulus and strength (a supplier's sheet is better).

- **simulate**: click **held here…** and click the spots on the model where it is held, then **pushed here…** and the spots where it is pushed (right-click or Esc ends picking); set the **force** in newtons (10 N is about 1 kg), its **direction** (into the surface, or along an axis) and the **detail**, then **simulate**. The part is cut into small bricks and solved: how far it moves, the peak stress, and the **safety factor** - the material's strength over that peak (2 or more: room to spare; 1-2: not by much; under 1: it would give way) - and how much it **stretches**: the most the material is pulled longer and squeezed shorter, in % of its length, beside the stretch where the material starts to give way (its strength over its stiffness). Pulled longer, a part also gets thinner, and squeezed it gets fatter - by the material's **Poisson's ratio**, which the simulation takes from the material: about 0.3 for metals, 0.35-0.4 for most plastics, nearly 0.5 for flexible ones like TPU (they keep their volume: a TPU bar stretched 1 % gets about 0.5 % thinner); the bent shape shows it. **In the view** colours the part by what **show** picks: **stress** - blue (relaxed), green, yellow, red (giving way); **movement** - how far each spot moves, blue still to red the most; **stretch** - red pulled longer, blue squeezed shorter, grey untouched (full colour at the stretch nearly all the part stays under, so one sharp spot by a hold does not wash the rest out). The peak stress is ringed. **Bent shape** draws the part the way the load bends it, in those colours, the unloaded part a ghost around it; the bend is exaggerated by the factor beside it (set after each simulation so the largest movement shows clearly; x 1 is true size). A first look, not a certificate: thin walls need a finer detail, and sharp inside corners read high.
- **casting** (its own window - the crucible icon in the group with Σ at the top right; the corner of the icon opens the row): where a cast or moulded part sets last, and whether every thick spot can still be topped up from the gate while it shrinks. Click **gate here…** and click where the metal or plastic comes in (a sprue, a gate, a feeder; pick several if the mould has them), then **check**. **Cut-off spots** - thick spots behind a thinner neck, which freeze off from the gate - are ringed red with their thickness and the neck's: shrink holes in a casting, sink marks on a moulding. Thicken the way to them or feed them from a gate of their own - or press **risers on the thick spots**: a riser stands on each (on the thickest spot when none freezes off), its diameter 1.3 times the thickness there, 1.5 diameters tall, with a narrower neck, as an ordinary + object you can move or resize; combine it with the part. **Vents where it fills last** puts a thin pin on each fill-last spot. A runner is a route from the gate (the route tool). The places that **fill last** are ringed yellow: where the air goes (vents) and where flows meet. With no gate picked, the thickest spot is ringed as a place for one, and every spot that would still need feeding of its own is shown. **Show** picks the view: **sets last** (thick spots red), **cut off** (red where it is cut off) or **fills last** (blue near the gate, red last). The thickest section is compared with the typical wall: a moulding sinks over spots about twice the typical wall or more. It works from the shape alone - no gravity, material or temperatures - so it is a first look.
- **seams**: **check seams** measures every edge between two faces of the selected part (or every part) on the solid itself and grades it: **G0** the faces meet at an angle, **G1** tangent (the normals agree, the curvature jumps - an ordinary fillet against a flat face), **G2** curvature-smooth (the

curvature agrees too). The seams are marked in the view - **green** G2, **yellow** G1, **red** a kink under 15°, a seam that looks smooth and is not - with the numbers on each. The proof that a blend is what it says: a plain fillet measures G1 along its edges, a smooth (G2) blend measures G2 on every one. For an AI app, `sweepercad_check` with `seams: "G1"` (no kinks) or `"G2"`.

- **walls and cavities**: set the **thinnest wall** and press **check**. Thin spots are dotted red on the part; **closed cavities** (sealed voids inside, which cannot drain or be cleaned out) are ringed with their volume. Check **each part, as built** (a combine counts as one), **each piece, before combining** (a piece thin on its own shows up even if it is thick once joined), or **the selected one**. It works on a voxel copy of the part, good to about a fifth of the wall asked for.
- **design rules**: one set of limits a design has to keep, checked on everything shown (or the selected one) and marked in the 3D view. Start from a set - **play equipment (typical EN 1176 values)** or **custom**, which starts empty - and change any value; *off* switches that check off. **Save...** writes the set to a small file (*.swrules*) and keeps it in the list; **Load...** opens one. Give the file to the people who design for you (a playground builder to his customers, say) and they design freely, checking against your limits.
- **No openings from ... to ...**: an opening a ball of the smaller size passes through and one of the larger size does not (a body slips through, a head stays caught) - a hole in a panel, the gap between two bars, a tunnel. It is done the way the probe test is done: balls of both sizes, pushed through the gaps. An inside corner is not an opening, and two long bars trap in their middle, not at a free end. Each opening found is measured on the part itself; the size is the widest ball that sits in its narrowest part (about - measure one that is close to a limit; those within 2 mm are said).
- **Guardrail above / barrier above / highest standing surface**: every surface facing up at least as big as **standing surface from** (0.04 m² is 20 by 20 cm) is measured from the lowest point of the parts, and marked when it wants a guardrail, a barrier, or is too high.
- **Thinnest material**: thinner spots dotted red. **No sealed cavities inside**: voids closed off in a part.
- **Material** and **it must take**: the load it has to carry - pushed sideways at the top, pulled or pushed along it, or pressed down on top, standing on its far end along an axis - with a **safety factor**: estimated from its sections for the material (typical values).
- **Nothing starts in mid-air** along Z, for printing.
- **Physics**: **no parts go into each other** (two separate parts that overlap, with how deep - make them meet at a face, or cut a joint into one), **nothing floats** (every part joined to something that reaches the ground; a board resting on nothing is marked) and **it stands** (the centre of mass of each piece over the outline of what it stands on; a design that tips over says by how much). Parts count as joined where they touch, all of one material, and the ground is the design's lowest point. A design in several separate pieces is said too.
- **Smallest joint**: two parts that touch over less than this area are hardly joined - an edge or a corner only. Each joint is measured (about).

- **Takes a push of**: a sideways push at the top each piece must take without tipping over, times the safety factor - weighed in the set's material (10 N is about 1 kg). It says what it does take.
- **No cups opening** up (or along another axis): pockets closed at the bottom and round the sides, open only that way. On an outdoor part they hold rain; a part dipped in, or lifted out of, a liquid that way holds it, or sucks. Each is listed with how much it holds and how deep it is; sealed hollows count too. A drain or vent hole at its lowest point, or turning the part, cures it.
- **Fits in** a box (mm) - your printer, the CNC bed, a plank, a kiln: **each part** or **the whole design**, a part turned at right angles if that makes it fit. *Off* leaves a side free.
- **Weighs at most** (kg): the whole design in the set's material.
- **Smallest gap**: parts that do not touch keep at least this far apart - print-in-place pieces, sliding fits, a lid that must come off.
- **Only resting**: in a part's panel, **held** says how it is held by what it touches - *fixed* (glued, screwed, welded; the default) or *only resting on what is under it*. A resting part must sit steadily: its centre of mass over where it rests, or it would tip off - or fall, when nothing under it carries it. A script sets it with `model.held(part, "rests")`. It is a check to design by, not a certificate: openings only partly bounded (a V where two parts meet), finger-sized openings on a large design and real loading are not checked, and the standard that applies is what counts. A big design takes a few seconds; the window stays usable.

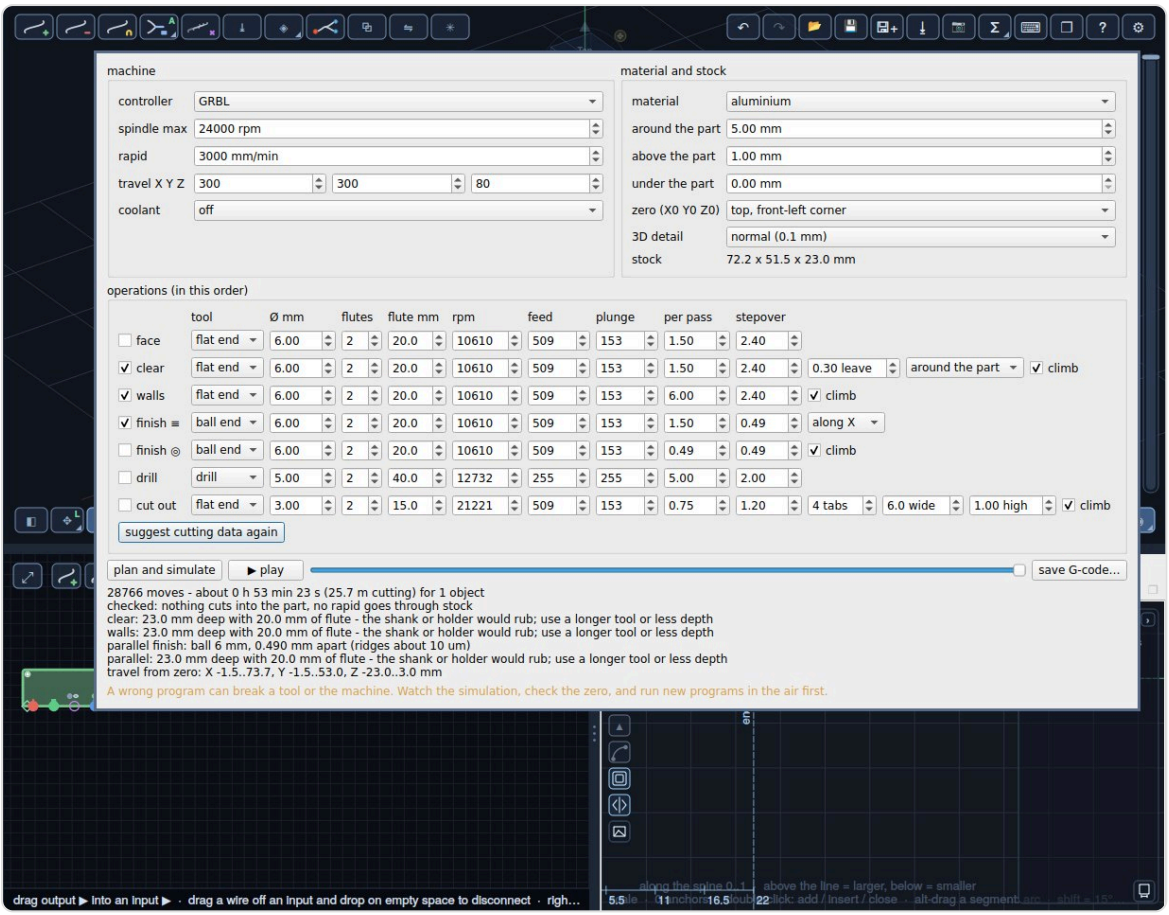
What the selected object measures

The **stats** panel (Σ in the 3D view) shows the selected part's **size**, **volume** and **surface area**. These come from the mesh on screen, so flat-sided shapes are exact and curved ones read very slightly low - about 0.3% on a 30 mm cylinder, 0.8% on a 10 mm one. They never read high.

Weight and material cost (now also bronze, sterling silver, 18k gold, cast iron, tough resin, beech, marble, concrete, and the plastics PC, POM, PP, HDPE, acrylic, rigid PVC, PS, PEEK, PEI, glass- and carbon-filled nylon, and casting wax). Pick a **material** (PLA, PETG, resin, aluminium, steel, wood and more) and the panel adds the part's **weight** and what its material costs, from the **price** per kg beside it. The prices are typical starting values; type what you pay and it is remembered for that material. Both are for a **solid** part: a 3D print with infill weighs and costs less, and the price is the material only - no machine time, and not what a print or CNC service would charge.

Machining (G-code for a CNC)

The machining icon (the hammer-and-pick, in the group with Σ at the top right - its corner opens the row) turns what is selected (or, with nothing selected, everything shown) into toolpaths for a 3-axis machine cutting from the top - a router or a mill on GRBL, LinuxCNC or a generic controller.



Machining: operations, tools and feeds, the toolpaths simulated on the stock.

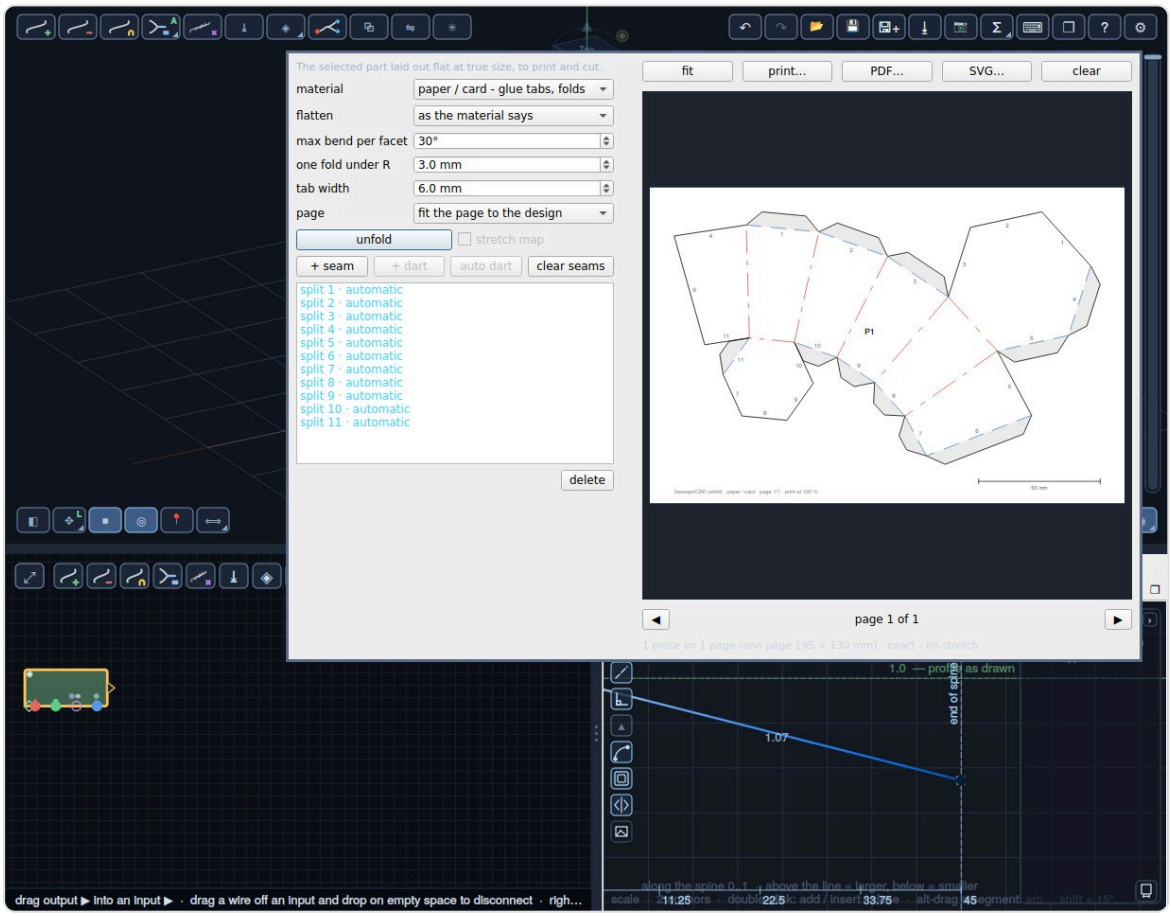
- **machine**: the controller, the spindle's top speed (on GRBL the S value is scaled to $30 - set them alike), the rapid rate (for the time), the travel (the program is checked against it) and coolant.
- **material and stock**: the material sets the suggested cutting data; the stock is the part's box plus what you give around, above and under it. **Zero (X0 Y0 Z0)** is where you zero the machine: the stock's top front-left corner, its top centre, or its bottom (the bed).
- **operations**, in this order, each with its own tool: **face** (the stock top down to the part), **clear** (level by level, everything that is not the part - *around the part*, or *pockets only* inside its outline - leaving a skin on the walls), **walls** (the walls to size), **finish** ≡ (3D: passes that follow the surface - along X, along Y, or both), **finish** ◎ (3D: contours at constant heights, for steep walls; the two together cover a curved part), **drill** (round vertical holes the drill fits, pecked; holes of another size are cleared instead) and **cut out** (the outline through the stock, with **tabs** that hold the part - last, so the part is held for everything before it). The finishes take a **ball end** (or a flat one); for a ball the stepover is suggested for ridges of 0.01 mm between passes. Each row suggests the spindle speed, feed, plunge, depth per pass and stepover from the material and the tool; change any of them. **Climb** milling is the default; untick it for conventional. Cutters ramp into material along their path (a helix on a small loop) and only plunge into air.
- **3D detail** (under the stock): how finely the surface is read for the finishes - *normal* (0.1 mm) or *fine* (0.05 mm, a few times slower). The finishing passes never cut into the part; they leave about that much more on curved and steep areas, so *fine* gets closer.

- **plan and simulate** draws the paths on the part (a colour per operation, rapids dashed) and the stock as they leave it; **play** or drag the slider to watch it cut. It is **checked**: a move that would cut into the part, or a rapid through stock that is still there, is drawn red and reported, as is a program that needs more travel than the machine has or a cut deeper than the tool's flutes.
- **save G-code…** writes the program. For **GRBL** that is one file per tool (GRBL has no tool changer; zero Z again after each change), with only what GRBL understands - no canned cycles. Other controllers get one program with tool changes.

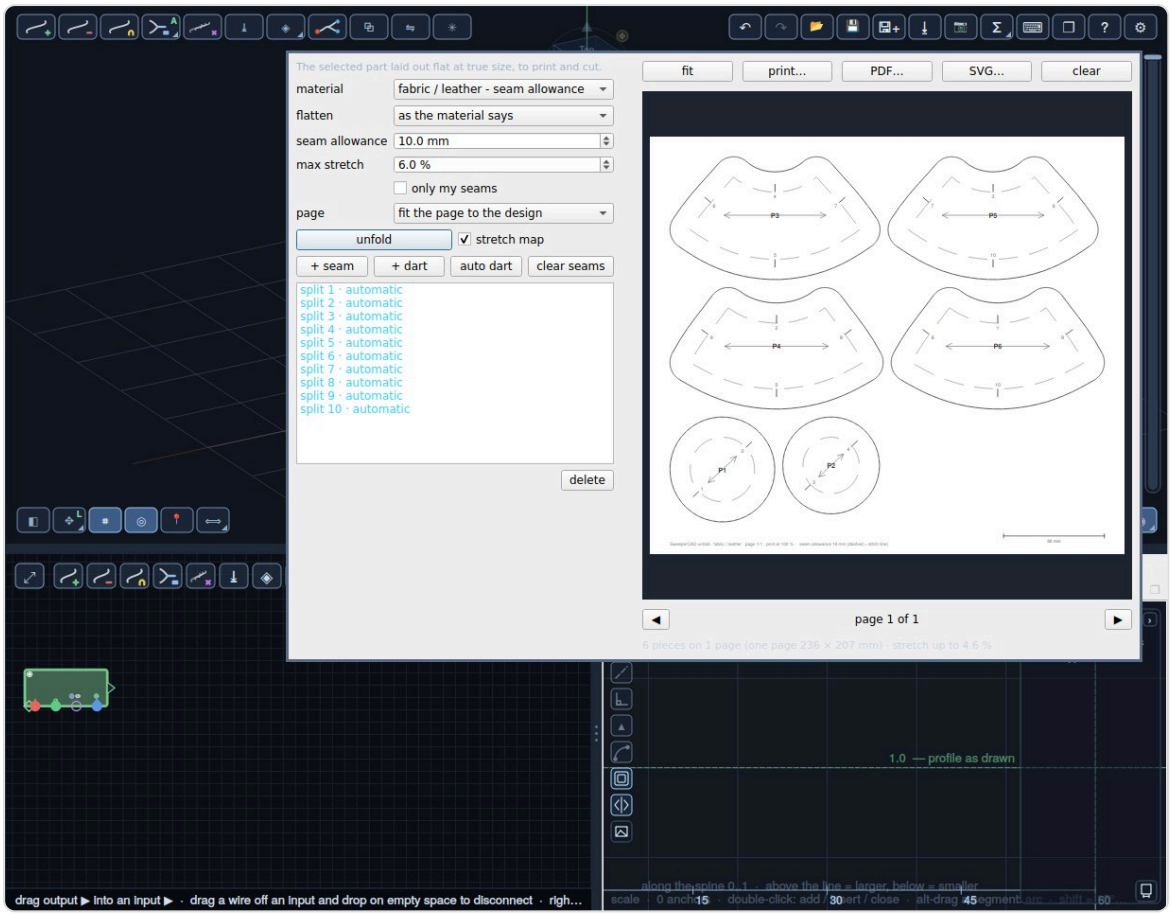
A wrong program can break a tool or the machine: watch the simulation, check the zero, and run a new program in the air first. The cutting data is a conservative starting point for a hobby machine, not a promise.

Unfold — lay it out flat to print and cut

The ✂ icon in the make-it row (Σ · casting · machining · unfold) opens the **Unfold** window: the selected part laid out flat at **true size**, in pieces, on pages to print — or as one full-size SVG for a cutting plotter or a laser. Pick the **material** first; it decides how the part is flattened and what is drawn around the pieces.



Unfold, paper: a lamp shade as one net with glue tabs and numbered seams.



Unfold, fabric: a vase in panels with seam allowance, notches and grain lines; the list shows every seam.

It works on the same **quad mesh** as the quad export: every four-sided face of the part is a regular grid — on a sweep the rows follow the spine and the columns go round the profile — so seams, darts and the automatic splits run along grid lines, never across a quad.

Exact (paper, card, sheet metal, foam): every face keeps its true shape, nothing stretches. The faces are unfolded hinge by hinge, and where a piece would overlap itself it is cut instead. Curved faces come out as strips: **max bend per facet** decides how many — a 90° fillet at 30° per facet is three strips (three small chamfers). A straight fillet smaller than **one fold under R** is drawn as a single fold along its middle; its strip still gives the material its true length.

Stretch (fabric, leather, felt): each region between sharp edges is flattened as one piece, with a little stretch where the surface curves both ways. Tick **stretch map** to see it on the part: blue is true, red is the **max stretch** you allow. A piece that would stretch more is split in two — first along the part's own faces (on a sweep those run along the spine, so a vase comes out as panels), then along the face's own grid lines — until it fits. Set max stretch to 0 to keep pieces whole and place your own:

- **+ seam**: click points on the part, as many as you like. The button turns green and reads **✓ end seam** — press it to finish (a right-click or Esc does the same). Each click snaps to the nearest grid point (a cross marks it) and the white dashed line shows the path the seam will take, along the grid lines, before you finish — two points on the same row or column give a straight seam along it. Seams work in exact mode too — that is how you choose where a paper net is cut.

- **+ dart**: click where the dart opens (on an edge), then where its tip goes. The slit runs between them and opens into a wedge when the piece lies flat, taking up the fullness. Click only the tip and press ✓ **end dart** for a dart from the nearest edge. The dart button stays on for the next dart until you press ✓ end dart.
- **Auto dart** tries a dart tip at three depths from the worst spot and keeps the one that lowers the stretch most — or says that no dart helps there (then a seam, or max stretch, will).
- **The list** under the buttons holds your seams and darts: select one to see it white on the part, untick it to leave it out without losing it, **delete** to remove it, **clear all** for all of them.

The seam list shows every seam: yours (seams, darts, splits you kept joined) and, in cyan, the splits the unfolder made itself — numbered as on the paper. Select one or more (ctrl / shift) to see them white on the part; untick one of yours to leave it out; **delete** removes the selected ones. Deleting an automatic split **keeps it joined** (it shows up as *kept joined*; delete that to let the unfolder cut there again). Paper and metal still have to cut a closed part somewhere to open it — keep a box edge joined and the cut moves to another edge; if it can't (it would overlap), the status line says so.

Only my seams (fabric / stretch): no automatic splits at sharp edges or past the max stretch — the pieces are exactly what your seams close off, and the stretch map shows what that costs. A closed piece still gets the one cut it needs to open.

Seams, darts and joins are saved with the project file.

In 3D, cyan lines show where the pieces split, magenta your own seams and darts (the one selected in the list white), and a white dashed line the path of a seam while you place it. Closing the unfold window clears all of it — the lines, the stretch map and the pages; your seams and darts stay with the part and come back when you open it again. Deleting the part clears it too.

What each material draws:

- **paper / card**: cut lines; folds red dash-dot (mountain — folds away from you, the printed side out) and blue dashed (valley); a glue tab on one side of every seam, and the same number on both sides.
- **sheet metal**: no tabs; bend lines with their number, direction and angle (*1: down 90°*). The model is read as the **outside** of the part: set the **thickness**, the **inside bend radius** of a sharp fold and the **K-factor** (where the neutral line sits, 0.33–0.45 typical). Each bend takes off its bend deduction, so the flat is cut to the neutral line and the bent part comes out at the size you drew. A fillet you modelled counts as the bend, its radius as the outside radius. Walls unfold from the largest panel (the base), as a press brake makes them — see **Sheet metal for the shop** below.
- **fabric / leather**: the cut line is the piece plus the **seam allowance**; the dashed line is where to stitch. Notches and numbers on matching seams, and a grain line along each piece.
- **foam / felt / laser**: clean cut lines, with an optional **kerf** (the cut line moves out by half the cut width).

Sheet metal for the shop. With *sheet metal* chosen, the flat is made ready to cut and bend:

- **metal**: steel, galvanised, stainless, aluminium 5052 / 5754, 6061-T6, 1050, copper, brass, or other. It sets a starting K-factor and what the checks use: the tightest inside radius the metal takes (6061-T6 cracks on tight bends), its springback and its strength. Starting values — test on an offcut.
- **bend reliefs**: a bend that ends against material tears or bulges there, so it gets a slot as wide as the sheet, through the bend zone and one thickness beyond — square or round-ended. Where two bends meet at a corner there is one corner relief over both.
- **corners**: where two flanges meet (a tray's corner) both walls would fill the same corner of the outside model. **Open** stops both short by the thickness plus the **corner gap**; **closed** lets the larger wall run through and stops the other at its inside face — ready to weld or rivet; **as modelled** leaves them.
- **+ hem**: click outside edges of the part; each is folded back flat on itself (**hem length**, outside, 4 × thickness when 0; a **hem gap** above 0 makes an open hem, flattened onto a spacer).
- **+ open face** (any exact material): click a face to leave it out — the lid of a box, so it unfolds as an open tray. It shows in the seam list; delete it there to bring the face back.

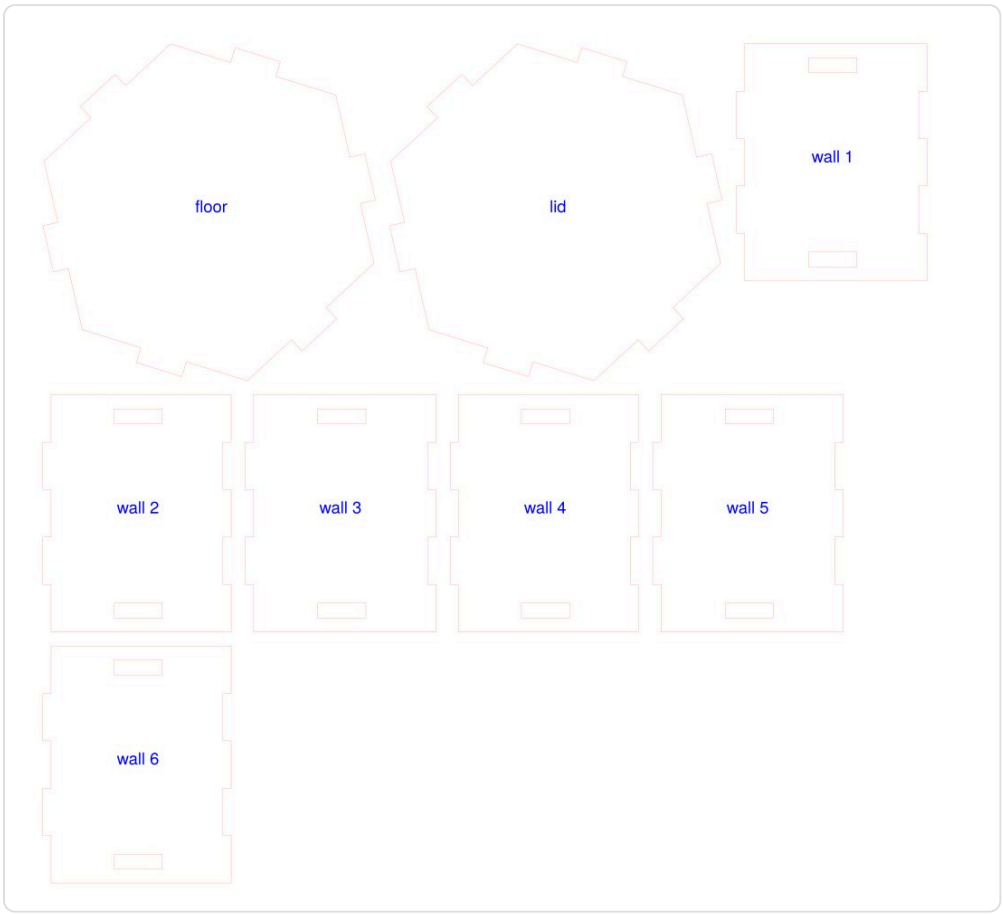
The status line gives the number of bends, the die, the reliefs and the first thing to check; hover it for all of them. **DXF…** writes the flat at true size in millimetres — outlines on layer CUT, bend lines on BEND_DOWN (red) and BEND_UP (blue), the numbers on TEXT — for a laser or plasma shop, CAD, or the brake's own software (it works for paper, fabric and foam too). **Brake sheet…** writes a PDF (or a CSV table): the flat with the bends numbered in the order to bend them (hems and outer lips first), and per bend the angle to **bend by** — springback included — which way the flange goes, its length, the **back gauge** (flat distance from the bend line to the edge against the gauge), the outside **flange**, and the force. The die is the usual one for air bending (about 6 × thickness, the nearest standard V), and the sheet says what to check before you bend: a radius too tight for the metal, a flange too short for the die (with the die that fits), a hole too close to a bend, a U whose walls may hit the top tool, and the inside radius that die really gives.

The page. Pieces are laid out at true size on A4, A3 or Letter, on a **custom size** you type (width × height in mm — a cutting mat, a plotter's roll), or with **fit the page to the design**: one page just big enough for all the pieces, the right thing for an SVG for a laser or a single sheet from a print shop. On the page: the **wheel zooms** about the cursor, a **drag** moves it, a **double-click** shows the whole page again. The toolbar above it — and a right-click on the page — has **fit**, **print…** (through your printer's dialog, always at 100 %), **PDF…**, **SVG…** (one full-size sheet, for a cutting plotter or laser) and **clear**, which empties the pages and the lines on the part but keeps your seams and darts (**clear seams** removes those). The SVG opens straight in the cutter's or laser's own program (Cricut Design Space, Silhouette Studio, ScanNCut, LightBurn, Inkscape): one colour per job - **black** cut, **red** mountain and **blue** valley folds (set those to score or perforate), **green** the numbers (draw them with a pen, or hide that colour). The lines are solid hairlines, the numbers outlines, and there is no page frame or tab

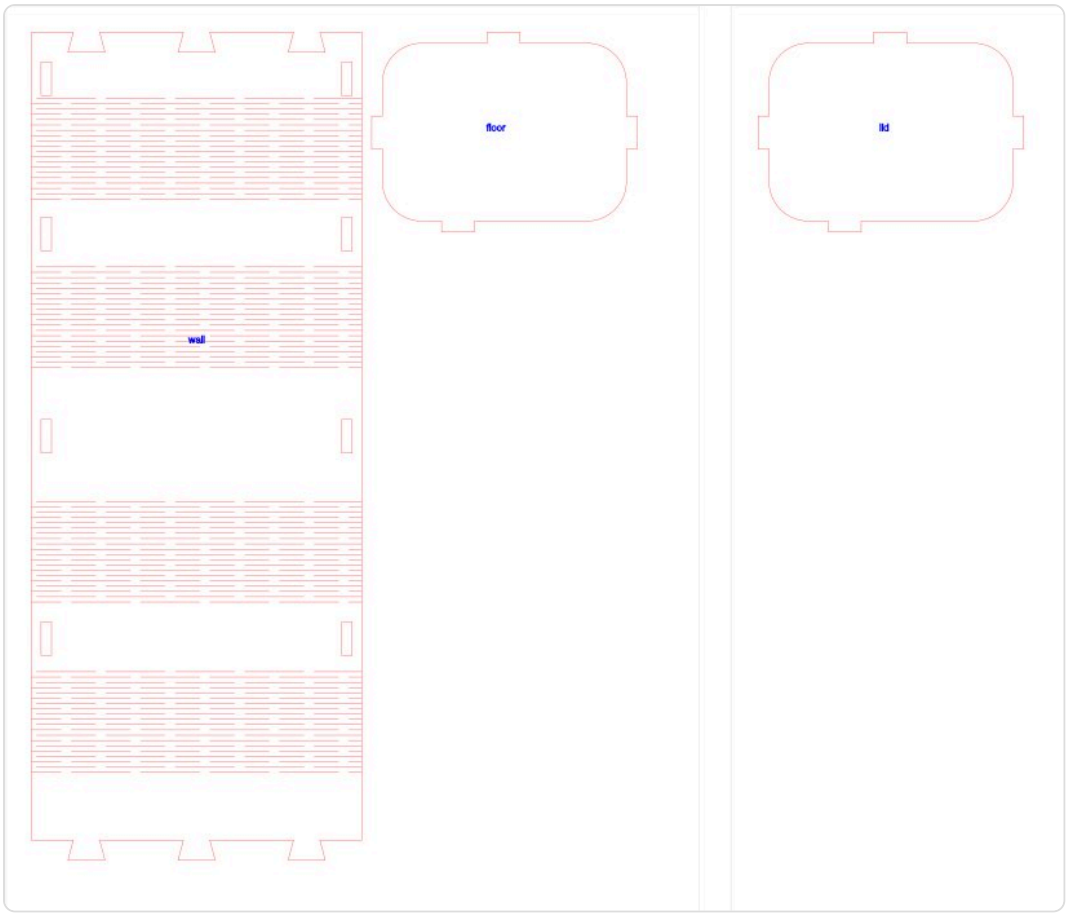
outline to cut by mistake. After printing, measure the 50 mm bar on the page to check the printer did not scale it. A piece bigger than the page gets a page to itself and a note — use a larger page or the SVG.

Stacked sheets

The part as layers of cardboard, plywood or acrylic, cut and stacked. Select the part and click the ☰ icon in the make-it group.



Box mode: a hexagon box - floor, lid and six walls with finger joints sized to the corners.



A rounded box: the wall unrolled with a living hinge where it curves.

Set the **thickness** of your sheet and the direction to stack in. **Each layer** is cut through its middle, or never smaller than the part (*outside* - sand down to the shape) or never bigger (*inside*). **Kerf** grows the outlines by half your laser's or knife's cut width. Holes in a layer are kept; a layer that falls apart into islands gets each piece numbered (12a, 12b).

Lining the layers up, any combination: **key slots and key strips** - a slot one sheet thick in every layer, and flat strips cut from the same sheet that slide through them and lock the stack; **dowel holes** for a dowel or skewer; and **the next layer's outline** drawn on each layer, as a gluing guide. Keys and dowels sit deep in the material; where the part changes too much for one to pass every layer, they run in overlapping sections that link up.

Waffle (*how it is built*): instead of stacked layers, upright slices across X and across Y, every so many mm, that slot together into an open lattice of the part - no glue. Where two slices cross, one gets a slot from the top and the other from the bottom, each half as deep as their shared height; every slot is labelled with the slice that goes into it.

Box (*how it is built*): for a part that goes straight up - any outline, the same from bottom to top (a prism). You get a **floor** (and optionally a **lid**) with tabs that go through slots in the wall, set in from the edge by the amount you choose, and the **wall unrolled** along its centre line, so it has the right length once bent. Where the outline has a sharp corner the wall is split into panels with **finger joints** (an odd number up each corner), sized to the corner's angle so the outside faces meet - a laser cuts square, so off a right angle there is a hairline wedge inside the corner, or a sliver to sand flush. Corners

sharper than about 70 degrees or flatter than about 150 (a triangle, a very shallow bend) are not jointed but **bent**: the corner is rounded a little and the wall bends round it on a hinge. Tick **bend round every corner** to do that everywhere - one piece of wall, no finger joints. Where the outline curves, the wall gets a **living hinge** - rows of slits, offset like brickwork, so the sheet bends. With no corner at all (a round or rounded box) the wall closes on itself with dovetails in its straightest place. Set the slit length, the bridge between slits and their spacing for your material - closer slits bend tighter; plywood likes a spacing of about half its thickness. You are warned when the outline curves tighter than the **tightest bend**, when the walls are not straight, and when there is no straight stretch for tabs (then glue the floor in). A wall too wide for the page is turned a quarter to fit. The hinge slits are cut lines (red) like the outlines.

The preview shows the stepped stack over the part, with the number of layers, pages and the sheet area. **Save PDF** gives pages at true size to print and cut by hand (with the 50 mm bar); **Save SVG** one file for a laser cutter or cutting plotter - red to cut, blue to engrave (the numbers), green the guides. It opens straight in the laser's own program (LightBurn, LaserGRBL, xTool Creative Space, the Glowforge app, or Inkscape printing to an Epilog or Trotec): true size in mm, hairline cut lines, the numbers as outlines so no font is needed, and no page frame. Give each colour its own power and speed there. **Save DXF** gives the same sheets for a plasma cutter, waterjet, laser or CNC router: true size in mm, outlines and slots on layer CUT, guides on GUIDE, the numbers as text on ENGRAVE.

Woodwork — the cut list, the stock, the cutting diagrams

⬢ **woodwork** in the make-it group reads the model the way a joiner reads a drawing. Every visible top-level object is a **part**: its **length, width and thickness** are the smallest box round it, however it is turned in the model - a leg standing at an angle is still 700 × 44 × 44. Parts alike in all three sizes are counted together. Keep boards as separate objects (not fused into one combine) so they are listed one by one; a board with holes or a rebate cut into it is one part, as it should be. *Selected only* lists the selection.

The **cut list** is the table: name, quantity, sizes, the grain (*length* - the long side runs along the stock's grain; *any* - the part may turn), and whether it wants a **board** or a **sheet** (a part wider than *sheets from* asks for sheet stock). Correct anything in the table; the numbers follow. Under it: the parts, the **board feet**, the square metres and the litres of wood, grouped by thickness.

Stock and saw is your yard: boards and sheets with their length, width and **actual** thickness (what the calliper says - the *nominal* is only a label), what a piece costs you, and how many you have (0 = as many as needed). It starts with the usual metric sizes; edit it, and the app remembers. The **kerf** is what every cut turns to dust; *thickness within* is how close stock must be to a part's thickness; *plane thicker stock down* lets a thinner part come from the thinnest stock that is thicker; *edge trim* takes a strip off every edge first.

The **cutting diagrams** lay every part on a piece of stock, the kerf between them, the grain respected: boards are cut along their length (two narrow parts side by side when they fit across), sheets in rows. Each piece says how many parts it holds and how much is waste; what no stock fits is said in red letters, not dropped.

Save CSV… is the list and the stock for a spreadsheet. **Save sheet PDF…** is one document for the bench: the table, then every board and sheet drawn with its parts and their labels. **Save diagrams SVG…** writes every piece at 1:1 in mm, one file each, for a laser or a plotter.

Joints between two boards

Joinery changes the parts, so it lives in the **profile menu ▸ joints (two boards)**, beside the machine parts. Select the two boards (the first selected is the one that *goes in* - the rail, the board with the tails; the second *receives* - the leg, the board with the pins; the form lets you swap them), pick the joint, set the few numbers, and both boards are cut - or added to - to match, as ordinary negative and positive objects in a combine that keeps the board's name. Everything is read off the two boards' own boxes: which end meets which face, the thickness, the width. The **fit** is how much bigger the hole is than what goes in it; change it later in the negative's own box. One undo takes a joint back.

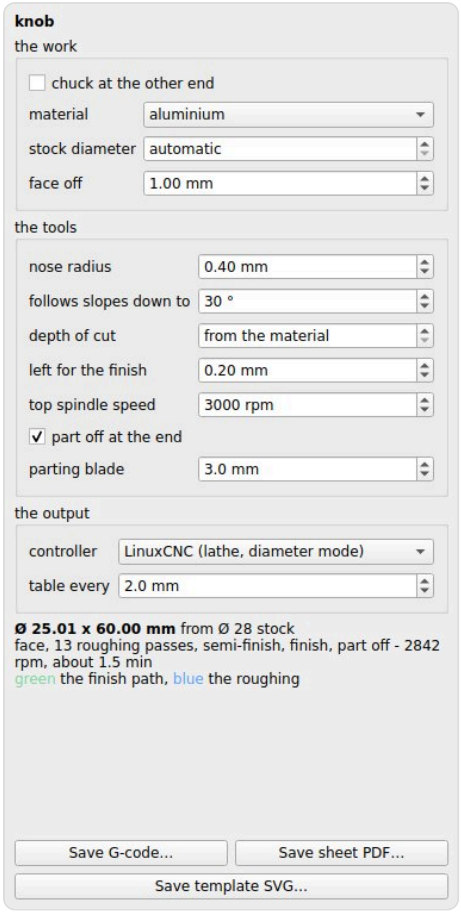
- **Mortise and tenon** - the tenon a third of the rail's thickness and most of its width, two thirds into the leg (or *through*). Boards that touch get the tenon **added** to the rail; a rail you already ran into the leg has its end **cut down** to a tenon as long as the run-in.
- **Dowels** - a row across the first board's end: the holes in both, and the pegs as parts of their own, so they turn up on the cut list.
- **Finger (box) joint** - a corner: the overlap cut into an odd number of fingers, the thickness wide unless you say. Boards that only butt are run on to the outer corner first, so the joint works however you modelled the corner.
- **Dovetail** - tails on the first board, pins on the second, half pins at the edges, slope 1:6 (softwood) or 1:8 (hardwood).
- **Half lap** - two boards flat on each other, crossing or meeting at a corner: each loses half its thickness over the overlap.
- **Housing (dado)** - the first board's end in a groove across the second, a third of the thickness deep; stopped or through.

The cut list sees the boards as they are after the joint - a rail with its tenons is listed at its full length, as a joiner cuts it.

Lathe

A part that is round about a straight axis - a round profile on a straight spine shaped by its scale graph, a revolve, an imported turned part - can be turned. Select it and click the ⌀ icon in the make-it group. The axis is found by itself (the part is sliced along it and every slice must be a circle); a part

that is not round says so.



The lathe panel: the work, the tools and the output.

Lathe coordinates: **Z** along the axis, 0 at the free end (the tailstock side), minus toward the chuck; **X** the diameter. Tick **chuck at the other end** to hold it the other way round. Set the **material**, the **stock diameter** (automatic: the part's largest diameter plus 2 mm), and how much to **face off** the end.

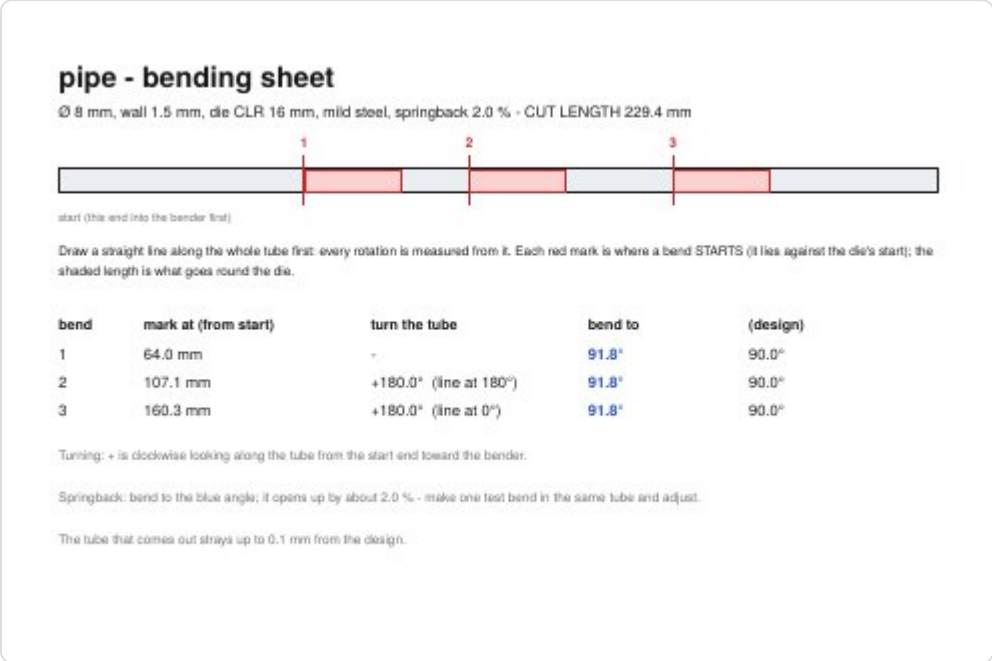
The tools. The turning tool's **nose radius** (the program gives the usual imaginary tip, touched off on the diameter and on the face, so tapers and curves come out true without nose compensation). **Follows slopes down to**: where the part narrows toward the chuck behind a bulge, the back of the tool hits the work - it can follow a slope only down to about its back clearance (a 35° insert: about 30°). What it cannot reach shows red and is reported, to finish with a neutral or left-hand tool. The **depth of cut** per roughing pass (or from the material), what is **left for the finish**, the **top spindle speed**, and **part off** with the blade width.

The program: face the end, rough in straight passes (entering a pocket gently), one semi-finish pass that follows the profile, the finish pass, then part off. Plain G0 and G1 only, so every controller reads it: **LinuxCNC** and **grblHAL** in diameter mode, a **generic / Fanuc-style** version, and plain **GRBL** with X as the radius. Speeds and feeds come from the material and are conservative; watch the first run in the air.

Save G-code… for a CNC lathe. **Save sheet PDF…** for a manual lathe: the diameters every few mm from the free end (and at every shoulder), and the profile **template** at 1:1 with the 50 mm check bar. **Save template SVG…** gives the same template for a laser cutter or plotter - hold it against the work as you turn. **Save template DXF…** is the template in mm, to cut it from sheet on a plasma cutter, waterjet or laser.

Bending

A tube or a wire is straights and bends. Select a swept tube - or several spines that continue each other (stacked spines, a route) - and click the ⌒ icon in the make-it group. The centre line is read from the spines, the **diameter** measured from the part (or typed), and you set the **wall** (0 for solid wire or bar), the **material** (it gives the **springback**: how much a bend opens up afterwards - one test bend tells the true number) and the bender: the **die radius** (its centre-line radius, CLR; the pin radius on a wire bender), the **shortest straight** the clamp can hold, and the **tolerance**.



pipe - bending sheet

Ø 8 mm, wall 1.5 mm, die CLR 16 mm, mild steel, springback 2.0 % - CUT LENGTH 229.4 mm

1 2 3

start (this end into the bender first)

Draw a straight line along the whole tube first: every rotation is measured from it. Each red mark is where a bend STARTS (it lies against the die's start); the shaded length is what goes round the die.

bend	mark at (from start)	turn the tube	bend to	(design)
1	64.0 mm	-	91.8°	90.0°
2	107.1 mm	+180.0° (line at 180°)	91.8°	90.0°
3	160.3 mm	+180.0° (line at 0°)	91.8°	90.0°

Turning: + is clockwise looking along the tube from the start end toward the bender.

Springback: bend to the blue angle; it opens up by about 2.0 % - make one test bend in the same tube and adjust.

The tube that comes out strays up to 0.1 mm from the design.

A bend sheet: the cut length, where each bend starts, how far to turn and to bend.

How it is worked out: a corner drawn sharp means "bend here" - the die rounds it. A curve as tight as the die is one bend; a gentler curve becomes several small bends close together, as many as the tolerance needs. Where the straights would meet are the bend points (yellow); the tube that comes out - those points rounded at the die radius - is drawn green over the design, and how far it strays is reported. Straights too short for the clamp show red, and you are warned when bends overlap, when a curve is drawn tighter than the die, and when the die is tight for a thin wall (wrinkles - use a mandrel).

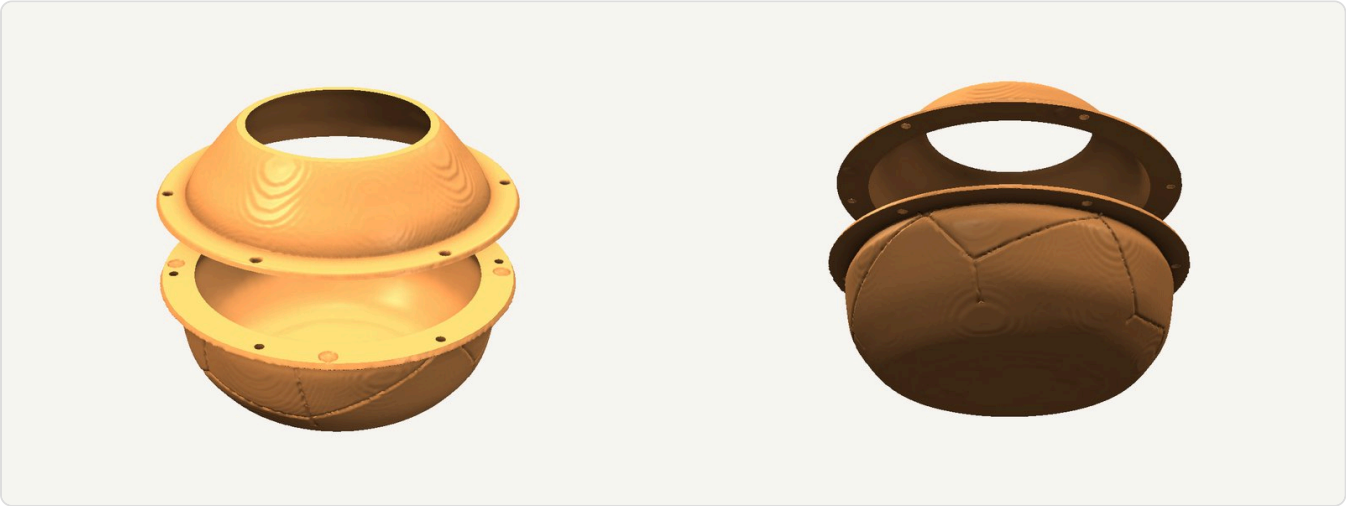
For a **flat bar** choose that section: the part is read slice by slice - its width and thickness, and how it lies along the spine - and every change of direction is one of three steps: a **flat bend** (across the thin side, the easy way), an **edgewise bend** (across the wide side, the hard way - roll it; tight ones are flagged because they buckle) or a **twist** about its length. With **holes along the bar** (pitch, first hole, diameter) every bend or twist that reaches into a hole is listed - bend on the bridges between holes, or the holes go oval. The sheet shows the bar straight with its zones and holes, then the steps in order: where to start, what to do, over how long.

For a **square bar** choose the section: it bends over its faces only, so between bends it may only turn in quarter turns - bends that would need it twisted are listed.

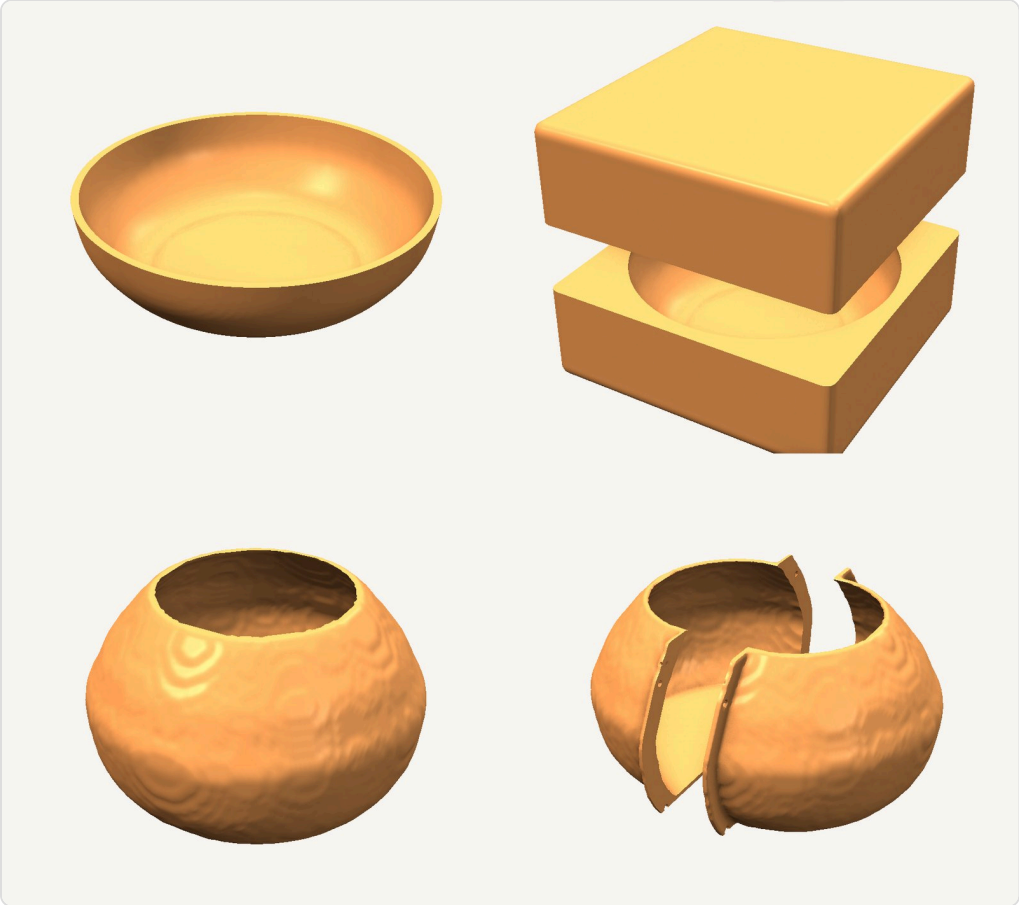
Every bender is told the same three numbers per bend: **Y** feed (the straight before the bend), **B** rotate (from the last bend's plane), **C** bend (the angle to bend to, springback included). **Save table CSV...** gives them, with the design angles, the arc lengths and the XYZ of the bend points - what bender software imports. **Save bend sheet PDF...** is for a manual bender: the **cut length**, the tube drawn straight with a mark where every bend starts, and per bend where to mark, how far to turn the tube (draw a line along it first) and the angle to bend to. **Save wire bender G-code...** drives a DIY three-motor wire bender: X feeds, Y turns the wire, Z swings the bend pin and back - set its units per mm and degrees per unit.

The eggshell mould

A thin breakaway shell around a part: print it, cast into it, and crack it away like an egg. Select the part and click the ⬯ icon in the make-it group (or add a lattice and choose *eggshell (breakaway mould)*).



A split mould: registration sockets in one flange, domes under the other; only the half that locks is cracked.



Made round the same part: a cradle, a fitted box, a pour box for silicone, and a pour box cut in two.

thickness is the wall, **count** the number of shards, **depth** how deep the crack grooves go from the outside, **crack width** how wide they are (automatic is a third of the wall), **seed** another crack pattern. The inside of the shell is the part's own surface turned inside out, so **the cavity is exactly the part**; the grooves always stop short of it. Add a pour funnel or flanges as ordinary objects, and export the shell as STL.

The mould settings turn the shell into a mould you can open and use again:

- **add pour funnel** puts a funnel on the part: a sprue (its **diameter**) rising from the middle of the part's top, sunk 1 mm into it, that opens into a cup with walls leaning 30 degrees. The funnel is the reservoir too: the cup holds the **%** of the part's volume you set (15 % by default), the material that keeps feeding the casting while it shrinks, and the place you pour into. It is an ordinary object, fused to the part in a combine that becomes the eggshell's input, so you can move it or drag its graphs like any sweep; the mould is opened at the top so the cup comes out open. Press it again with another diameter or share and the same funnel is resized.
- **open at the top** cuts the shell just below its very top, so whatever reaches the top comes out open: a pour funnel, a sprue, risers or vents. Fuse those to the part first (the casting check's *risers* and *vents* buttons place them) and make the eggshell of the result; the **cavity holds** line gives the volume and weight to pour, with them included.

- **cut across X, Y or Z** cuts the mould on a plane square to that axis, through the middle of the part or so many mm from it, and **tilted** and **turned** to any angle (the two degree fields). Up to **three cuts**, each on the pieces the others made - two crossing cuts give four pieces. ◰ next to a cut places it **by hand, with the section plane**: the plane jumps to the cut, and as you drag and turn it with the gizmo the cut follows - anywhere, at any angle (the row then says *cut on a placed plane*). Click ◰ again to leave it there. The same works for the part in pieces. **Pie** cuts it into 2 to 12 pie pieces round an axis through the middle of the part (Z, X or Y, the first cut turned where you like), each pulled straight outward. Each half is closed and ready to print. A **flange** stands out along the cut, both halves together twice the wall thick (at least 4 mm), with **holes** through it for pins, bolts or clamps (3.2 mm takes an M3 bolt; 0: none). **The flange stays within the mould's height**: it never reaches below the shell's lowest point, so a mould with a flat bottom stands on it, nor over where it is opened - untick it to run the flange all the way round. Flange 0: no flange at all. **Pieces apart** lays them out with a gap so you can see and export them.
- **Registration** lines the pieces up: on every cut, low domes stand out of one side's flange, between the holes, and the other side has their hollow counterparts, sockets 0.2 mm larger (0: none). A dome that would not sit fully on the flange (over the opening, off its edge) is left out.
- **Crack only where it locks** (with a split): each piece is pulled straight off, away from the cuts around it. Where the part hooks into a piece - an undercut, its surface facing back against the pull, like a belly wider than the cut above it - that half gets the crack grooves there, so it breaks away just where it must; everywhere else it lifts off whole. **Crack everywhere** scores the whole shell, and the crack **depth** at its least gives no grooves at all.

The cuts are made on the shell itself, exactly: the cavity stays the part's own surface, and the flat faces of the cut are closed watertight.

More from the mould panel:

- **add vents** finds where the air is trapped when you pour from the top - every high point that is not the top itself: the tip of an ear, a tower lower than the rest, a flat ceiling - and puts a thin rod (its **diameter**) from each straight up to the funnel's top, so they come out open with it. One vent per high area. A high point with part above it gets none (a straight vent would run back into the part) and is counted, to vent by hand. Add the funnel first; when you move or resize it, the vents follow.
- **show** (next to the cracks) colours the part red where it locks into its piece - an undercut against the way that piece is pulled off - and says how much of the surface that is. It follows the cuts as you change them. With no cut at all, everything that faces down locks: a closed shell can only be broken off. **place cut 1 where it locks least** slides the first cut along its direction (its tilt and turn kept) to where the least of the part hooks in.
- **engrave what it holds on the flange** cuts the cavity's volume and its weight in the material chosen ("516 cm3 4.5 kg bronze") into the outer face of the first flange, between two holes, bent along it.

- **hollow, wall**: a core for a hollow casting. The core is the part that far in from its surface, with a **print** - a peg of the diameter set - from inside it down through the bottom and 8 mm out. The print's socket is in the mould, so the mould holds the core in place; the core is laid out in front of the mould, to print or to make in plaster. The casting gets a hole where the print was.

make chooses what is made round the part:

- **eggshell** - the breakaway or split mould above.
- **cradle** - a holder shaped to the part, for painting, gluing, soldering or engraving: the shell, **fit** mm clear of the part, up to the share of its height set (**up to**), with a flat bottom to stand on.
- **fitted box, in two halves** - a box with rounded corners, **room** mm of wall round a cavity shaped to the part (**fit** clear), cut in two across Z unless you set other cuts, with domes and sockets in the cut faces. For storing or shipping the part.
- **pour box for silicone** - walls that follow the part, **room** mm off it (the silicone's thickness), with a floor the part stands on and open at the top, ending just under **room** over the part. Following the part saves silicone; the line underneath says how much it takes. Cut it like any mould to get it off the silicone block.
- **pour box for plaster (slip casting)** - the same, starting at 25 mm of room: a case to pour a plaster mould in, round a master, for slip casting ceramics. Cut it in two or more, with domes, so the plaster mould comes out in pieces.
- **the part in pieces, joined, to print** - no mould: the part itself is cut, for a part bigger than your printer. Give your **bed** size (X, Y, Z) and it is cut square to the axes as often as it takes for every piece to fit - a tenon's length included - or leave the bed at 0 and use the cuts above, at any angle. On every cut the **joints**: **tenons and mortises** (a square peg out of one face, its pocket in the other, **fit** mm clear all round and a little deeper, so it cannot twist), or **dowel holes** in both faces for a printed pin, a length of filament or a wooden dowel, or none. They go where the cut face is widest, as many as fit (or the number you set), their size from the face (or yours), and only where the piece behind is deep enough to hold them with a wall round them. The line underneath says how many pieces and whether they all fit the bed.

add a support box puts an open box round the mould, 15 mm clear of it, wall 3 mm: stand the mould in it and fill round it with sand or plaster, so a heavy pour does not burst a thin shell. It is two plain objects (an outer box with an inner one cut out), so resize it like any object.

Lenses for glasses (a study tool)

Not medically approved. These are lens *shapes* for designing a frame around - never make lenses from them to look through. Real spectacle lenses need optical material and are medical devices: an optician orders them from a lens lab.

The ⚭ icon in the make-it group opens the lens panel. Type the prescription per eye - sphere, cylinder, axis, PD (from the nose line to the pupil), prism and its base, and the **fitting height** (*fit h*: the pupil above the bottom of the lens shape, as the optician measures it with the frame on - the optical centre goes there) - and the lens and fit: material (polycarbonate and Trivex do not shatter), front curve, minimum thickness, eye-to-lens distance (and the distance at the eye test, so the power is corrected), tilt and face wrap.

Every change redraws both lenses in the 3D view a moment later, with the pupils marked. **The face check** turns each lens green when it fits and red when it does not, with the numbers: too close to the face, or the temple corner sticking out. It measures against a face curve, or against your own face scan (select it and click *use the selected object as the face*).

The lens shape is how the optician cuts the lens: round, oval, rounded rectangle, cat-eye, or your own outline (draw it as an object's profile at real size, the temple side to the right, select it and click *take the outline*). Width A, height B and the bridge DBL, and the edge: a V-bevel for a full rim, a groove for a half rim, flat for rimless.

Fit to the face - nose and head. The pupil is rarely in the middle of the lens: the box (A wide) starts DBL/2 from the nose line and the pupil sits at the PD, so it usually lands a few mm towards the nose, and the fitting height puts it above or below the middle. The panel says by how much. Then the measurements for the frame itself: the **nose** - its width where the pads sit, how high its top (where the bridge rests) is above the pupils, how far down the pads touch, how far it stands in front of the eyes, and its **splay** and **frontal** angles (how its sides open towards the back and downwards); and the **head** - its width at the temples, above the ears, and how far back and how high the ears are, and the **hinge height** (where the temples start on the frame front, above the pupil line). With **show the guides** they are drawn live in the 3D view and follow every number at once: the lens boxes and pupils (grey and blue), the nose as an orange wedge with the bands where the pads rest, and the head width with the temples' path to the ears (purple). Design the frame against them. Below the numbers the panel says what they mean - how much room there is for the rim at the nose, how wide the end pieces get, how long the temples are - and warns when something will not fit. **Fit the lenses to the face** sets A, B and DBL for these measurements while the pupils stay exactly where they are: DBL becomes the nose width at the pads plus the *frame beside the nose* on both sides; A grows out towards the temples, but only until the pupil is 3 mm from the lens centre - the limit an optician aims for, so the lenses stay thin and even and fit the blank - and wider end pieces (at least *end pieces at least*) take the rest of the head width; B keeps the shape's proportions and the fitting heights keep each pupil's height in its lens. The panel also says how big a blank each lens needs (the farthest corner of the shape from the pupil, twice, plus 2 mm) and warns when that is more than the blank diameter. **The nose is a ridge**, not a block: from its top it runs down and comes forward at the **nose slope** (its angle from vertical, seen from the side), and its sides open backwards at the splay angle (and a little more downwards, the frontal angle) so that it is exactly as wide as you measured where the pads touch it. Tick **the nose runs on through the frame front** to see its outline carried forward through the frame - where the frame crosses it, it has to stand in front of the nose. **Shape the lenses round the nose** (on at first) pushes each lens's nasal side out of the nose wherever it would come closer than **lenses**

clear of the nose by - usually the lower inner corner, as real frames do; the cut lenses, the cutters, the data sheet and the OMA file follow it. Without it the panel warns how far the lenses reach into the nose, and a red cross marks the spot. **Check the selected frame against the nose** measures how far the objects you selected - your frame - reach into the nose, and marks those spots red. *Also add the guides as objects* makes the nose as a negative block (in a combine with your frame it keeps the frame off the nose), the temples' paths as thin rods and the ear points. While the lens panel is open, these measurements also go with the prompt in *Make it with AI*, so a frame an AI writes leaves room for the nose.

Add to scene makes the cut lenses as parts, with their 3D edge. Options add: *cutters for the frame* (the lenses grown by a clearance, already negative - put them in a combine with your frame and they cut the opening and the groove), *the frame surface* (a curved sheet through the front of both lenses at your wrap - a base for the frame front) and *markers at the pupils*.

Save data sheet writes an A4 PDF with the prescription, the lens and fit, the thicknesses, the frame numbers and both shapes at true size (with a 50 mm bar to check the print), and beside it an OMA shape file - the format optician edging machines read. Check it with your optician's software before relying on it.

Exporting

- **STEP** — exact surfaces, no meshing. Use this for other CAD. Flat faces are written as true planes, so they can be mated to and sketched on at the other end.
- **IGES** — exact too, for older CAD that will not take STEP.
- **BREP** — exact, OCCT's own format: FreeCAD, CadQuery and anything else built on the same kernel read it with no translation.
- **STL** — triangles, for printing, milling, and most fabrication.
- **3MF** — the newer printing format: it states millimetres, keeps the mesh indexed (so a slicer can check it is watertight) and carries **colour**.
- **OBJ** — triangles, with a material per object colour - **or quad patches**: every CAD face becomes its own grid of quads on the true surface, watertight, like MoI's output. Chosen in the export dialog, with an **aspect ratio limit** and **avoid smaller than**.
- **PLY** — triangles with a colour on every vertex (MeshLab, Blender).
- **GLB** — glTF for the web, Blender and the system viewers, with vertex colours. It is converted to metres and Y-up, as that format requires.
- **Drawing…** — a workshop drawing of the model with **hidden lines**: tick the views (**front, top, right**, left, back, bottom and an **iso**), visible edges solid, hidden edges **dashed** (not in the iso, unless you tick **hidden lines in the iso and the 3D view's angle too**). Tick **as seen in the 3D view** for one more view at exactly the angle the 3D view shows now: line art of the model, hidden lines dashed if you want them. Each view has its overall size under it, at **1:1** when it fits

the page (A4, A3 or Letter, landscape) or the nearest standard scale - or the scale you pick. **First-angle** projection (ISO: the top view below the front, the view from the right on its left) or **third-angle** (US). A **section** cuts the part at a plane along X, Y or Z (the middle is offered) and draws it seen square on, the cut face **hatched** and the bore left clear. The title block carries the title, scale, projection, date and SweeperCAD. Saved as **PDF** (a page to print), **SVG** (true size in mm, for Inkscape or a browser) or **DXF** (true size, on layers VISIBLE, HIDDEN, HATCH, TEXT - for CAD). The lines come from the kernel's own hidden-line algorithm on the exact solid: outlines of round faces included, smooth seams left out. From a script: `model.drawing("part.pdf", views=("front","top","right","iso"), sections=[("y", 15)])`; from the command line, `--out part.pdf`.

UVs, at real size. OBJ and GLB carry texture coordinates where one UV unit is 100 mm everywhere, so a tileable material is the same size on every part - good for renders. Quad patches get UVs along the surface (around the profile, along the spine); triangles are box-mapped.

Polygon parts and CAD files. STEP, IGES and BREP are exact surfaces, so an imported **mesh** has no place in them. With polygon parts in the model the export asks: **Leave them out** (a cut made by a mesh is left out too, and the status line says how many parts were skipped) or **Include as faceted solids** (every triangle becomes a flat CAD face).

Lattices are always exported at **full detail**, whatever the view shows. They go into STL, OBJ, 3MF, PLY and GLB as meshes.

A picture of the model. The camera icon (📷, top right) saves a PNG of the model as it is shown - any view mode, chrome included - at 1× to 4× the view's size, with a **transparent**, the view's own or a white background, and trimmed to the model if you like. Only the model: no grid, gizmo, dials or selection. Made for sharing.

Line art. In the same window, **save as: line art** keeps exactly the 3D view's angle but writes the model's **edges** instead of the shading - taken from the exact solid, at **true size in mm**, with **hidden lines dashed** (or left out). No page, frame or title block: just the lines, as **SVG** (Inkscape, Illustrator, a browser), **PDF** or **DXF** (layers VISIBLE and HIDDEN). For a full workshop sheet with several views, use Export ▸ Drawing....

With reference objects. When a reference (a scan, an imported mesh) is loaded, the export menu offers **+ with reference objects**. Ticked, every visible reference is written as its own STL next to the exported file - `part - ref scan.stl` - in the same coordinates, so any other program loads them in the right place relative to each other, as separate bodies ready for a boolean.

Mesh settings, and a look first. Exporting anything that has to be meshed — STL, OBJ, 3MF, PLY, GLB — opens a small panel: a slider from draft to very fine, triangles or quad patches where the format takes them, welding, and roughly how many triangles and megabytes that will be. **Preview in the 3D view** draws the actual mesh at those settings, so you see the faceting before you commit; closing the panel puts the model back. STEP, IGES and BREP carry the exact surfaces and need none of this.

A curve on paper. The small sheet icon in a graph's bottom-right corner - and **Export curve as PDF…** in the File and export menus - draws that graph: the curve at a stated scale (1:1 where it fits, otherwise a proper drawing scale), its overall width and height, every rounded corner and every arc labelled with its radius, the points marked, the origin shown, and a title block with the object, the graph, the scale and the date. Vector, so it stays sharp at any zoom - for the workshop, for checking, or to keep with a project.

When a sweep would fold. The spine turns **red** and says the bend is too sharp for the profile; the profile turns **amber** and says what it was built at. Both clear as soon as the profile fits the bend.

When a sweep would fold. A profile that reaches further than the spine's bend is sharp would pass through itself on the inside of the curve. The app scales the profile down until it clears - keeping its shape, only smaller - and says so in the status line. To keep your intended size, widen the bend, slim the profile, or use the scale graph to pull it in only through the curve.

Separate bodies. After an export the status line says when the file holds more than one body: a **note** when you exported several objects together, a **warning** when one object or combine is itself in pieces - parts that do not quite touch, or a union that did not fuse - which a slicer would print as loose parts. A hollow part (a closed cavity inside) is one body and is not flagged.

Colour in exports. Every face takes the colour of the object it came from — end caps included, and also after a fillet or a chamfer. The fillet or chamfer surface itself belongs to neither object, so it is neutral grey. A combine given its own colour is that colour throughout.

After an export, the status line offers **Get this made**: a link to sweepercad.com for having the part fabricated — printed, cut, milled or otherwise. Opening it is a choice; the app itself never connects to anything.

What gets exported

Every top-level result, not only a final combined one. Three objects you never combined export as three solids in one file — they are all part of what you made.

- **Hidden** objects are **left out**. Hiding is how you say "not part of this", so it applies to export too.
- **Locked** objects **are** included. Locking only stops you selecting or dragging something; it does not mean it is not there.
- Objects consumed by a combine or an array are not exported separately — they are already inside their result.

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8. Scripts and AI

Made to be driven by a program, not only by a person

SweeperCAD saves plain readable JSON, keeps geometry as named curves rather than baked triangles, and its kernel runs with no window at all. **The app you downloaded is also the command line** - the same program, started with a word after it (on a Mac the program is inside the app: `SweeperCAD.app/Contents/MacOS/SweeperCAD`; run it from a terminal):

```
SweeperCAD build part.sweeper                    build and report
SweeperCAD build part.sweeper --out part.step    ...and write it
SweeperCAD build part.sweeper --set wall=3 --set lid.length=40 --out lid.stl --save
lid3.sweeper
SweeperCAD params part.sweeper                   what the file lets you --set
SweeperCAD run make.py --out part.3mf            a script, as the script editor runs it
SweeperCAD run tweak.py --open part.sweeper      ...on a project opened first
SweeperCAD build part.sweeper --out part.pdf     a drawing (also .svg, .dxf)
SweeperCAD selftest                              proves no window is needed
```

`--set` changes a number before the build: one of the file's **names** (`wall=3`) or a part's own number (`lid.length=40`, `post.draft=2` - the ones the parameters chapter lists as drivable), with maths and names allowed (`wall*2`). Every field bound to that name follows, so one `--set` can resize a whole design; `--save` keeps the changed project. `--out` is given once per file: `.step .iges .brep .stl .obj .3mf .glb .ply`, or a drawing with hidden lines as `.svg .pdf .dxf`. `--json` makes the report machine-readable. A wrong name says what the file has. The build reports faces, volume, surface area, bounding box, centre of mass, warnings and the error text if it failed - so a script, a build server or an AI can write a part, build it, read what came out and correct itself without anyone in the loop. (From the source, the same through `python cli.py ...`.)

AI_GUIDE.md documents the format, with a minimal example the check suite builds every run, so it cannot quietly stop being true.

Scripting

Script → Script editor... (Ctrl+R). The document is available as `model`, and `math` and `numpy` are already imported.

```python a = model.add("+", "body") model.profile(a, model.ngon(6, 20)) model.twist(a, 90) model.round_rims(a, 0, 2) # the top rim rounded 2 mm
```

for i in range(8): # a bolt circle h = model.add("-") model.profile(h, model.circle(3)) ang = i * 2 * math.pi / 8 model.move(h, x=30*math.cos(ang), y=30*math.sin(ang)) ```

Useful calls: `add`, `combine`, `duplicate`, `delete`, `find`, `link`, `unlink`, `objects`, `move`, `rotate`, `scale`, `profile`, `spine`, `scale_graph`, `twist`, `offset`, `mirror`, `volume`, and curve helpers `line`, `circle`, `polyline`, `ngon`, `star`, `curve_from`.

The newer tools have calls too, so a script - or an AI - can use them:

```
python c = model.combine(a, b, "joined") model.blend(c, 3) # its seams rounded 3 mm ("chamfer" for a bevel) model.bridge([arm, hand], "wrist") # parts apart joined by one smooth piece row = model.array(peg, 5, model.line((0, 0), (100, 0))) # 5 in a row along X ring = model.array(tooth, 8, model.circle(40)) # 8 in a ring round it model.hollow(vase, 2) # 2 mm walls, closed all round for f in model.faces(a): print(f["name"], f["what"]) # its faces, by name model.push(f"n{a.id}.seg1", 5) # a side pushed out 5 mm model.round_edge(f"n{a.id}.seg0", f"n{a.id}.seg1", 3) # the edge between them rounded model.net(f"n{a.id}.cap1", 3) # a 3 x 3 net on its end... model.net_point(f"n{a.id}.cap1", 1, 1, 6) # ...its middle lifted 6 mm model.cage(ring, 3) # a box cage round the whole ring... model.cage_point(ring, 1, 1, 2, z=10) # ...its top middle point 10 mm up part = model.import_file("bracket.step") # a CAD file as a part
```

A part's faces are named `n<id>.cap0` (its start), `n<id>.cap1` (its end) and `n<id>.seg<k>` (its sides, in the order of the profile), where `<id>` is `part.id`.

Everything goes through the same calls the buttons use, so a scripted model is identical to a drawn one.

`model.link(b, "profile", a)` makes b's profile follow a's. **Loops are refused** — if a cycle existed, each rebuild would overwrite one drawing with the other and the original would be gone with nothing to undo it back from.

A script runs with full Python access. That is stated plainly rather than pretended otherwise — a restricted namespace is not a security boundary, and one that looks like a sandbox invites misplaced trust. Only run scripts you would be willing to run in a terminal.

Make it with AI

Any AI you already use - Claude, ChatGPT, a local one - can build things in SweeperCAD for you. Open **Script → Make it with AI...** (or the ⌨ icon) and follow the steps at the top:

1. **Describe it** - "a vase, 120 mm tall, wavy rim, 3 mm wall" - and press **Copy for AI**.
2. **Paste it into your AI** and copy the answer it gives you.

3. Press **Paste the answer & run**. The object appears in the 3D view.

Didn't work? Press **Didn't work? Copy the problem for AI**, paste that into the same AI, copy its new answer and do step 3 again. That also goes for a script that runs but builds something wrong (a cut the geometry kernel couldn't make, an offset too big for the shape): once the 3D view has built it, the problem is listed under the script and the button lights up.

Rebuild a mesh. Press **Mesh to rebuild…** and pick an STL or OBJ. It shows see-through in the 3D view and is never part of your model. Say what it is - "rebuild this bracket" - and the AI gets its measurements: its size, its outline wherever it cuts it, a part straight from an outline. Each round of *Do it for me* checks how close the result comes - how far the surfaces stand apart, how much they share, where the biggest misses are - and is done when it is within the distance set beside the button. The result is ordinary parts: change them, or ask the AI to, like anything else. In scripts: `model.reference(path)`, `model.section(axis, at)`, `model.along(axis, n)`, `model.part_from_section(axis, start, end)` and `model.compare(tol)`.

Want it different? Step 4 lights up once something is made. Type the change - "make the rim wavier and add a foot" - and press **Copy for AI**. That copies the script as it is, everything you asked so far, the change - and what the last script actually MADE, measured: every result's size, where it sits, its volume, whether it is one solid or pieces that do not touch, and anything the build reported. Press **Copy pictures** too and paste them into the same chat after the text: the result from the front, the side, the top and at an angle, in one picture. So the AI checks its own work against what you asked - the loop closes without you having to describe what went wrong.

Examples from your shelf (optional, off at first). Tick *Add similar parts from my shelf as examples* and SweeperCAD looks on your asset shelf for parts with a similar name or note - your own, the starter shapes, and what you took from the online library - and puts up to two of them in the prompt, as scripts that build them. The AI does not remember anything between conversations, but this way it starts from what already worked. An example can also carry its mistakes into the new part, so use it when your shelf holds parts you trust. Only what is already on your computer is used.

Pictures of what you want. Under step 1, **Add…** up to four photos, sketches or screenshots of what you are after. They go with the prompt: **Copy pictures** puts them in a band marked *what I want* above the four views of the result, to paste into the chat after the text; when it runs by itself they are sent with every round, so the AI compares its result with them. The prompt tells it that your words and measurements count for more than what a picture seems to show.

Sketch it. Sketch… (beside *Add…*) opens a sketch pad: draw what you want with the mouse or a pen tablet, like on a napkin - a side view or an outline says most. **Free** draws free-hand; **Line** draws straight lines (press, pull, let go) that snap level and upright and onto the end of an earlier line, so an outline closes. **Curve** draws a smooth B-spline: click its control points (the curve is pulled towards them and runs through the first and last), double-click or right-click to end it, or click the first point again to close it; Undo takes the last point back while you draw, Escape drops the curve. Eraser, three widths, undo and a faint grid; or **Draw over a photo…** to trace a real napkin sketch or an object. **Draw**

on the model puts the 3D view as it is now under the paper, to give directions on what is there: circle a part that should change, draw an arrow where something should move or grow, a line where it should be cut - marks on the model are drawn in red, and the AI is told what red means. Turn the view to the side that shows it best first. **Use this sketch** adds it to the pictures. The sketch shows the shape; the sizes still go in your words. (An AI on your own computer needs to read pictures for this - not every local model can.)

Let it run by itself (only if you want). SweeperCAD makes no network calls. In **AI settings…** you can switch on talking to an AI directly: Claude or ChatGPT with your own account and API key (kept on this computer only), or an AI running on this computer (Ollama, LM Studio) - then nothing leaves your machine at all. Before anything is sent off the computer you are told exactly what goes (what you describe, the script, the measurements, the pictures - nothing else) and asked to agree. While it is on, the status bar says ● **AI on**. Then ▶ **Do it for me** does the steps for you: it asks, runs the script, measures and pictures the result, sends that back and lets the AI improve it, up to the number of rounds you set - every round is built in front of you, **Stop** stops it, and Ctrl+Z goes back a round. With a change typed in step 4, it changes what is there.

Checked against the design rules. Tick **Check it against the design rules each round** (**Design rules…** opens them in the stats window, to set, load or save). Every round, what the AI made is checked against the set - openings, heights, wall thickness, cavities, strength under a load, printing - with the same analyses as the stats window; what fails, with the numbers, where it is and what would make it pass, goes back to the AI with the measurements. It is not done while a check fails, even if the AI says so: the failures go back to it. The strength figures are estimates, a first look and not a sign-off.

Setting up the AI, step by step. Open the script editor, press **AI settings…**, tick **Let SweeperCAD talk to an AI**, then one of these:

Claude (Anthropic). The API is paid per use and billed apart from a Claude chat subscription. Sign in at **platform.claude.com**, add some credit under billing, then go to **Settings → API keys** and press **Create key**. Copy the key (it starts with `sk-ant-` and is shown only once) and paste it into **API key**. Leave **model** empty for the default, or type another Claude model name. Tip: set a monthly spending limit in that account.

ChatGPT (OpenAI). Also paid per use and billed apart from a ChatGPT subscription. Sign in at **platform.openai.com**, add credit, open **API keys**, create one and paste it into **API key**. Type the **model** name from your account - one that reads images, if you want it to see the pictures.

An AI on this computer - nothing leaves it. Install **Ollama** (from ollama.com), then in a terminal fetch a model that can read images, for example `ollama pull llama3.2-vision`, and leave Ollama running. In AI settings choose **an AI on this computer**, type that model name exactly, and leave the key and address empty. With **LM Studio** instead, start its local server and set the address to `http://localhost:1234/v1/chat/completions`. A model on your own computer is free and private, but smaller ones write weaker scripts - let it run more rounds.

The key is kept in SweeperCAD's settings on this computer only, never in your project files. To stop, untick the box (or clear the key). Then copy its answer and do step 3: the new version *replaces* the last one, it does not pile up next to it. Repeat as often as you like; Ctrl+Z brings back the previous version.

What is copied in step 1 tells the AI everything it needs: how SweeperCAD builds shapes (profile, spine, scale), which way is up, the calls it may use, a worked example, and what is already in your document - so "add a handle to the selected cup" works too. Nothing is sent anywhere by the app; you choose what to paste where. Like any script, only run an answer you are happy to run.

Using SweeperCAD from an AI app (MCP). An AI app on your computer that speaks MCP (the Model Context Protocol) - Claude's desktop app, several code editors, local AI apps - can use SweeperCAD as a tool: read how SweeperCAD models, run modelling scripts, measure the result, look at it, export it (STEP, STL or a SweeperCAD file) and send it to the open SweeperCAD window. It can also **check** what it made with SweeperCAD's own analyses - a set of design rules by its name or from a *.swrules* file, or wall thickness, sealed cavities, openings, heights, strength under a load (estimated, or simulated with where it is held and pushed) and printing one by one - and gets each check back as passed or failed, with where and what would make it pass, so it can change the design and check again. **Settings > AI > Connect an AI app (MCP)...** shows the few lines to paste into that app's MCP settings; press **Copy**, paste, and restart the AI app. SweeperCAD itself sends nothing anywhere this way: the AI app starts a SweeperCAD helper on your computer and talks to it directly. What the AI app does with what it is shown is up to that app - one with an AI in the cloud sends it there under your account, one with an AI on your computer does not. A model the AI app sends to the window is offered there (**Add to this scene**, **Open it** or **Not now**); untick **Settings > AI > Let AI apps send models to this window** to stop that.

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9. When something goes wrong

What the app notices while you draw

Let go of the mouse after editing a curve and it is checked — in a few milliseconds, before anything is built. If something is worth knowing, the status bar says so:

- **crossings** — the curve passes through itself
- **tight bends** — a radius small enough to pinch the sweep
- **very short segments** — they cost sections and add no shape

It also says when the three curves together are asking for something expensive — *"this shape wants 226 sections (~904 faces) — a cut with it takes about 5s"*. That is the same sectioning analysis the build runs, without building: about 20 milliseconds to find out, against seconds to discover the hard way.

The same applies to **cuts between finished shapes**, which no curve can predict: before each rebuild the app counts the faces of the pieces it is about to cut and holds if that will take seconds. Working it out costs a fraction of a millisecond, because those pieces are already built.

If you switch the hold on (Settings ▸ Guardrails ▸ *Hold rebuilds predicted to take seconds* - it is off by default; a heavy build goes to draft mode instead, and pause is yours), past about three seconds the rebuild does not start at all. The app pauses itself and says what it found: *"held — this shape wants 226 sections (~904 faces) — a cut with it takes about 5s. RESUME to build it, or undo to step back"*. Telling you a build takes five seconds while the five seconds are already running is no help; asking first is.

Both of these live under **Settings ▸ Guardrails** and can be switched off independently — the warnings, the hold, or both. The warnings are on by default and the hold is off; both are switchable, because they are judgements about what is worth interrupting for, and you may not agree.

It never refuses anything. A crossing profile is usually a mistake and occasionally exactly what you meant, and the app cannot tell the difference — so it says what it sees and leaves the decision to you. Ordinary shapes say nothing at all: a circle, a square, a star and a cog are all quiet.

When a feature says a package is missing

Running from source, the app uses a few libraries besides Qt: one for offsets, two for the heavier maths (lattices, strength and casting checks and more) and one for images. If your Python lacks one, the app says so when it starts - which package, what needs it, and the exact line to type in a terminal

(it uses the same Python that runs the app, so the package lands in the right place). Everything else keeps working. `pip install -r requirements.txt` in the SweeperCAD folder installs the lot.

When something is taking too long

Past a second and a half, the status bar counts: *working… 12s*. A red **CANCEL** button appears next to PAUSE and disappears when the work finishes.

- **Cancel** stops what is running and leaves everything else alone. The last good result stays on screen.
- **Pause** is different: it stops rebuilding altogether until you resume, which is what you want while making a series of heavy edits.

A cancel lands at the next object rather than instantly — one object's build cannot be interrupted once the geometry kernel has started it.

When something will not build

The status bar explains rather than just failing. A sweep is refused if the profile crosses itself, and an offset is refused if it would collapse the curve — in both cases the message says which and why.

Warnings that begin *"auto-fixed"* mean the shape was built but something had to give: the bend was too tight for the profile, so the sweep would have folded through itself. New objects **ease the bend**: the part keeps its size and the spine is widened only where it is too tight (below). What easing cannot reach - a profile wider than the bend has room for - is then shrunk, and the warning says so. The other choices (object panel, **fix by**): *shrink all sections* - the same amount everywhere, the shape keeps its proportions and comes out slimmer; *squash along bend* - only the sections that bind are narrowed; *ease + shrink* - a little easing, then a little shrinking.

Ease the bend widens the spine only where it is too tight for the profile (a sharp break between two handles counts as the tightest bend): - A **rounded corner** gets a bigger radius and stays an exact arc. - Any other tight stretch is replaced by the arc a ball of the needed size would roll through.

The rest of the spine stays exactly as drawn. When no arc fits where it is needed (a U-turn whose legs are too close, the tightest bend right at an end, a profile wider than the spine is long), the **whole spine is relaxed** instead: its two ends stay where they are and every bend that is too tight opens up, as far as it takes. *Ease the bend* never shrinks the profile - but a big profile on a short, curly spine can come out nearly straight. *Ease + shrink* splits the work: the bend is widened only part of the way and the profile is shrunk for the rest - the spine stays closer to your drawing, the part comes out a little slimmer. So on the same tight bend *ease the bend* keeps the most of the part, *shrink all sections* the least, and *ease + shrink* lands between them.

Easing is never allowed to make things worse: an eased spine that is not clearly wider than the drawing is not used, and the warning says **bend not eased** (the shrink then does it, as a safety net).

Easing happens on the **built** spine; your drawing is untouched. The spine graph shows the eased spine **dashed**. **Bake the eased bend into the spine** (object panel) writes it into your drawing so you can edit it: a grown corner keeps being a corner, and a smoothed stretch replaces those points. Guides and constraints on the spine are removed then, because they counted the old points.

Before a slow build starts, it says so. A scale graph that doubles back many times, or a combine between shapes with a great many faces, is predicted rather than attempted: over a few seconds you get a warning naming the cause; past a minute it is refused and the model is left exactly as it was. Fewer points on a profile is the quickest way down. Tick **allow slow builds** on the object or combine to run it anyway.

If a combine leaves nothing — a negative that covers the positive exactly — the object vanishes for a correct reason, and the status line says so.

Pause stops a rebuild that is running, not just the next one. If the app is busy on something you did not mean, pause it.

Loft mode is *ruled* by default. It is the fastest by a wide margin and exact wherever the surface really is ruled, which is most sweeps.

Auto, hybrid and mitre loft in pieces. Wherever the spine or the scale graph changes character — a straight stretch meeting a curve, a corner — the object is split at its own points. Straight stretches get one section per point and are exact; curved stretches are lofted smooth on evenly spaced sections; a corner is mitred, and next to a curve the sections turn into the mitre gradually so the surface never runs back on itself. The status line says so: *"lofted in 3 pieces (2 straight, 1 smooth), 1 mitred corner"*. An end profile, a twist and an asymmetric scale all go through the same pieces, and so does a closed spine — a ring or a frame has no end caps, it simply meets itself.

Straight and circular sweeps are built exact. A straight spine at a constant size is an extrusion of the exact profile: a circle or an arc in the profile gives a true cylinder, a straight side a true plane. A spine that is a circle or one circular arc - a ring, an elbow - is turned about its centre: tori, cones, planes. The status line says *"built exact"*. This is what makes fillets take: the kernel's fillet is far steadier on true cylinders and planes than on approximations of them, and a fillet that will not take on the exact solid is tried again on a NURBS copy of it automatically. (A "circle" drawn with the preset is four curves 0.03 % outside a true circle; built exact it IS the circle, so its volume reads 0.03 % less than before. That is the drawing becoming what it says.)

A smooth closed spine is one seamless ring. When the spine has no corner anywhere (the point where it closes included), each side of the profile becomes ONE surface all the way round - no joins, nothing to see in reflections. The status line says *"lofted as one seamless ring"*. (A rounded frame - straight runs into arcs - stays in pieces: its bend starts abruptly at each joint in the drawing itself, and the pieces meet exactly there.) A twist works too when turning the profile by it lands the profile on

itself: a square twisted 90 degrees, a hexagon 60 or 120, any whole turn, and a round profile at any twist. A frame with corners, a scale that is different where the ring meets itself, or a twist the profile does not repeat under still goes through the pieces as before.
